

Ionia County Community Health Overview

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Ionia County Health Department

2026

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1 Introduction

This document serves as an overview of health and related socioeconomic factors pertaining to Ionia County and its residents. When possible, metrics are put in the context of state averages and other counties. The materials and analysis provided are meant to highlight key indicators of community health and strengths or areas of growth.

1.1 County Map

Ionia County is the 30th most populated county in Michigan out of 83. Ionia County is considered a rural county, as 61% of the population live in low density areas, yet it sits in between two of the largest cities in Michigan: Grand Rapids and Lansing.

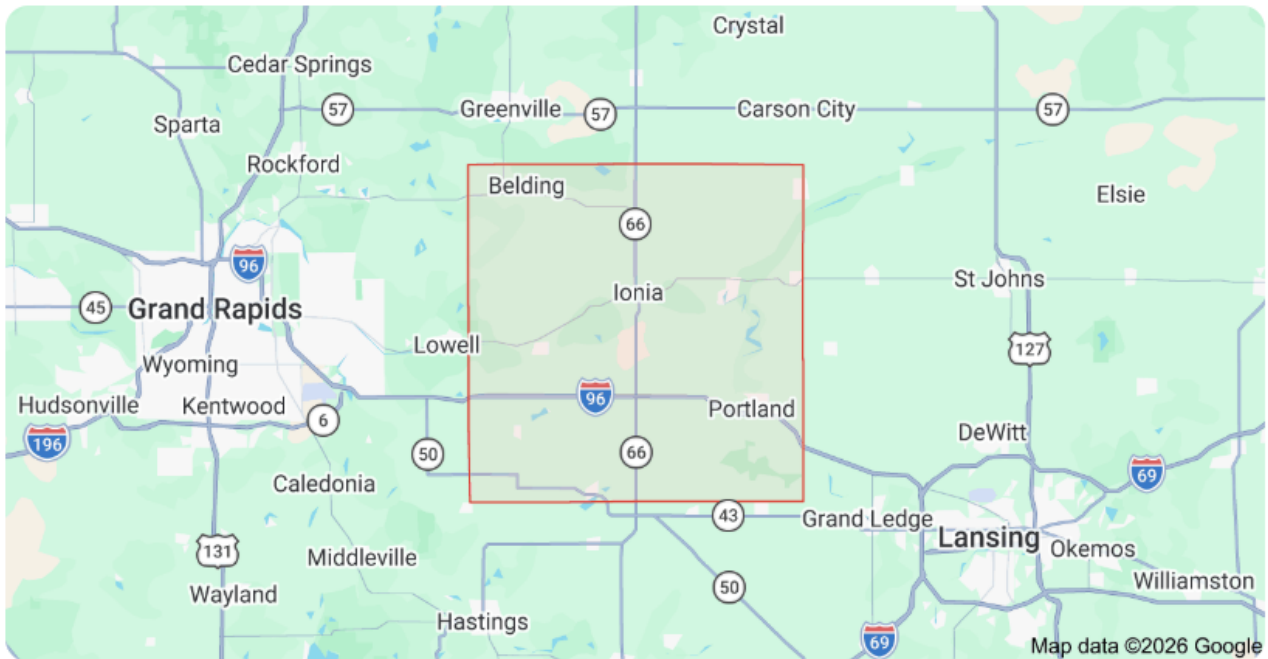


Figure 1: Ionia County

1.2 Census Tracts & Cities, Villages, and Townships

Ionia County has 18 (13 aggregate) census tracts, including 1 enclosing the state prisons located in Ionia County (9822). There are 3 cities, 7 villages, and 16 townships within Ionia County.

2 Demographics

2.1 Race & Ethnicity

Ionia county has a non-institutionalized population slightly over half male (51.6%), and the most common race is white (85.3%).

Gender	
Male	51.6%
Female	48.4%
Race (Non-Hispanic or Latino) N = 63,035	
White	85.3%
Black	3.3%
American Indian	0.5%
Asian	0.5%
Some Other Race	0.3%
Two or More Races	4.8%
Ethnicity (Hispanic or Latino of any race) N = 3,539	
Mexican	4.1%
Puerto Rican	0.3%
Cuban	0.1%
Other Hispanic or Latino	0.8%

Table 2: Ionia County Demographics

The most common race in both the State of Michigan and Ionia County is white (89.3% & 73.9%). The second most common racial identification in Ionia County is two or more races (7.2%), while in the State of Michigan the second most common race is black (13.1%).

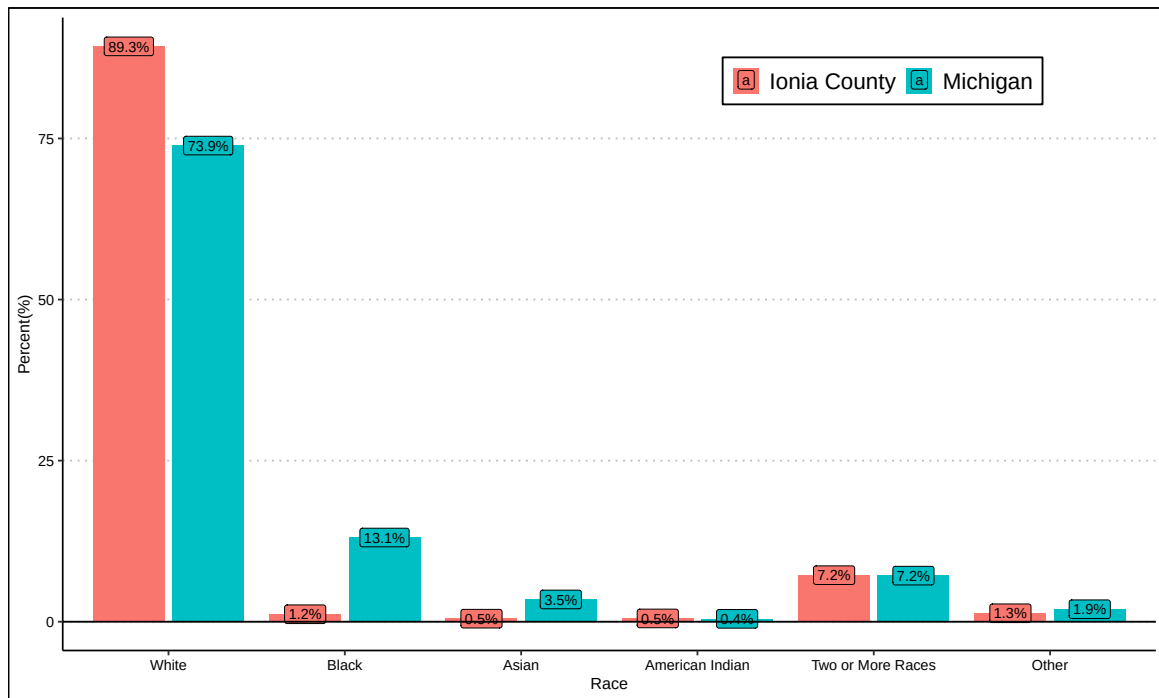


Figure 4: Racial Demographics: Ionia County vs. Michigan

2.2 Age

Ionia County has similar age demographics to the State of Michigan overall.

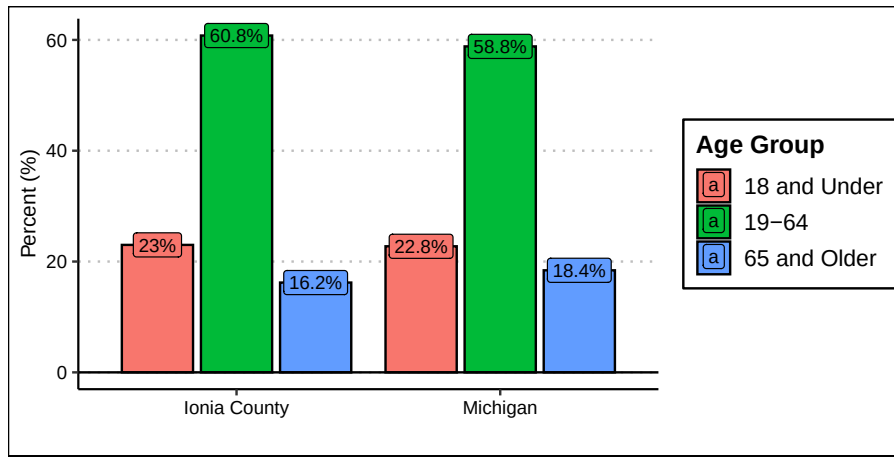


Figure 5: Age Demographics: Ionia County vs. Michigan

Census tract 313.02 (South Boston and Campbell Townships) has the highest proportion of the 65 years and older demographic (22.9%), while tract 317 has the lowest proportion of the 65 years and older demographic (9%). Census tract 301.02 (Lyons and southern North Plains Township) has the highest proportion of the 18 years and under demographic (31.7%), while tract 316 (North of Ionia City) has the lowest proportion of the 18 years and under demographic (15.5%).

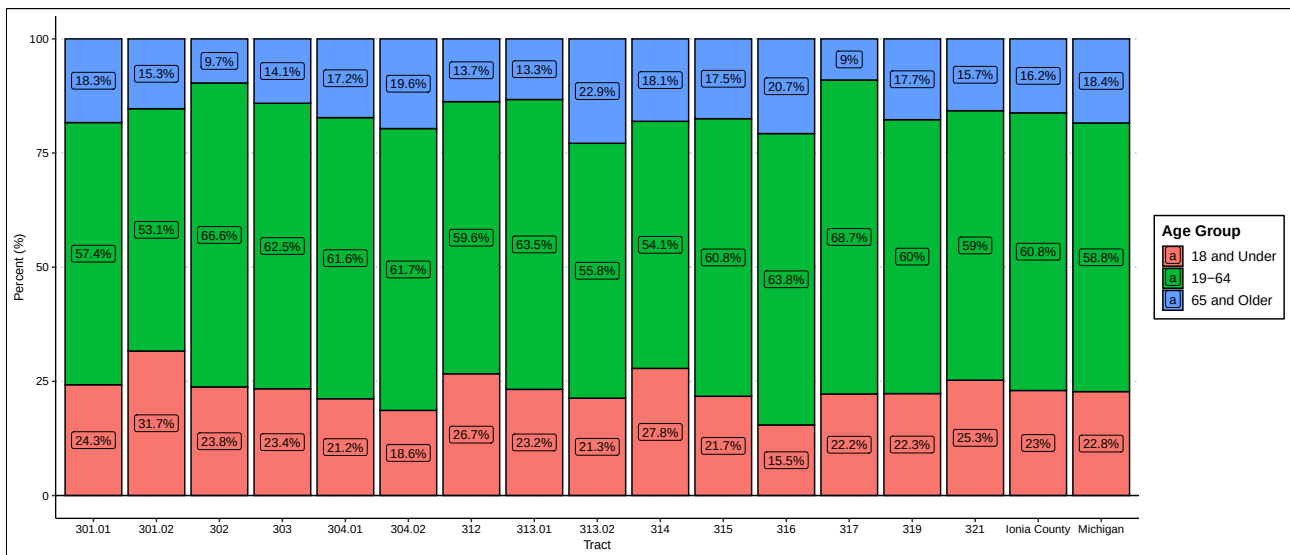


Figure 6: Age demographics by Census Tract

Two ZIP codes have much greater percentages of the population under 18 years old compared to the state and county averages: Pewamo (39%) and Palo (32.7%).

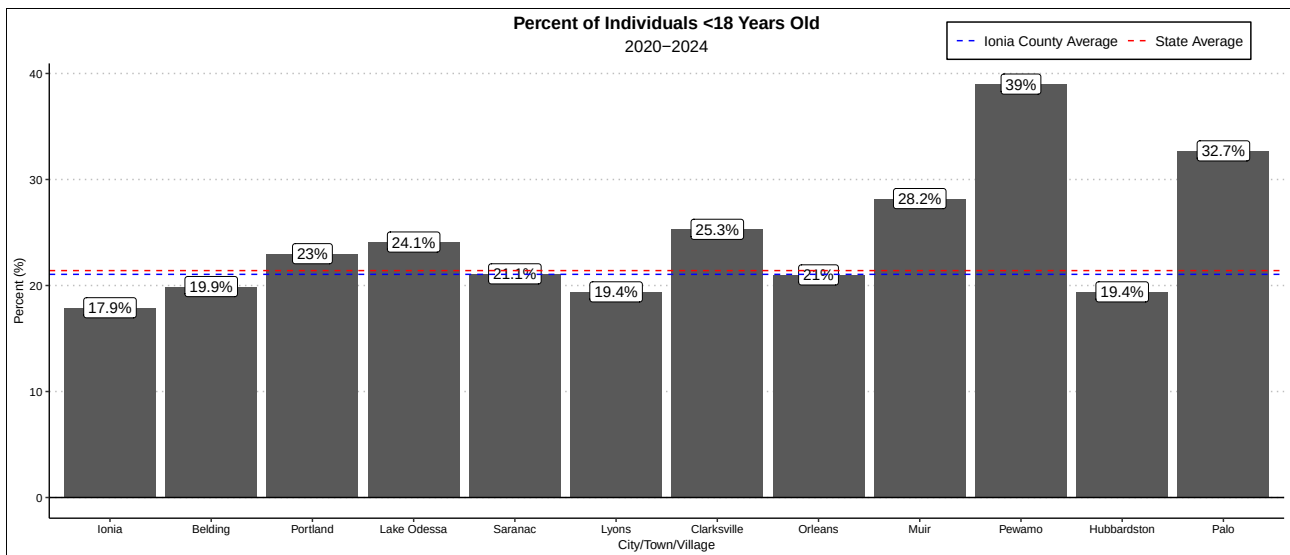


Figure 7: Minor Population by Ionia County ZIP Code

Inversely, Palo has a very low percentage of individuals aged 65 and older (12.9%). Two ZIP codes have higher percentages of individuals aged 65 and older compared to state and county averages: Lyons (23.3%) & Hubbardston (24.7%).

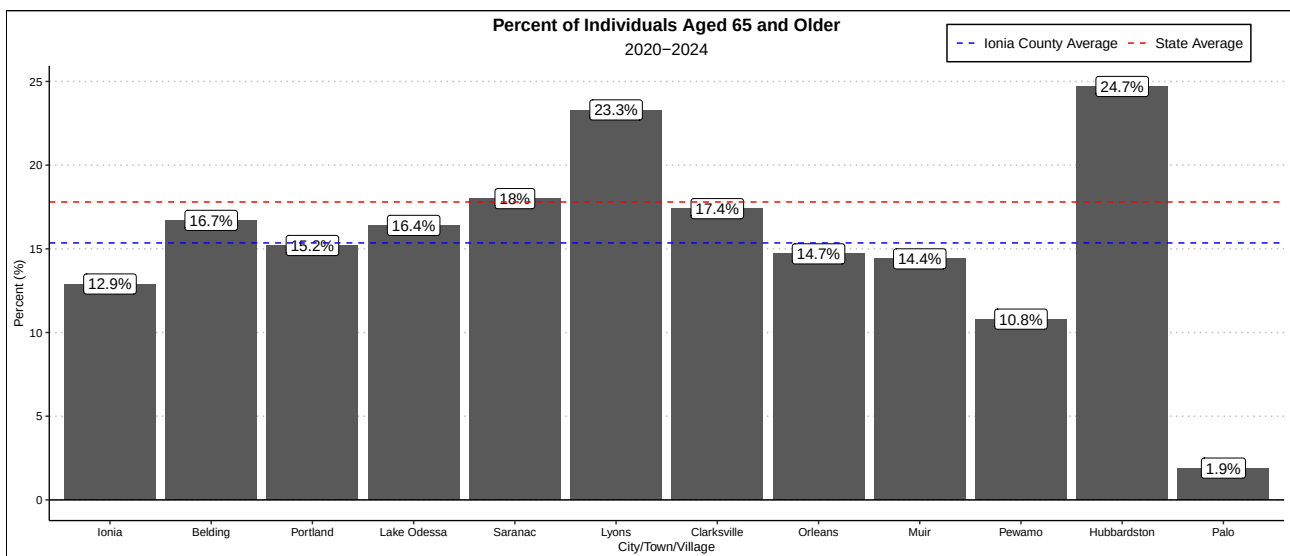


Figure 8: Older Adult Population by Ionia County ZIP Code

2.3 Veterans

Ionia County has the 18th lowest percentage of veterans out of the 83 counties in Michigan. However, the 6.5% of veterans for Ionia county is greater than the state average of 5.7%. Typically, counties with higher populations have a smaller percentage of their population who are veterans. This is why most counties are above the state average for percent of veterans.

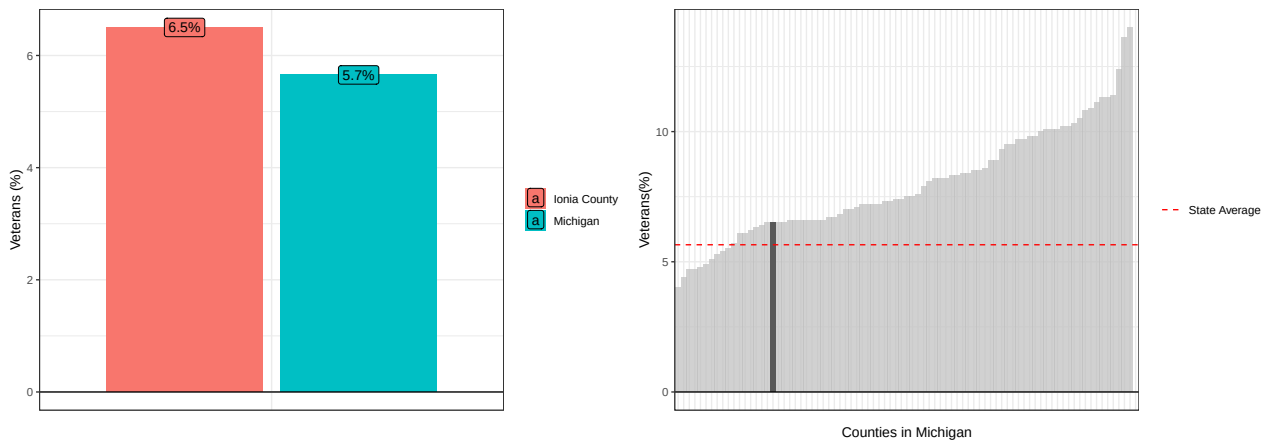


Figure 9: Percent Veterans across Michigan Counties

The census tracts with the highest percentage of veterans are 312 (Odessa & Sebewa Townships) & 319 (Mostly Berlin and Orange Townships), while tracts 302 (East Belding), 314 (Portland City & Township), and 315 (Danby Township) have the lowest percentage of veterans.

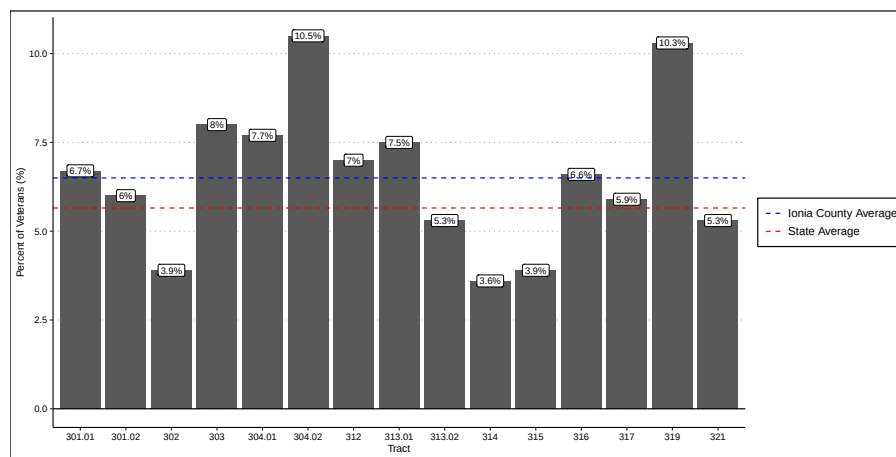


Figure 10: Percent Veterans by Census Tract

2.4 Disability

Ionia county has a similar disability rate (14.4%) compared to the State of Michigan (14.5%):

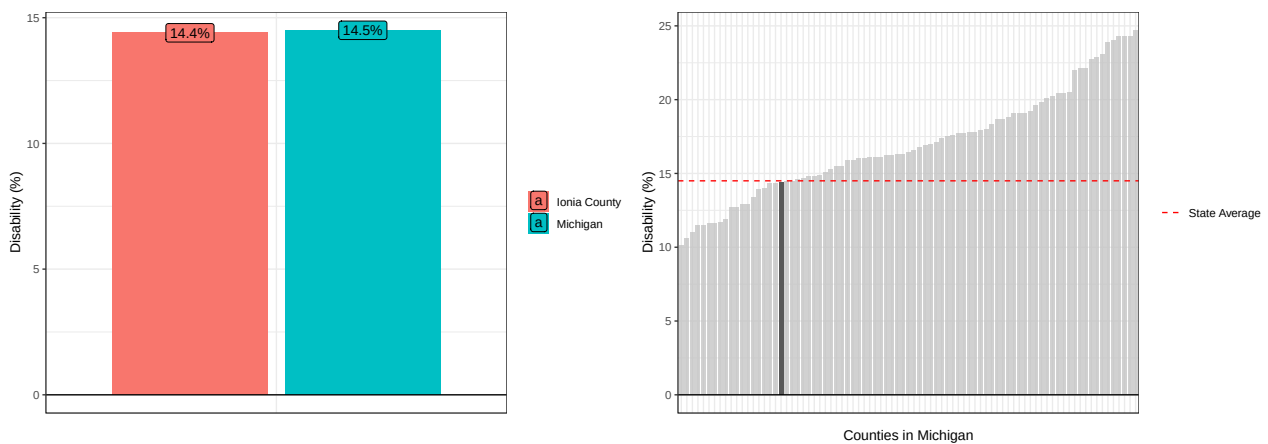


Figure 11: Ionia vs. Michigan: Disability Rate

Disability rates are variable between census tracts across Ionia County. Census tracts 302 & 303 (Belding

City) have the highest rates of disability, while tract 314 (Portland City & Township) has the lowest rate of disability.

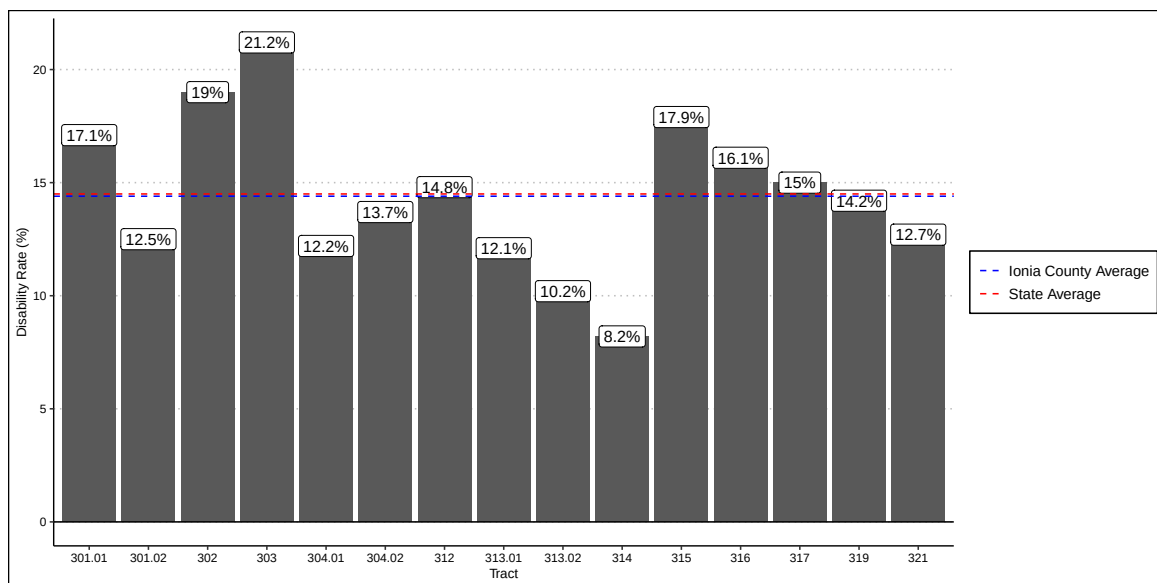


Figure 12: Disability Rate by Census Tract

Ionia County typically has lower rates of disability than the State of Michigan when adjusting for age and disability type. This is the case for ambulatory disability, cognitive disability (except for the 65 and over population), independent living disability, self care disability, and vision disability.

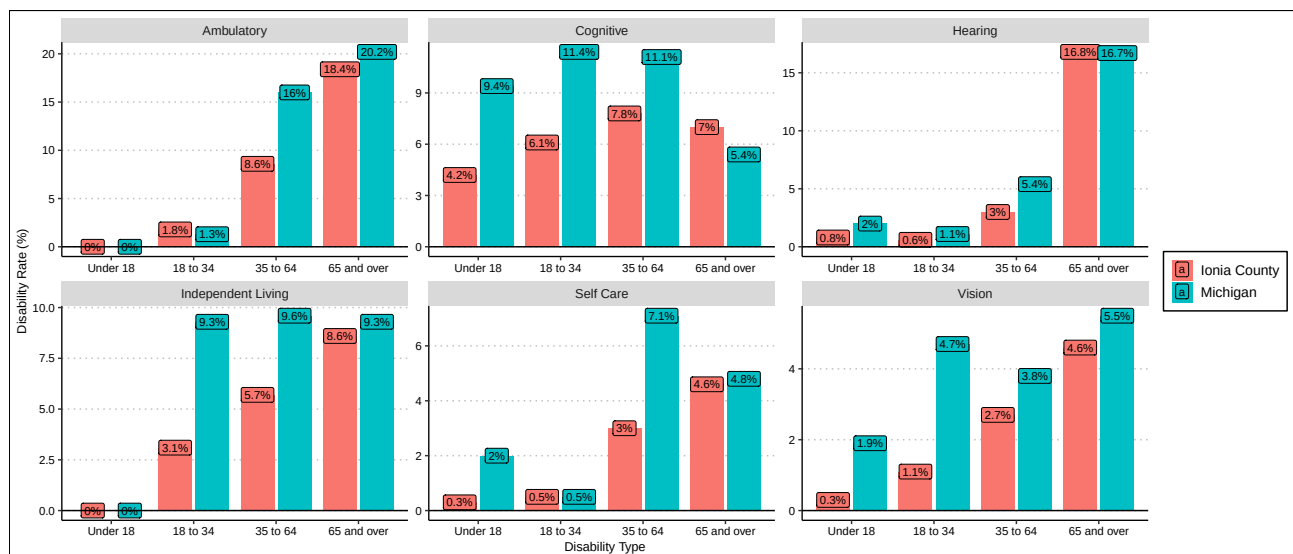


Figure 13: Disability Rate by Age and Type

3 Educational Attainment

Compared to the State of Michigan, Ionia County has a smaller percent of the population with an educational attainment of an associate's degree or more. Also, compared to the State of Michigan, Ionia County has a higher percent of the population with no high school diploma.

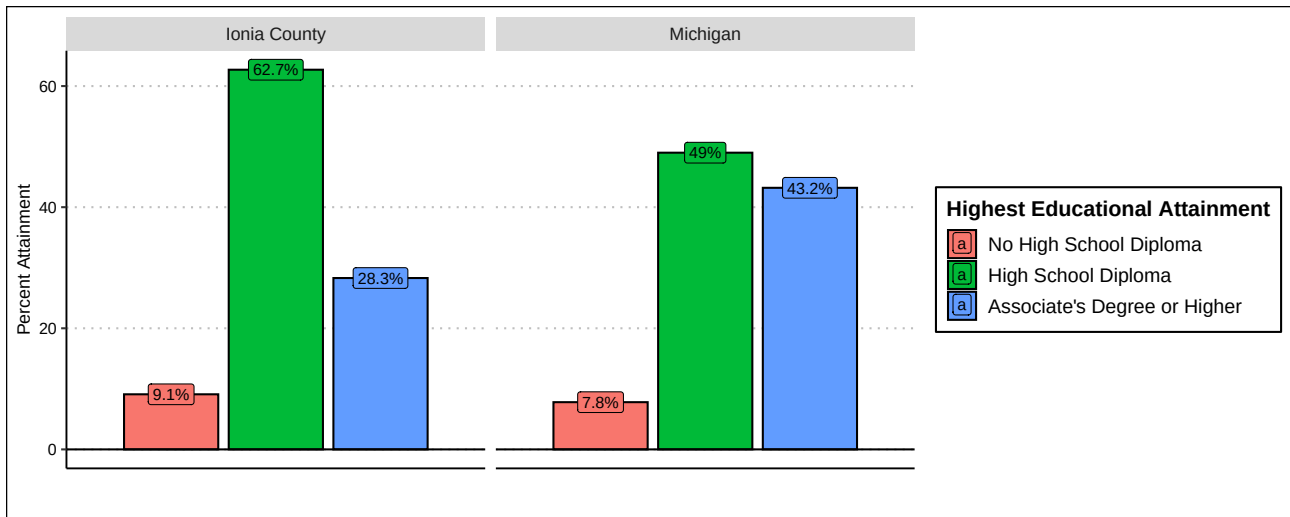


Figure 14: Educational Attainment: Ionia County vs. Michigan

Educational attainment varies between census tracts.

Tract 314 (Portland City and Portland Township) has the highest proportion of the population with an associate's degree or higher (54%), while tract 301.02 (Lyons and southern North Plains Township) has the lowest proportion of the population with an associate's degree of higher (19.9%).

Tract 316 (North Easton & Ionia Townships) has the highest proportion of the population with no high school diploma (14%), while tract 314 (Portland City & Portland Township) has the lowest proportion of the population without a high school diploma (2.3%).

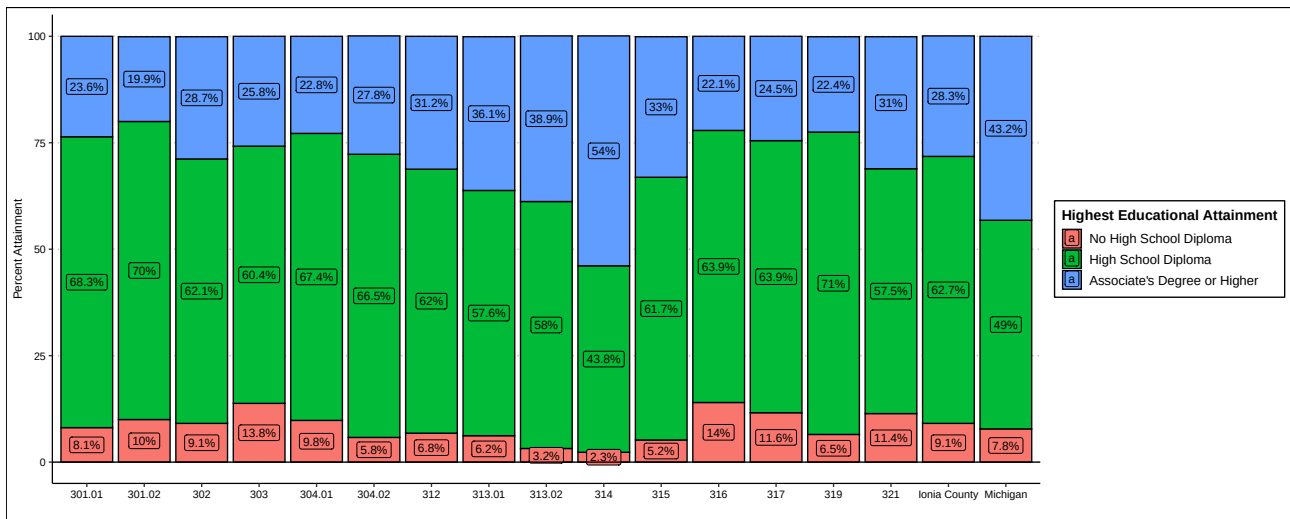


Figure 15: Educational Attainment by Census Tract

4 Housing

4.1 Multi-Unit Housing Structures

Compared to the State of Michigan, Ionia County has a lower percentage of housing structures with 10+ units (6.4% vs. 9.3%). Ionia County has the 22nd-23rd (tied) highest percentage of housing structures with 10+ units in the 83 counties.

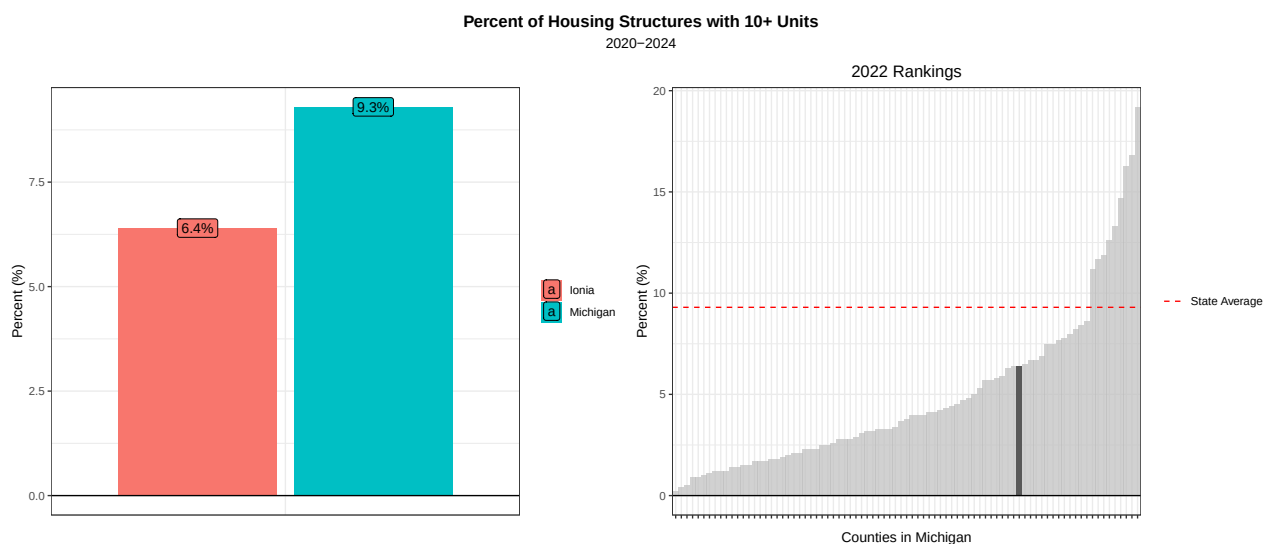


Figure 16: High Unit Housing Structures in Ionia County

There's only one ZIP code in Ionia County, Ionia city, that has a higher percentage of housing structures with 10+ units than the State of Michigan.

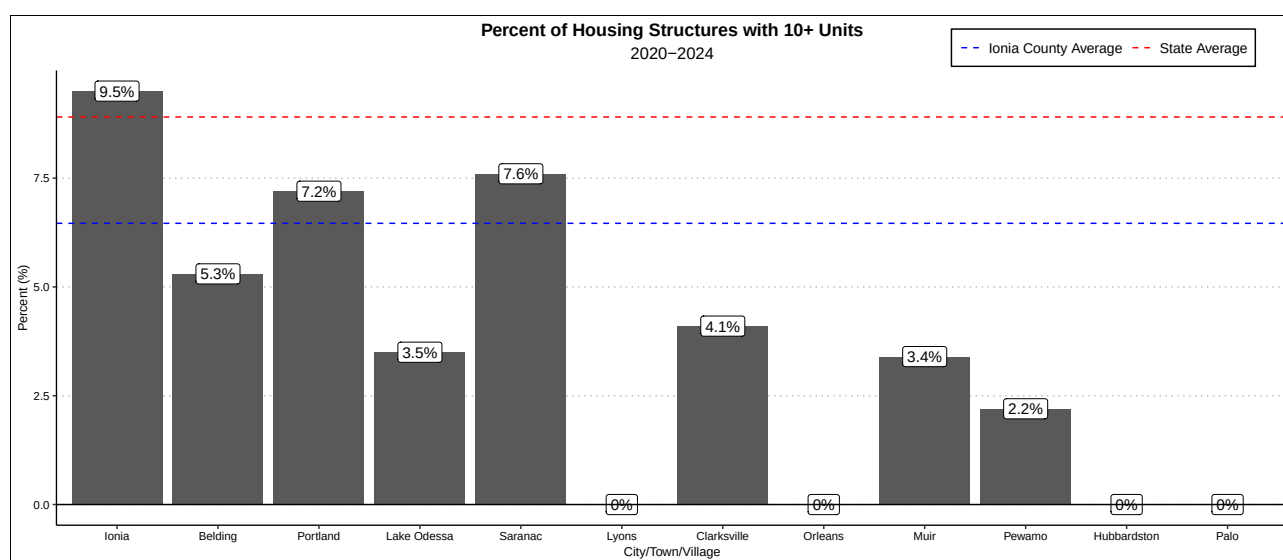


Figure 17: High Unit Housing Structures in Ionia County ZIP codes

4.2 High Housing Cost Burden

Compared to the State of Michigan, Ionia County has a lower percentage of households with high housing cost burdens (19.6% vs. 24.5%). Ionia County has the 67-68th (tied) highest percentage of households with high housing cost burdens in the 83 counties.

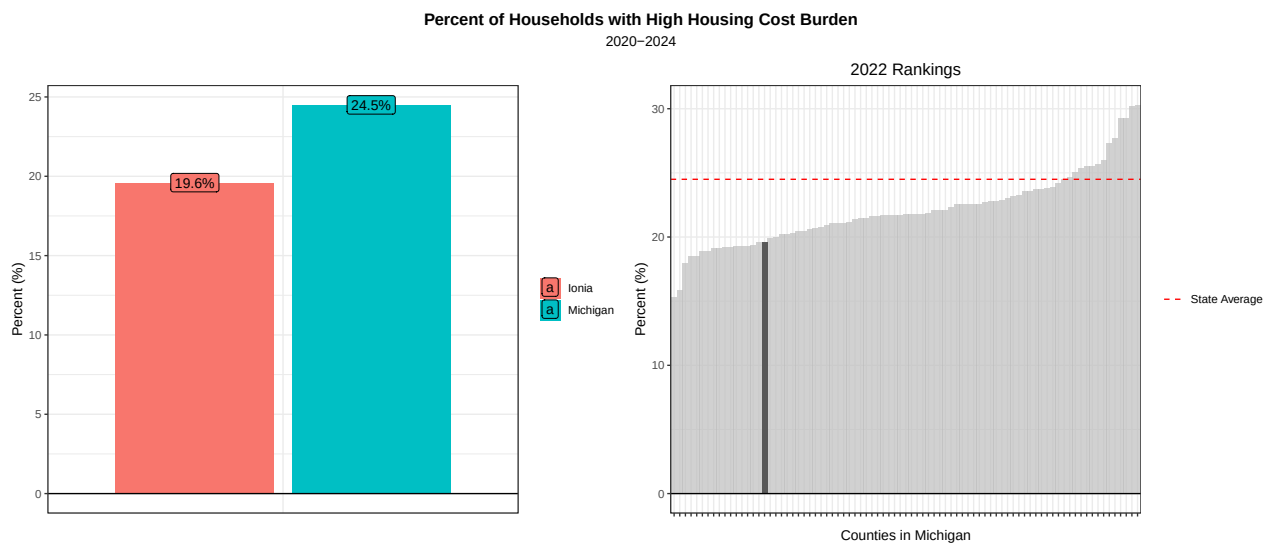


Figure 18: High Housing Cost Burden in Ionia County

There's only one ZIP code, Lyons, that has a noticeably higher percentage of households with high housing cost burden than the State of Michigan.

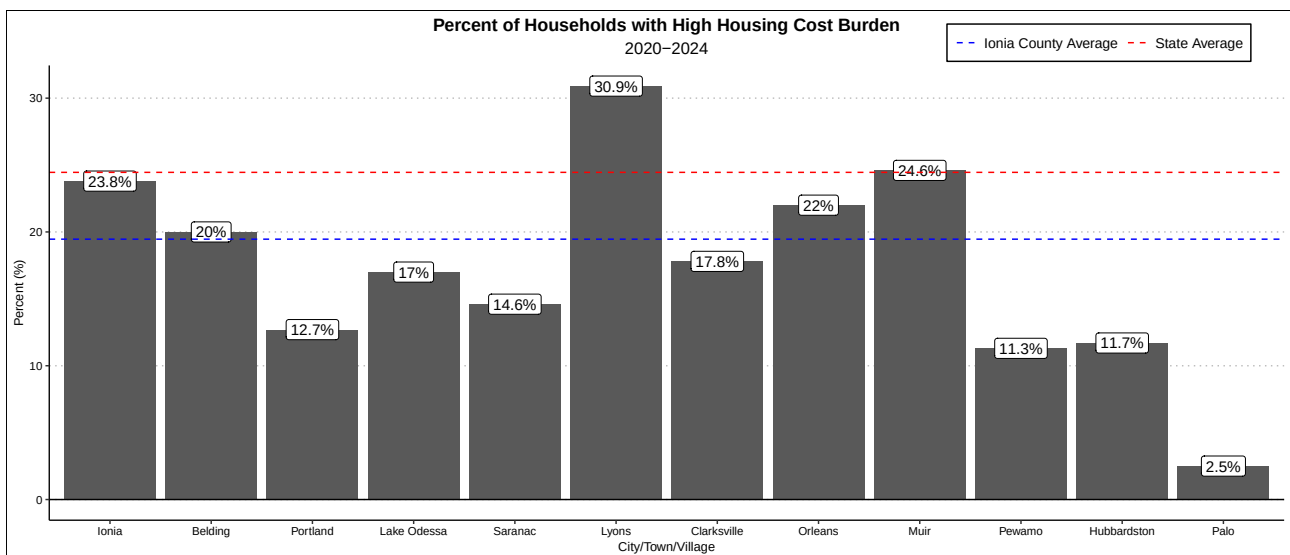


Figure 19: High Housing Cost Burden in Ionia County ZIP Codes

4.3 House Size Too Small

Compared to the State of Michigan, Ionia County has a similar percentage of households with more people than rooms (1.5% vs. 1.6%). Ionia County has the 40th highest percentage of households with more people than rooms in 83 counties.

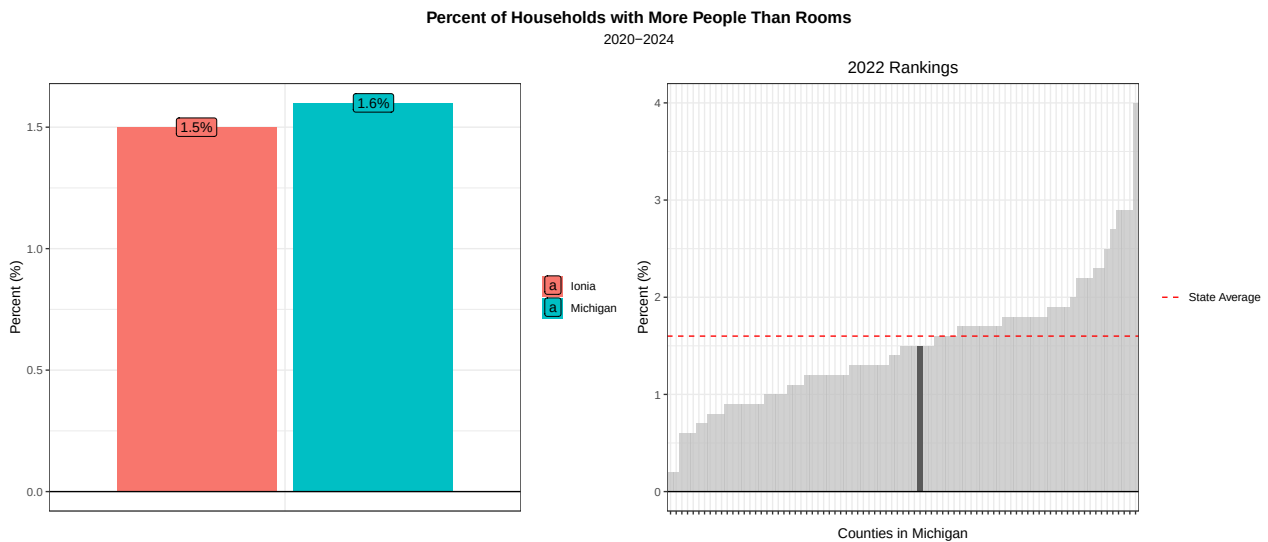


Figure 20: More People than Rooms in Ionia County

Two ZIP codes, Pewamo and Hubbardston, have a significantly higher percentage of households with more people than rooms than the state average. Clarksville and Orleans also have a noticeably higher percentage of households with more people than rooms than the State of Michigan.

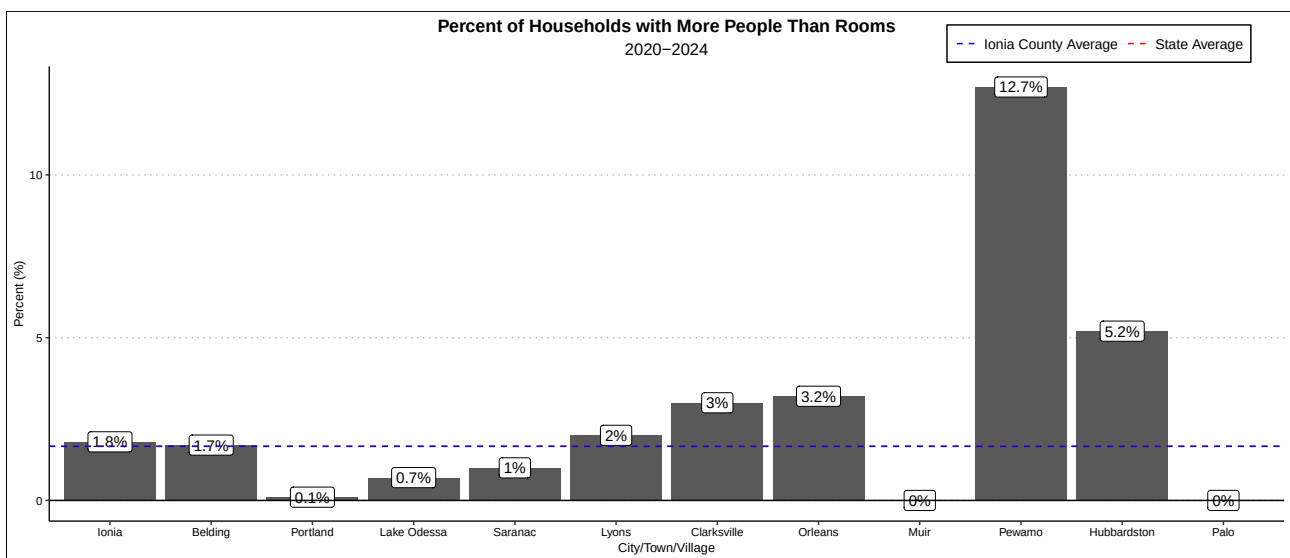


Figure 21: More People than Rooms in Ionia County ZIP Codes

4.4 Living in Group Quarters

Note: Group quarters refers to living in a group arrangement in a facility outside of a typical household, ran by another organization. This can include correctional facilities, college dormitories, rehabilitation facilities, etc.

compared to the State of Michigan, Ionia County has a higher percentage of individuals living in group quarters (5.1% vs. 2.1%). Ionia County has the 14th highest percentage of individuals living in group quarters in 84 counties.

Remark: Ionia County has a correctional facility population between 3,000-4,000. This will be especially apparent in Ionia City, where the correctional facilities are located.

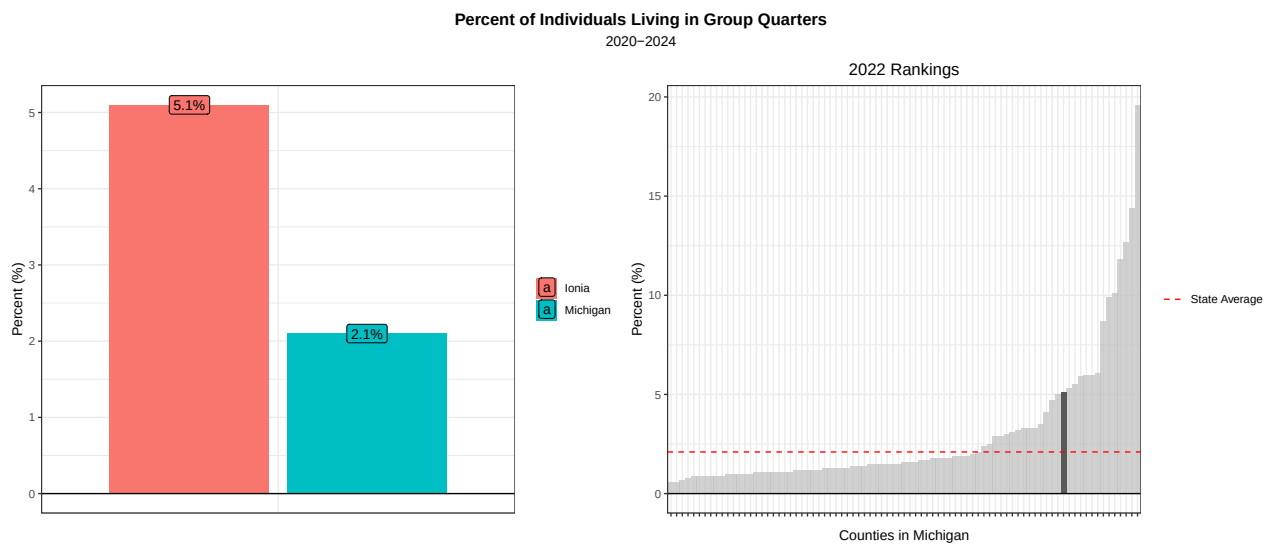


Figure 22: Group Quarter Living in Ionia County

Compared to the State of Michigan, as previously mentioned, Ionia city has a large percentage of individuals living in group quarters. Hubbardston is also notably higher for individuals living in group quarters.

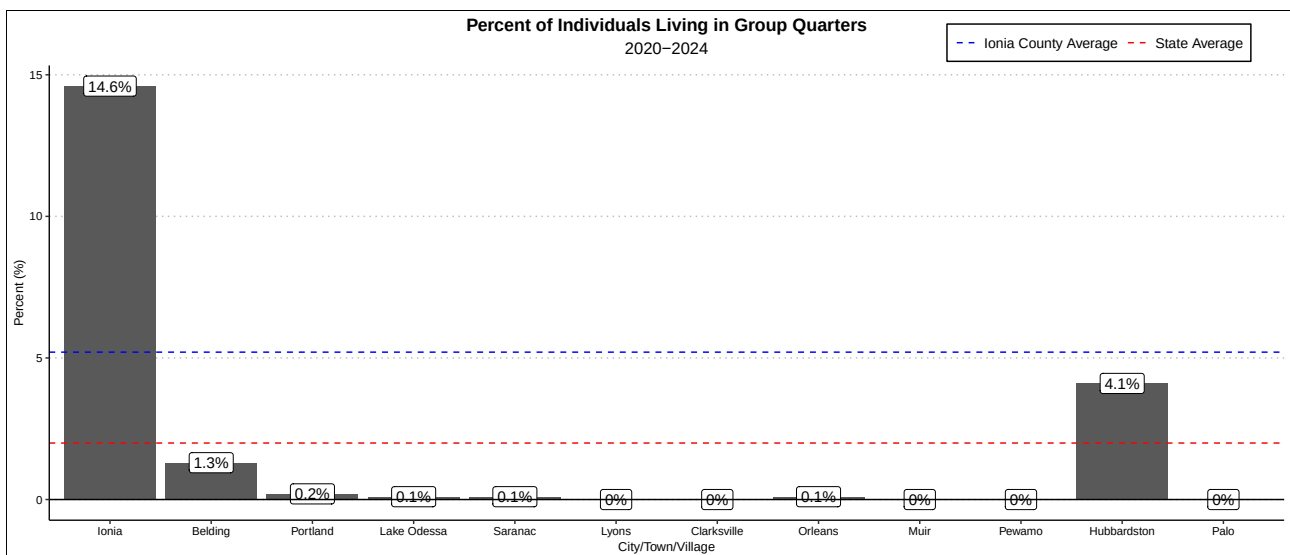


Figure 23: Group Quarter Living in Ionia County ZIP Codes

4.5 Mobile Homes

Compared to the State of Michigan, Ionia County has a much higher percentage of housing units that are mobile homes (11% vs. 4.9%). Ionia County has the 17th highest percentage of housing units that are mobile homes in 83 counties.

Note: Mobile Home refers to any home within the County considered a mobile home, not just those within mobile home parks.

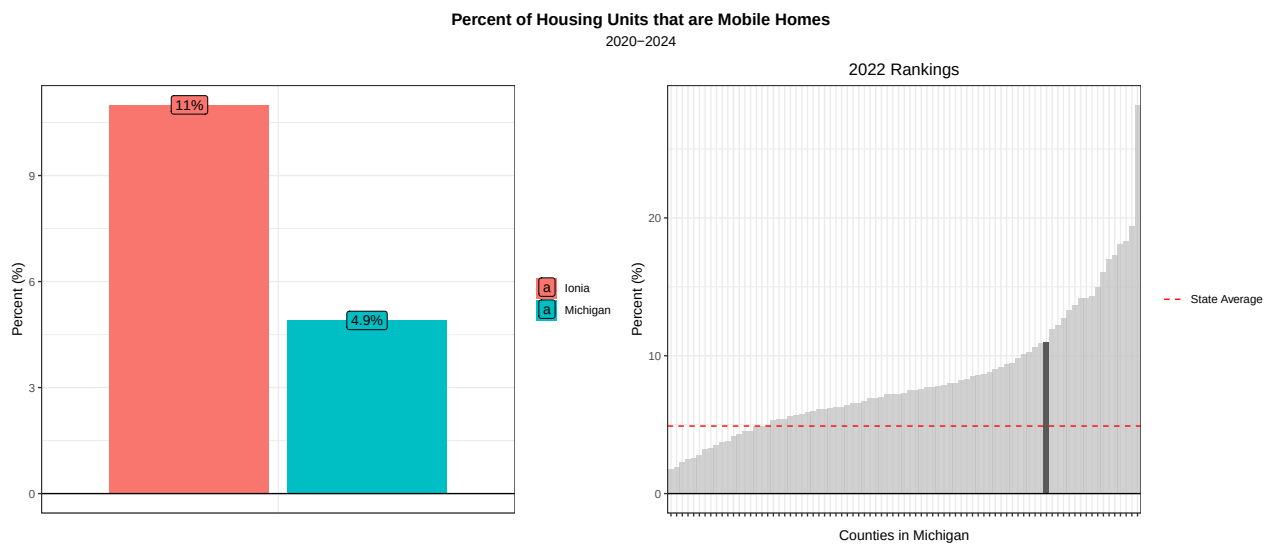


Figure 24: Mobile Homes in Ionia County

Almost every ZIP code except two, Clarksville and Palo, have a higher percentage of housing units that are mobile homes compared to the state average. Orleans has a much higher percentage of housing units that are mobile homes compared to the state average as well as the county average.

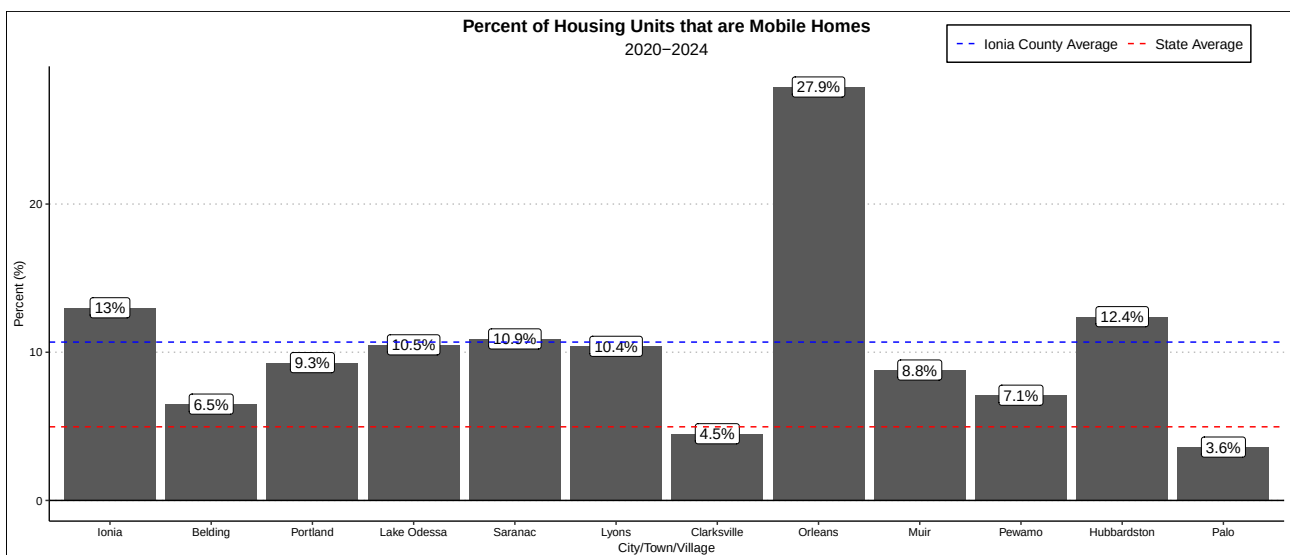


Figure 25: Mobile Homes in Ionia County ZIP Codes

4.6 Households without Computer & Internet Access

Compared to the State of Michigan, Ionia County has a slightly higher percentage of households without a computer with broadband internet (10.4% vs. 9.2%). Ionia County has the 51st-53rd (tied) highest percentage of households without a computer with broadband internet access in 83 counties.

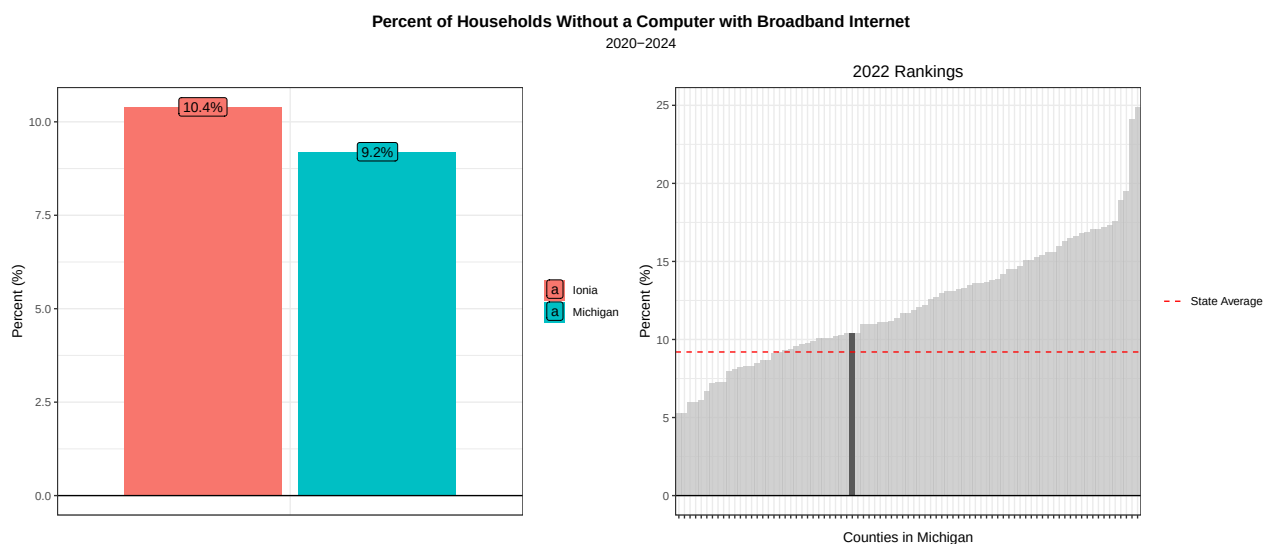


Figure 26: Computer & Internet Access in Ionia County

4.7 Single Parent Households

Compared to the State of Michigan, Ionia County has a similar percentage of households with a single parent and children (6% vs. 6.1 %). Ionia County has the 16th highest percentage of households with a single parent and children in 83 counties.

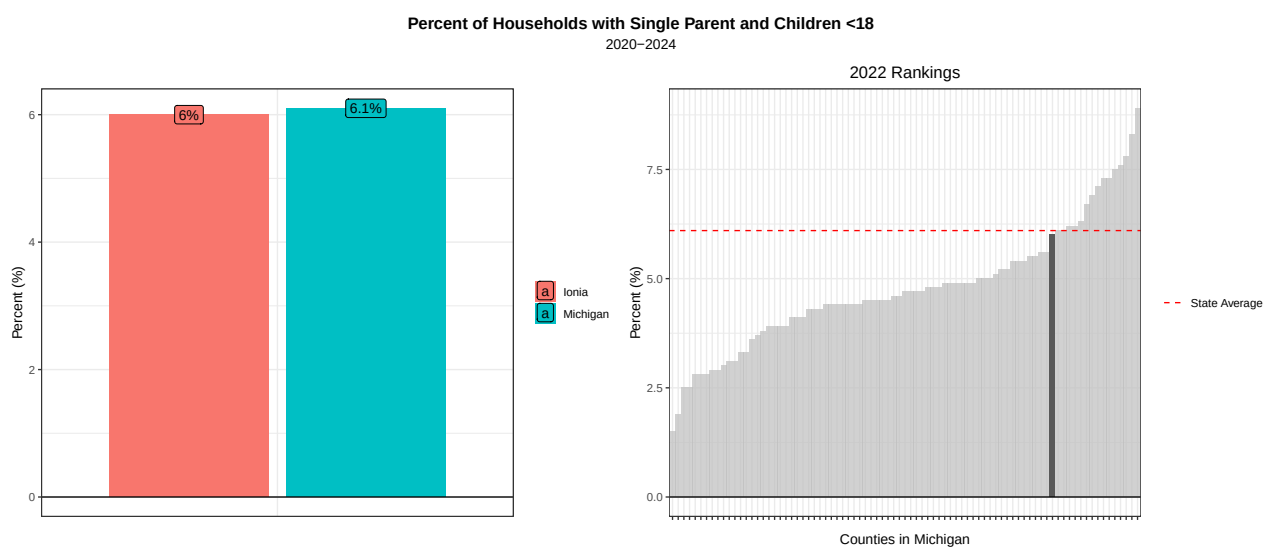


Figure 27: Single Parent Households in Ionia County

Two ZIP codes, Ionia city and Muir, have noticeably higher percentages of households with a single parent and children than the State of Michigan.

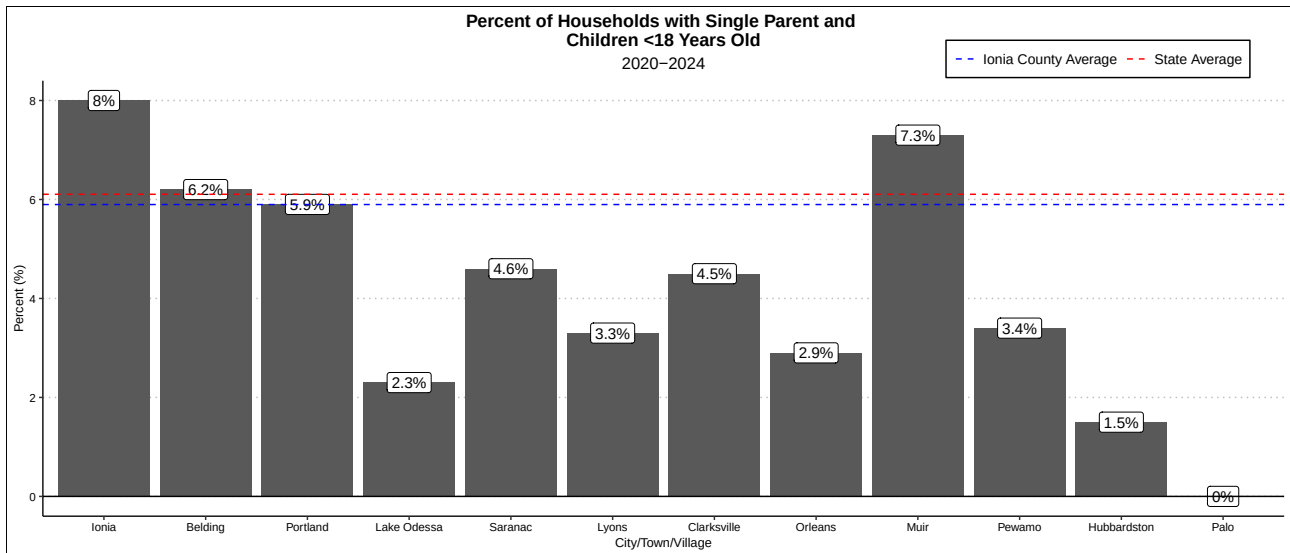


Figure 28: Single Parent Households in in Ionia County ZIP Codes

4.8 Housing Stress

See: Section 8.2 apart of the Unmet Needs report.

5 Income & Poverty

5.1 Household Income

Compared to the State of Michigan, household income distribution is similar in Ionia County. In general, Ionia county has a higher percentage of households in the \$35,000-\$75,000 range and a lower percentage of households in the \$200,000 and over category than the State of Michigan.

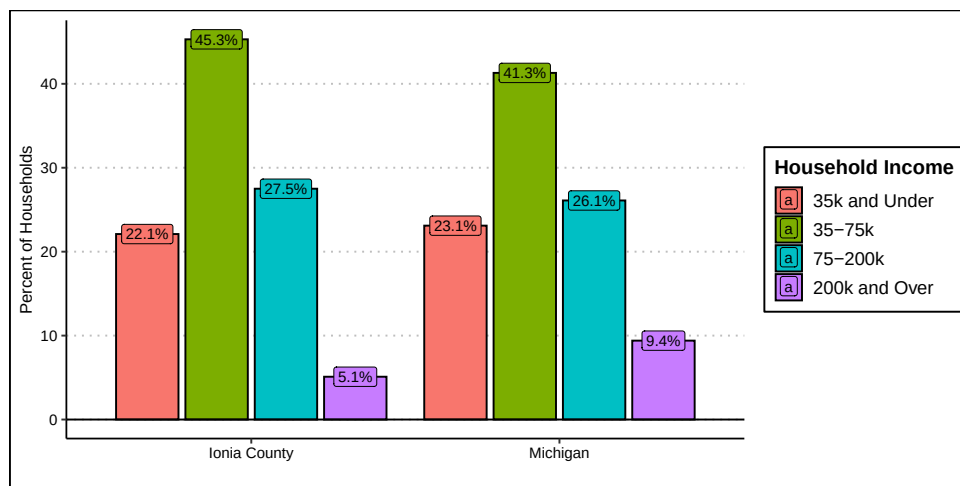


Figure 29: Income by Census Tract

Percentage of households in the highest and lowest income levels vary throughout the census tracts in Ionia County. There are two tracts, 302 (East Belding) & 317 (East Ionia City), with no reported households in the \$200,000 and over income level. The tract with the highest percentage of households in the \$200,000 and over income level is 314 (Portland City & Township).

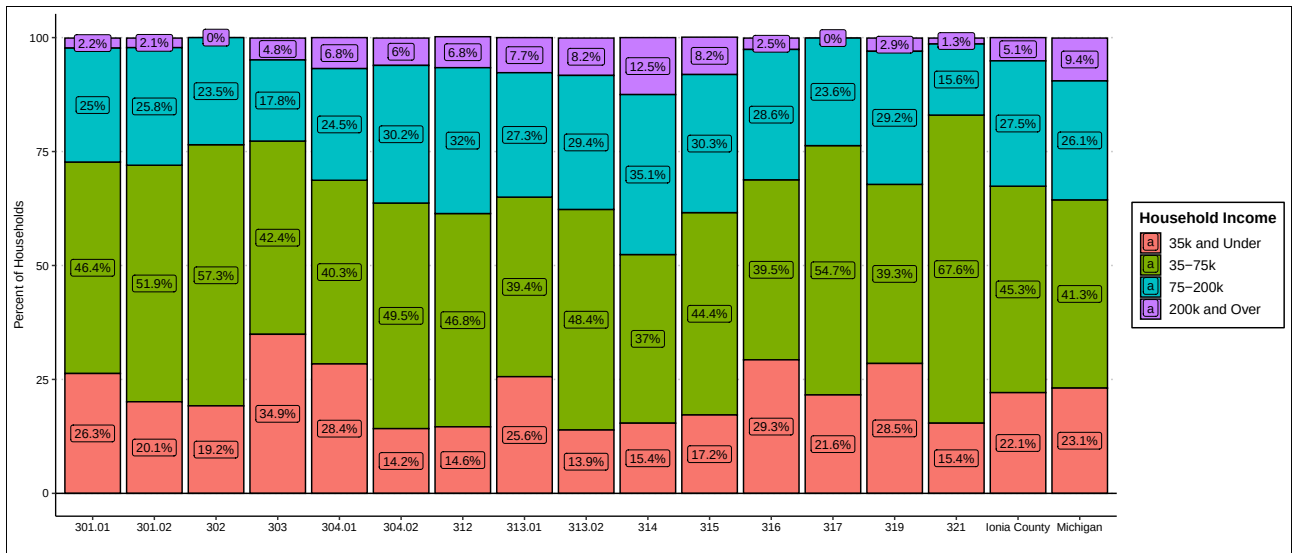


Figure 30: Income by Census Tract

5.2 Unemployment

Compared to the State of Michigan, Ionia County has a slightly lower rate of unemployment among the 16 years and older population. Ionia County has the 44-48th (tied) highest rate of unemployment among the 16 years and older population in the 83 counties.

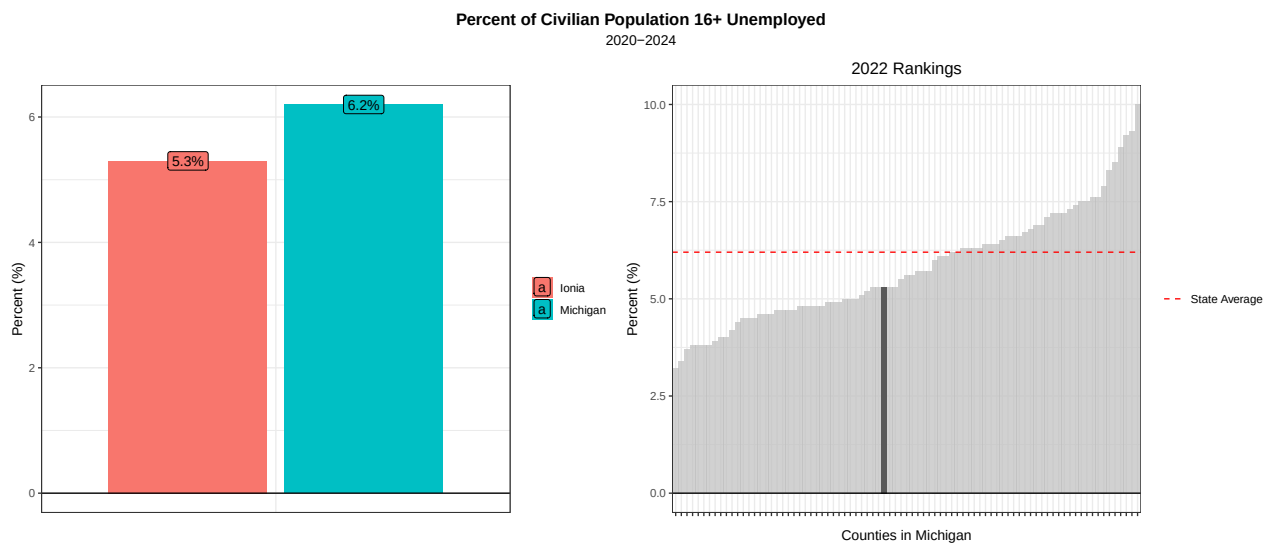


Figure 31: Unemployment in Ionia County

Two ZIP codes, Lyons and Hubbardston, have considerably higher unemployment in the 16 years and older population.

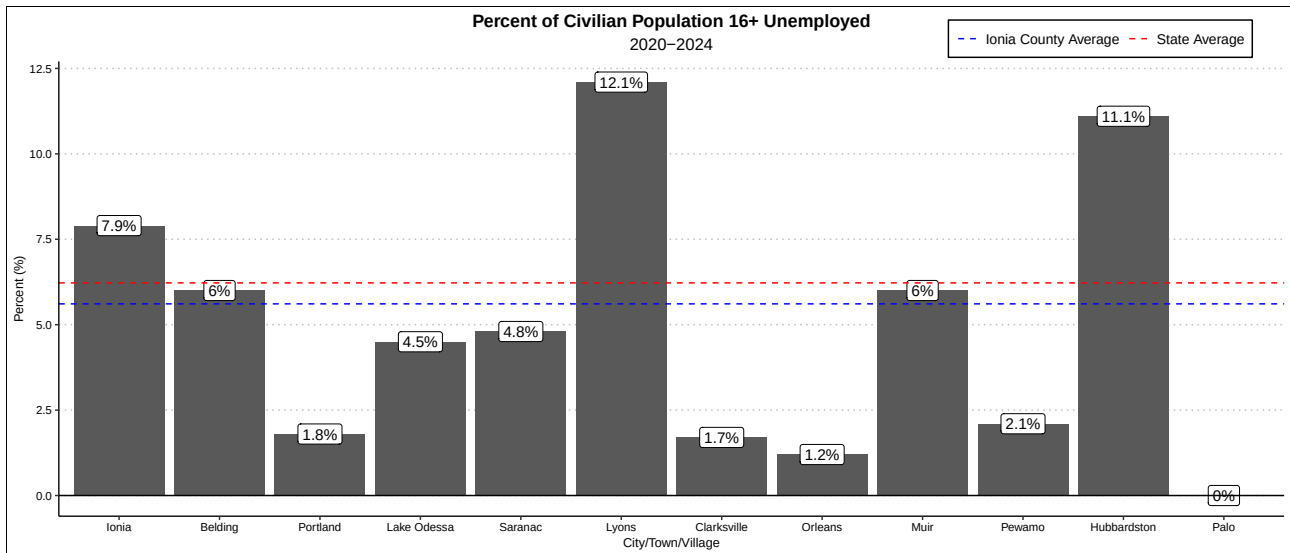


Figure 32: Unemployment in Ionia County ZIP Codes

5.3 Poverty Rates

Ionia County has a 10.6% poverty rate overall, the 21st lowest out of the 83 counties in Michigan.

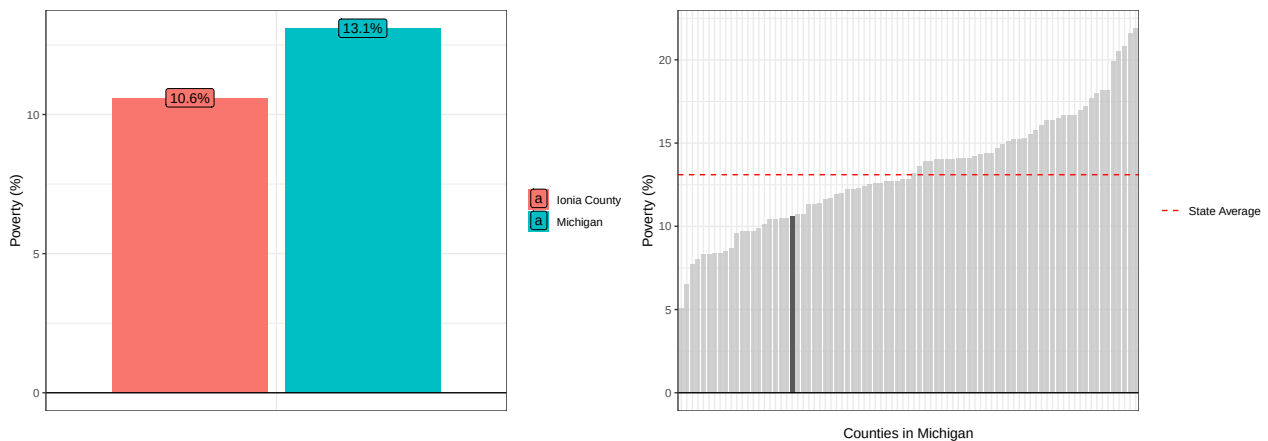


Figure 33: Poverty by County in Michigan

5.3.1 Poverty by Age

Poverty rates are highest for the population under 18 years old in Ionia County (13.0%) compared to the lower rates for 18-64 years old (10.1%) and 65 years and older (9.6%). For the State of Michigan, poverty rates for under 18 years old is even greater than Ionia County's rate for the same age (17.8%). Poverty rates for the population between 18-64 years old is also higher than Ionia County's rate for the same age (12.5%). Lastly, the poverty rate for adults 65 years and older is similar to Ionia County's rate for the same age (9.3%).

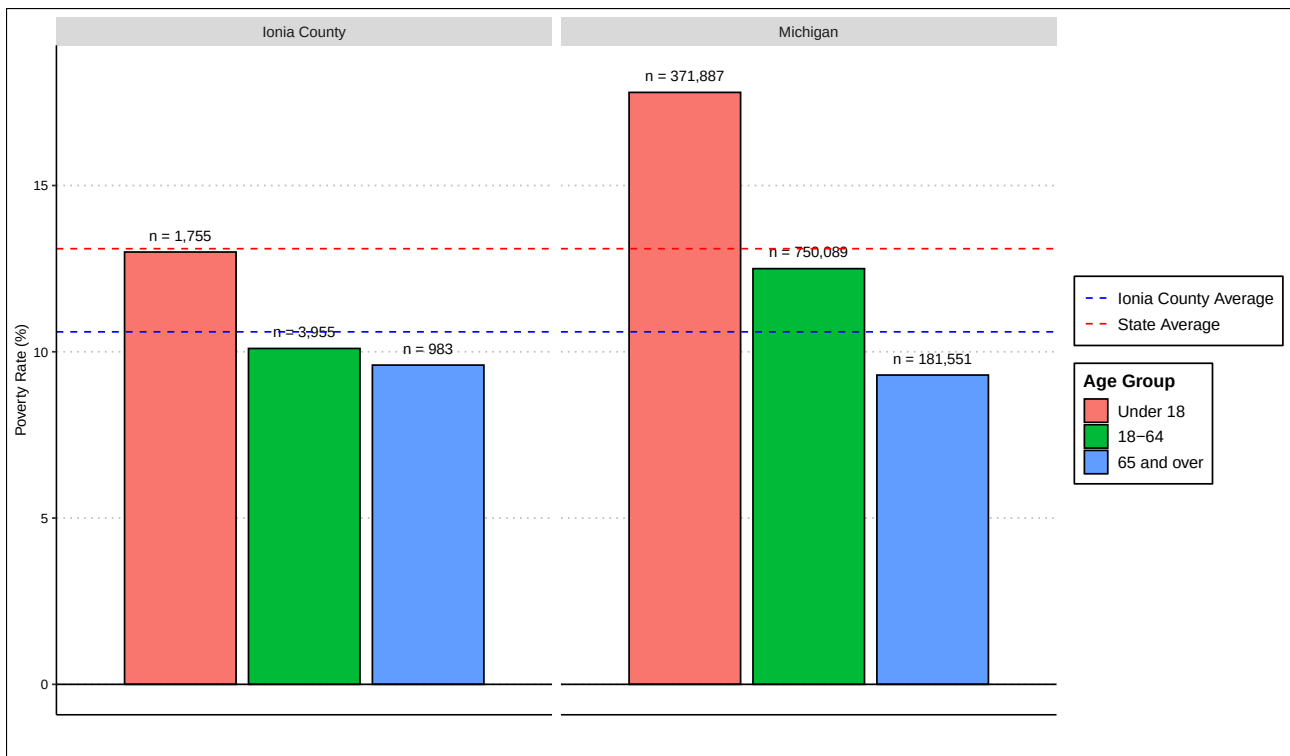


Figure 34: Ionia County vs. Michigan: Poverty by Age Group

Poverty rate among age groups varies greatly between census tracts in Ionia County. Tract 319 (mostly Berlin and Orange Township) has a very high childhood poverty rate compared to adults 65 and over, while tract 302 (East Belding) has its highest poverty rate age group as adults 65 and older, and a comparatively low childhood poverty rate.

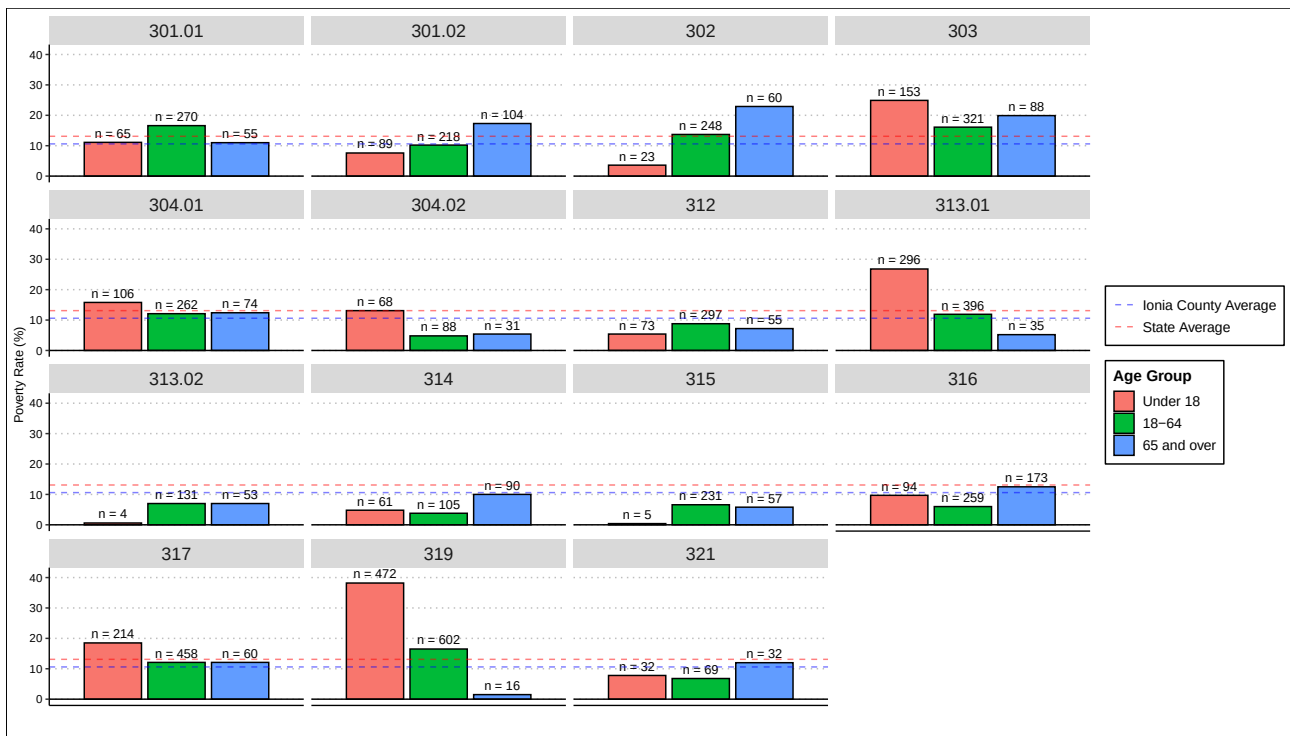


Figure 35: Poverty Rate by Age Group & Census Tract

6 Healthcare

6.1 Health Insurance Coverage by Age

Ionia County has similar rates of health insurance coverage to the State of Michigan across age groups:

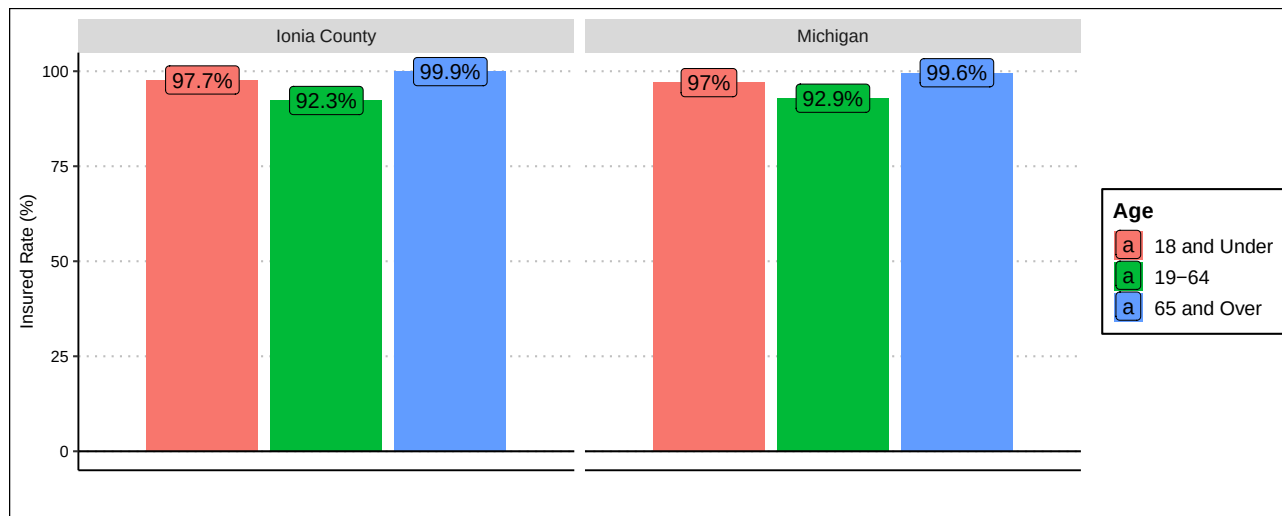


Figure 36: Ionia County vs. Michigan: Insured Rates by Age

The percentage of adults in Ionia County without health insurance has been on a decreasing trend in recent years. The uninsured rate has fallen from 18% in 2009 to 7% in 2022.

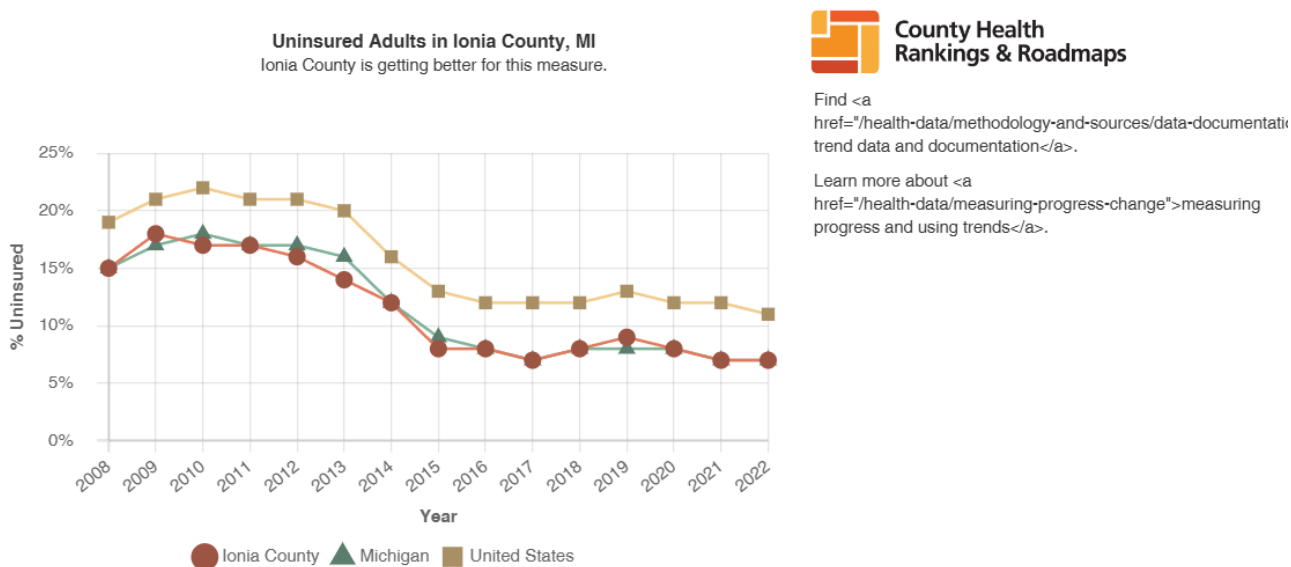
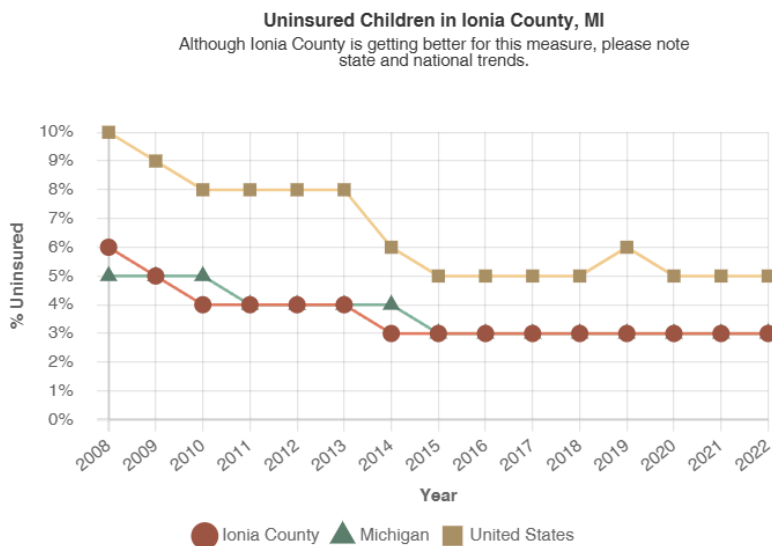


Figure 37: Ionia County Adults Insurance Rates Over Time

The percentage of children in Ionia County without health insurance has stayed at the same rate since 2014 (3%). This is consistent with statewide trends, whereas the national rate of uninsured children is slightly higher.



County Health Rankings & Roadmaps

Find trend data and documentation.

Learn more about measuring progress and using trends.

Figure 38: Ionia County Children Insurance Rates Over Time

Insured rates across age groups in census tracts are mostly consistent with county and state-wide trends. there's only one census tract, 301.01 (northeast Ionia County) where children and adults 19-64 years old have lower rates of insurance coverage compared to the State of Michigan and Ionia County.



Figure 39: Insured Rates by Age and Census Tract

6.2 Healthcare Provider and Facility Access per 100k People

Healthcare provider and facility access data sourced from the 2025 Area Health Resources Files compiled by the Health Resources and Services Administration (HRSA).

6.2.1 General Medicine & Dentistry Providers

For general medicine and dentistry providers, Ionia county has fewer providers per 100k people in every category:

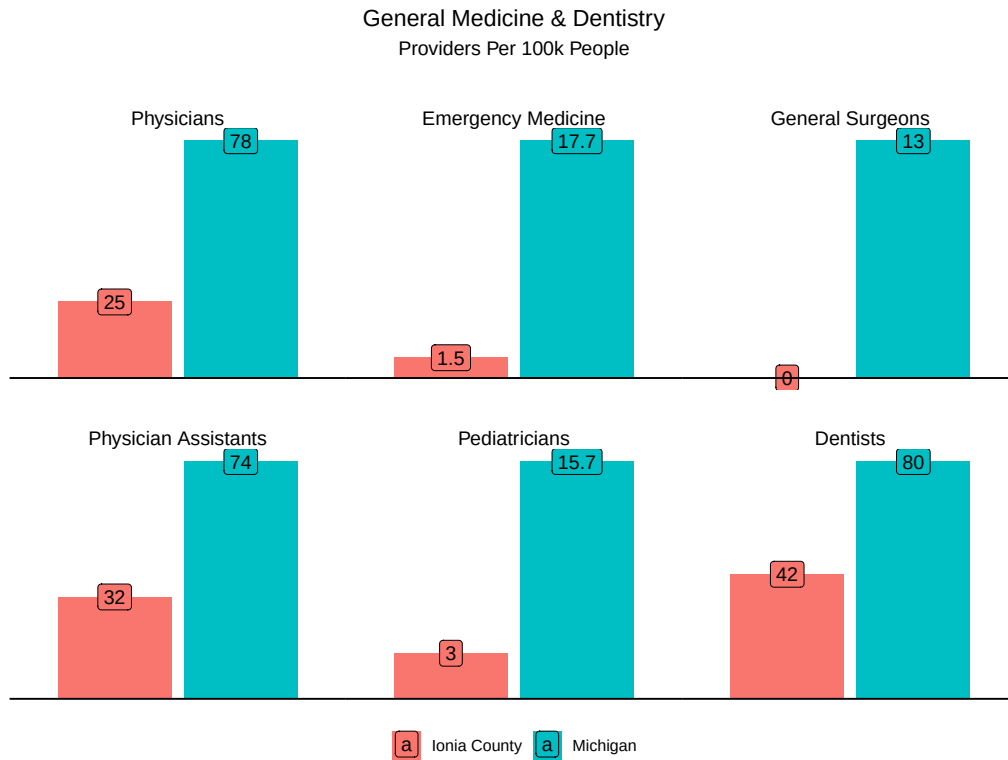


Figure 40: Ionia County vs. Michigan: General Medicine & Dentistry

The number of people in Ionia County per primary care physician has been increasing gradually since 2012 from 3360:1 to 3950:1 in 2021.

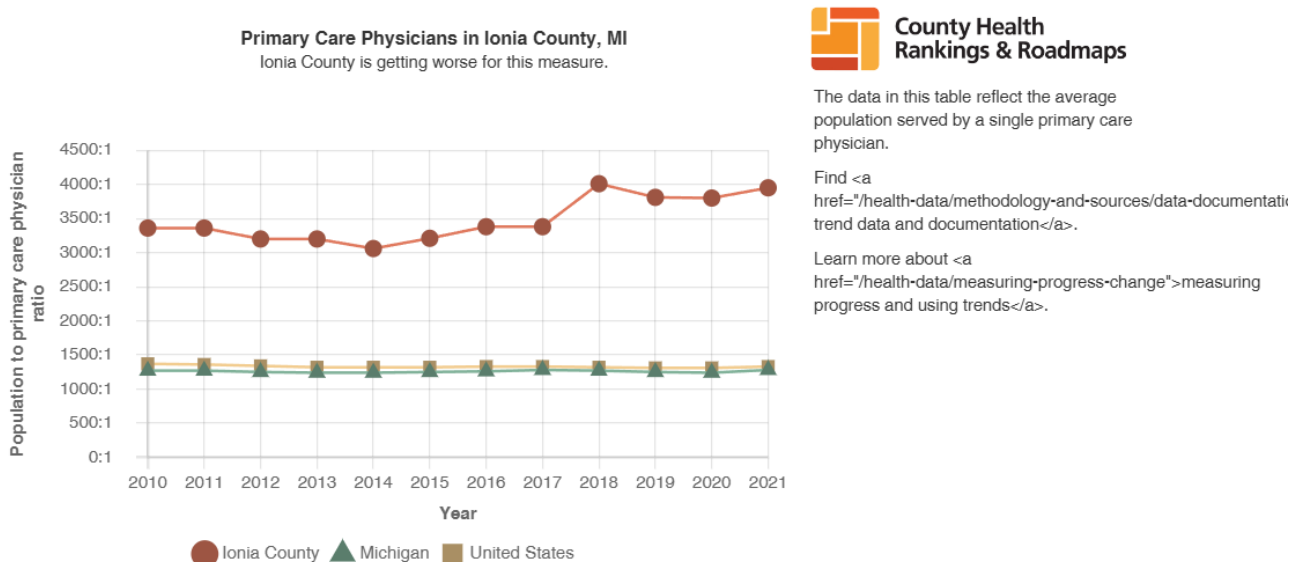


Figure 41: Ionia County Primary Care Physicians Over Time

6.2.2 Hospitals & Pharmacists

Ionia county has about the same number of hospitals (per 100k people), but has less than a third of the pharmacists (per 100k people) as the State of Michigan:

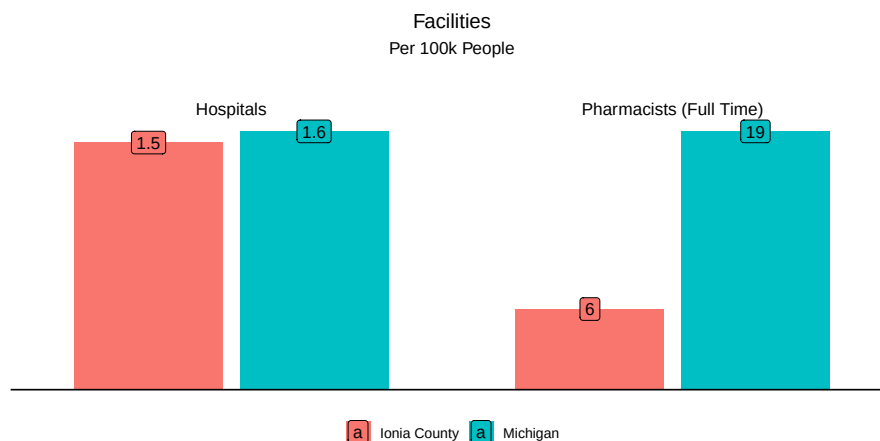


Figure 42: Ionia County vs. Michigan: Hospitals & Pharmacists

100% of the population in every Ionia County ZIP code lives within a 30 minute drive of a hospital.

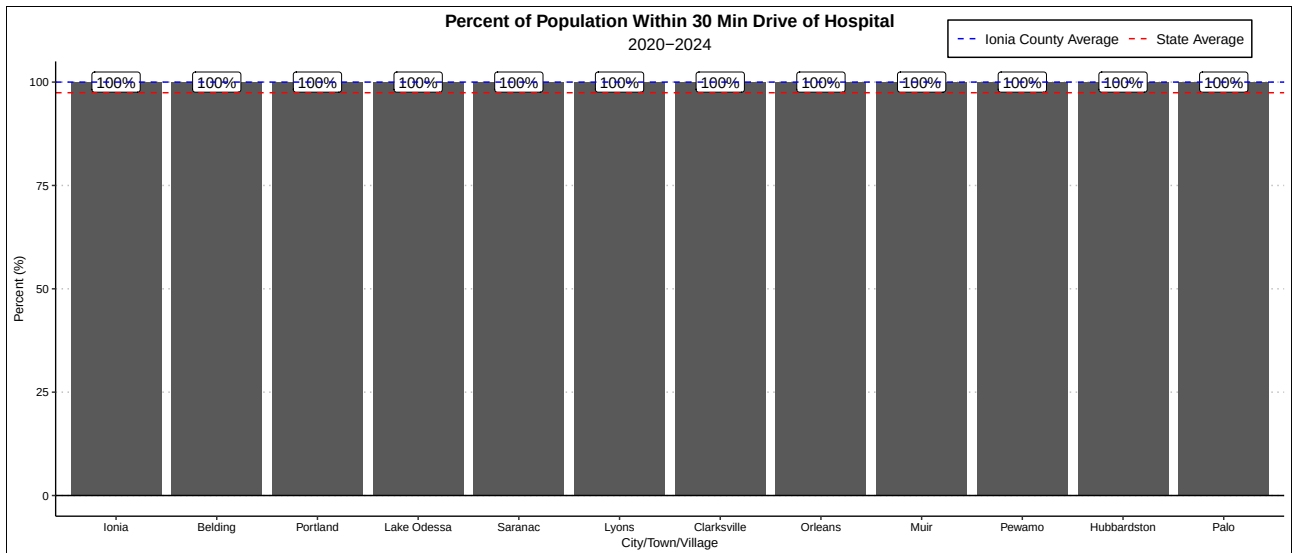


Figure 43: Hospital Driving Access across Ionia County ZIP codes

Most Ionia County ZIP codes are near the state average of % of the population within a 15 minute drive of a pharmacy. Three ZIP codes, Clarksville, Muir, and Pewamo have a noticeably lower percent of the population within a 15 minute drive of a pharmacy.

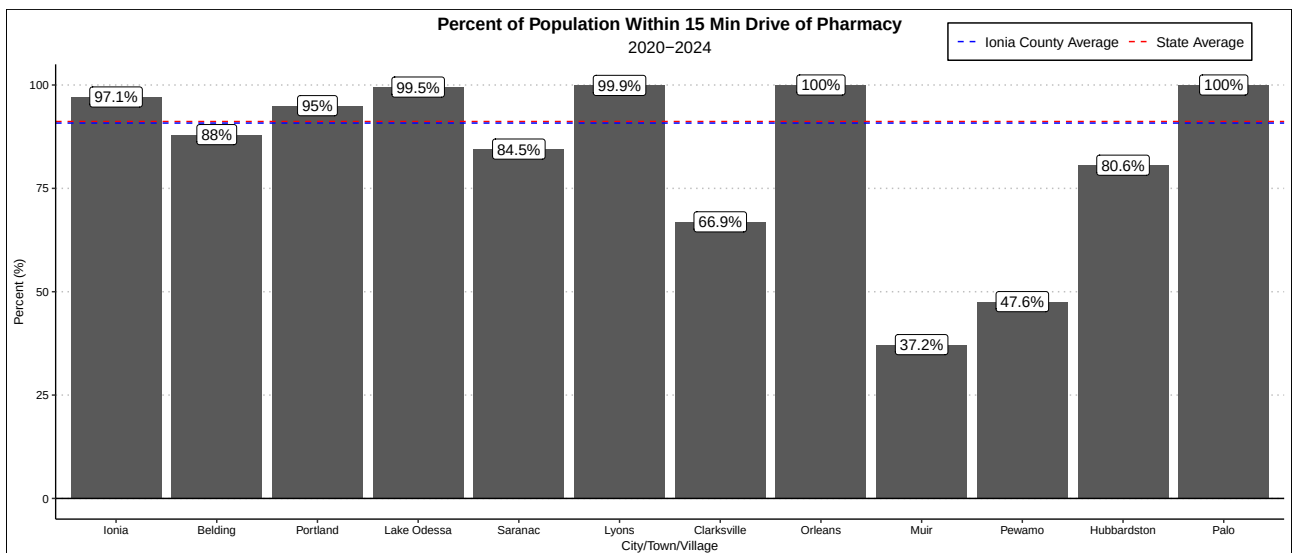


Figure 44: Pharmacy Driving Access across Ionia County ZIP codes

6.2.3 Specialty Providers

Ionia County has fewer specialty medicine providers (per 100k people) in every area compared to the State of Michigan, with many specialties having no reported providers in the County:

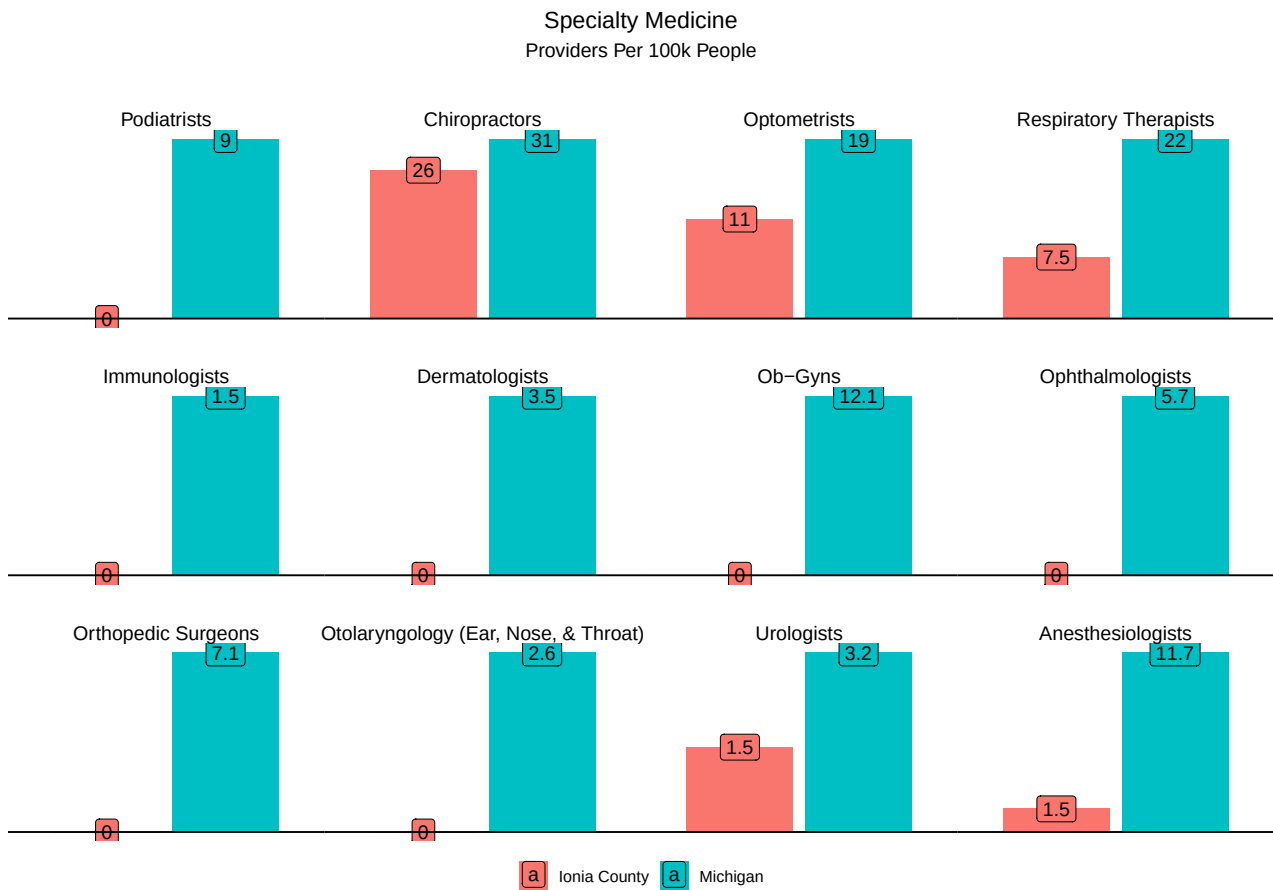


Figure 45: Ionia County vs. Michigan: Specialty Providers

6.2.4 Mental Health Providers

Ionia County has less than half the mental health professionals (per 100k people) compared to the State of Michigan, as well as no reported psychiatrists within the county:

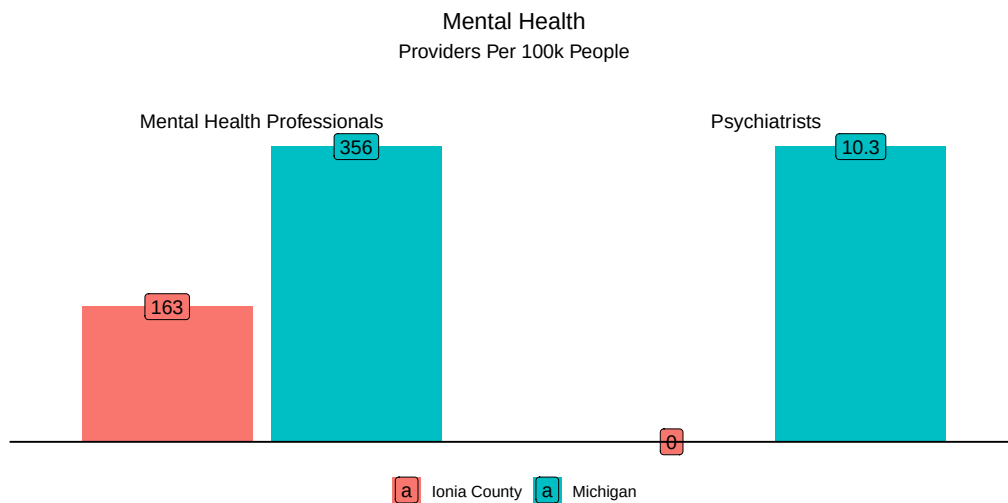


Figure 46: Ionia County vs. Michigan: Mental Health Providers

6.3 Vaccination Coverage

Vaccination coverage data comes from the Michigan Department of Health and Human Services (MDHHS) 2026 Q1 Ionia County Quarterly Immunization Report Card and the Report Card FAQ for descriptions.

Overall, the MDHHS ranks Ionia County as having the 25th best coverage rank out of Michigan's 83 counties.

Shown below is the coverage statistics for vaccines and vaccine combinations, comparing coverage rates for Ionia County, the State of Michigan, and the United States.

6.3.1 Children 19-35 Months Old

Ionia County has similar vaccination coverage for children 19-35 months old as the State of Michigan, though Ionia County and Michigan tend to have less coverage than the U.S. overall. Vaccination coverage for children 19-35 months old tend to be within 50-75% for all types of coverages.

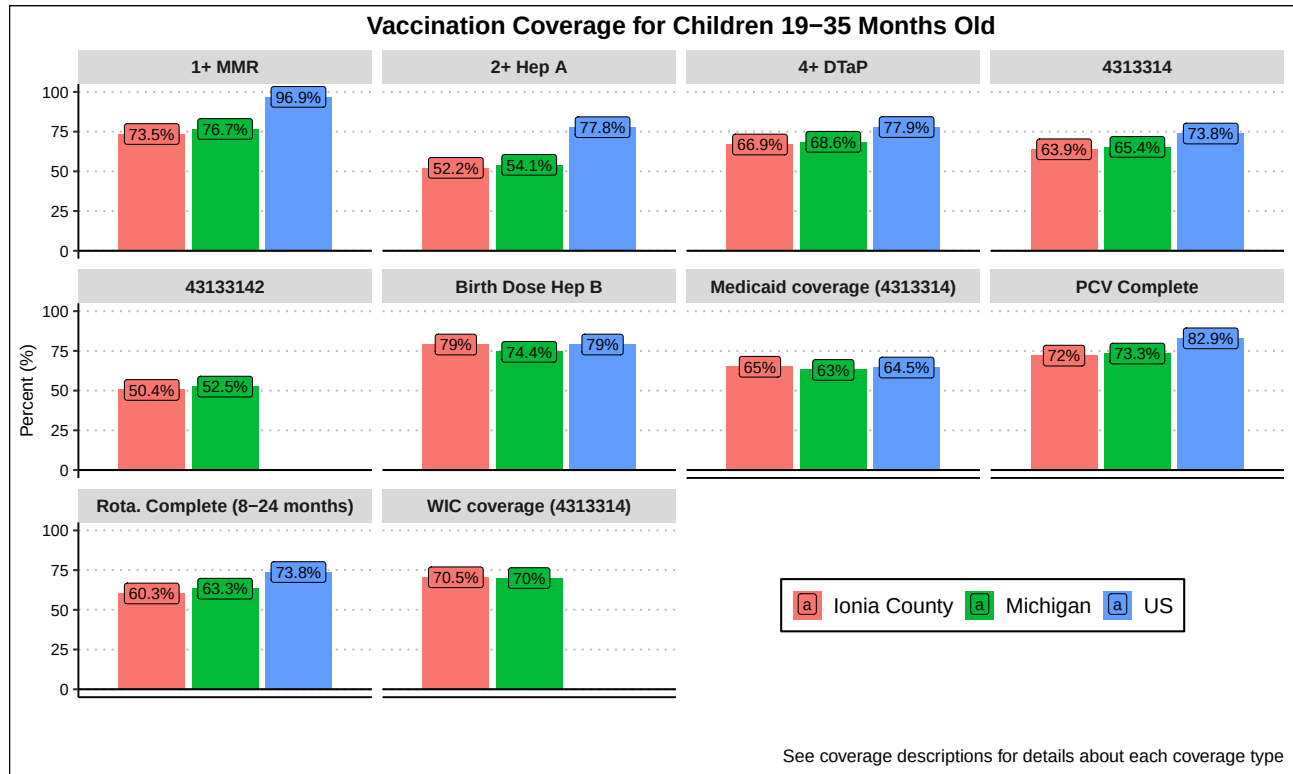


Figure 47: Vaccination Coverage for Children 19-35 Months Old in Ionia County

See: Coverage descriptions for Children 19-35 Months Old.

All vaccine coverages in Ionia County for children and newborns have been mainly decreasing since 2024.

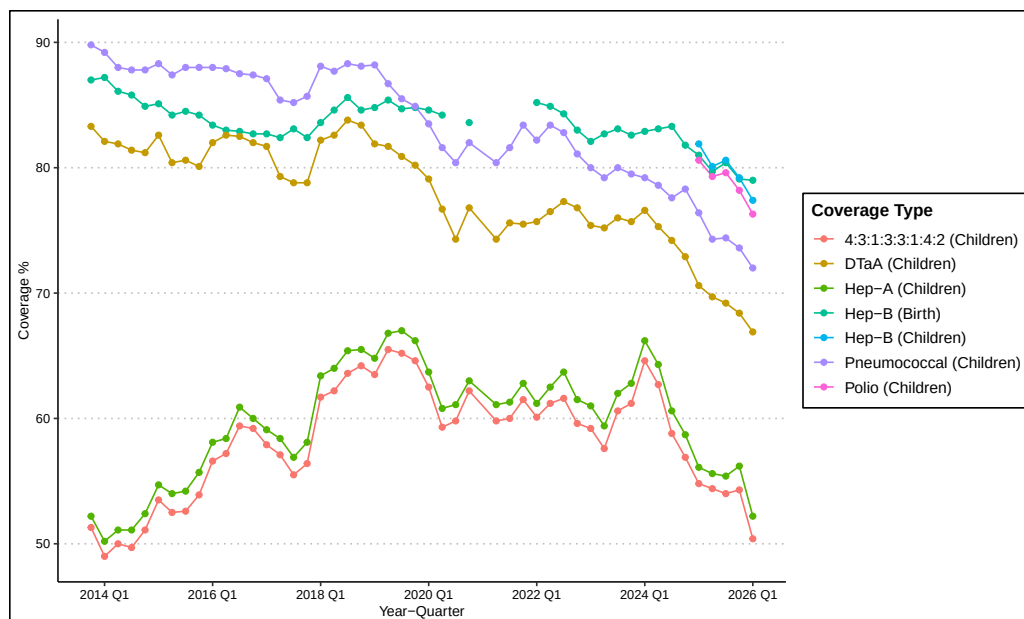


Figure 48: Vaccination Coverage Trends for Children 19-35 Months Old and Births in Ionia County

6.3.2 Children 13-17 Years Old

Ionia County has similar vaccination coverage to the State of Michigan, except for a noticeably lower coverage for 1+ MenB (16-18 years old) for Ionia County than Michigan. Both Ionia County and Michigan have slightly lower coverage than the U.S. for vaccination coverages with national data available.

Coverage for 1+ MenB(16-18 years old), 1323213, 2+ MenACWY (17 years old), and HPV complete vaccination types are considerably lower compared to the other vaccination types.

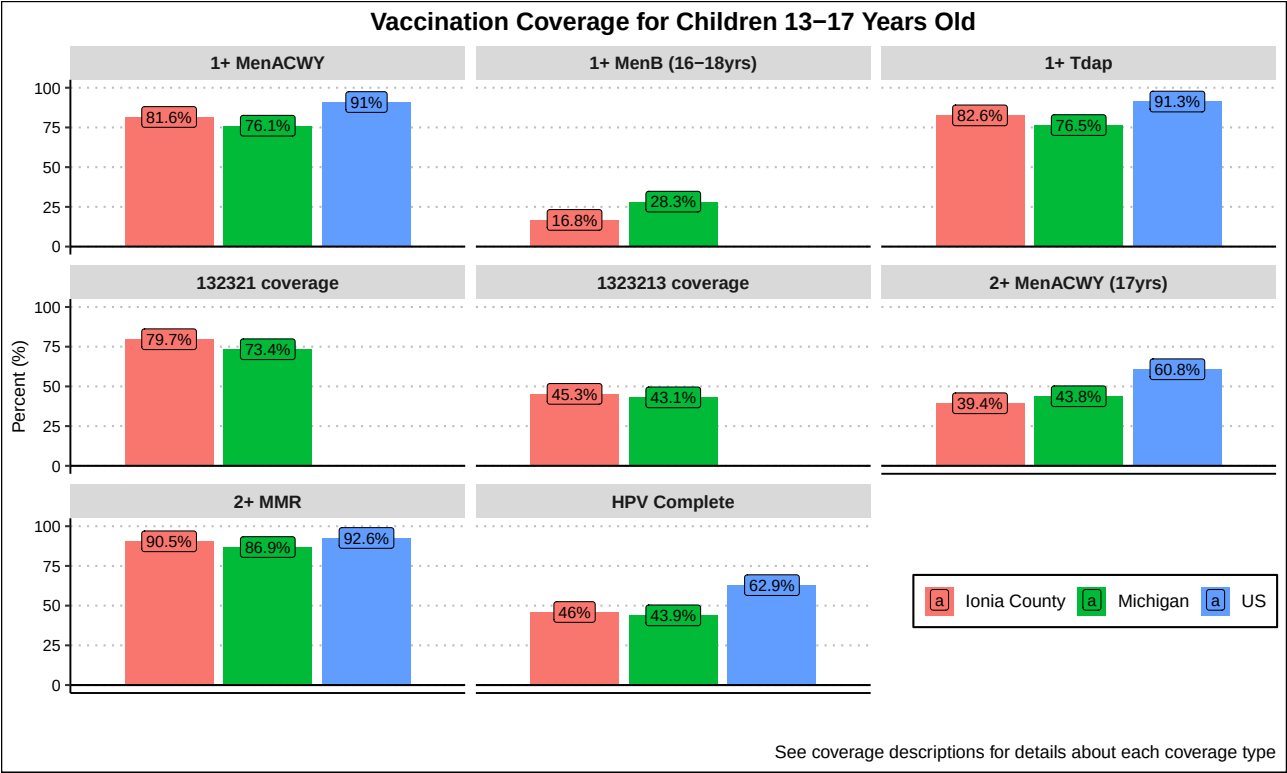


Figure 49: Vaccination Coverage for Children 13-17 Years Old in Ionia County

See: Coverage descriptions for Children 13-17 Years Old.
Vaccination coverages in Ionia County for children 13-17 years old have changed little since 2018.

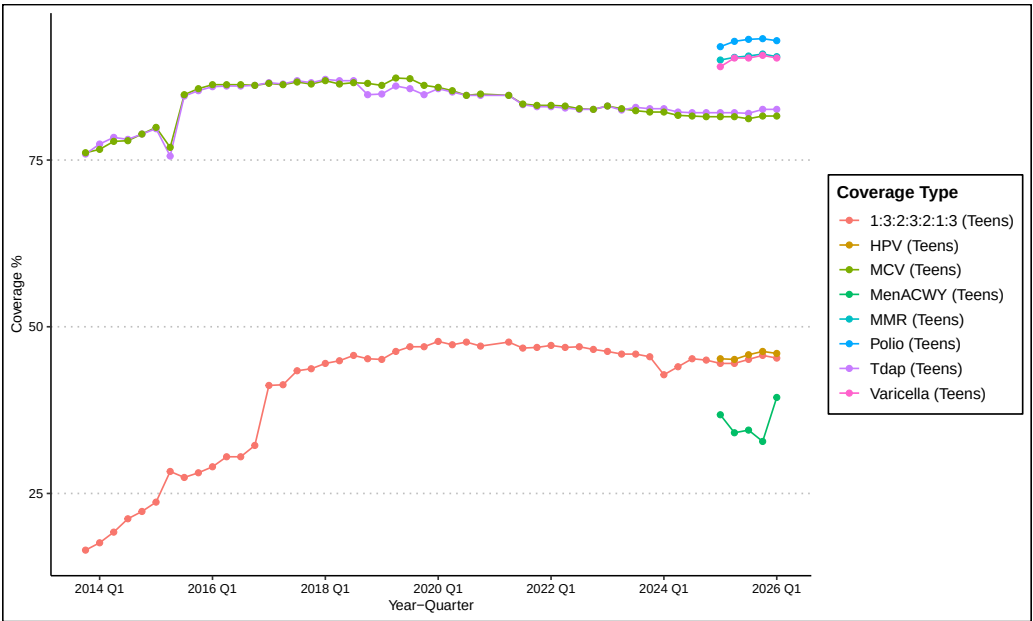


Figure 50: Vaccination Coverage Trends for Children 13-17 Years Old in Ionia County

6.3.3 Adults 18 Years and Older

Ionia County has similar vaccination coverage for adults than the State of Michigan for adult vaccinations. Ionia County and Michigan, while typically having less vaccination for children compared to the U.S., have similar or greater vaccination coverage compared to the U.S. for adults.

The Composite Measure (19+ years old) vaccination coverage is noticeably lower than other vaccination coverage types.

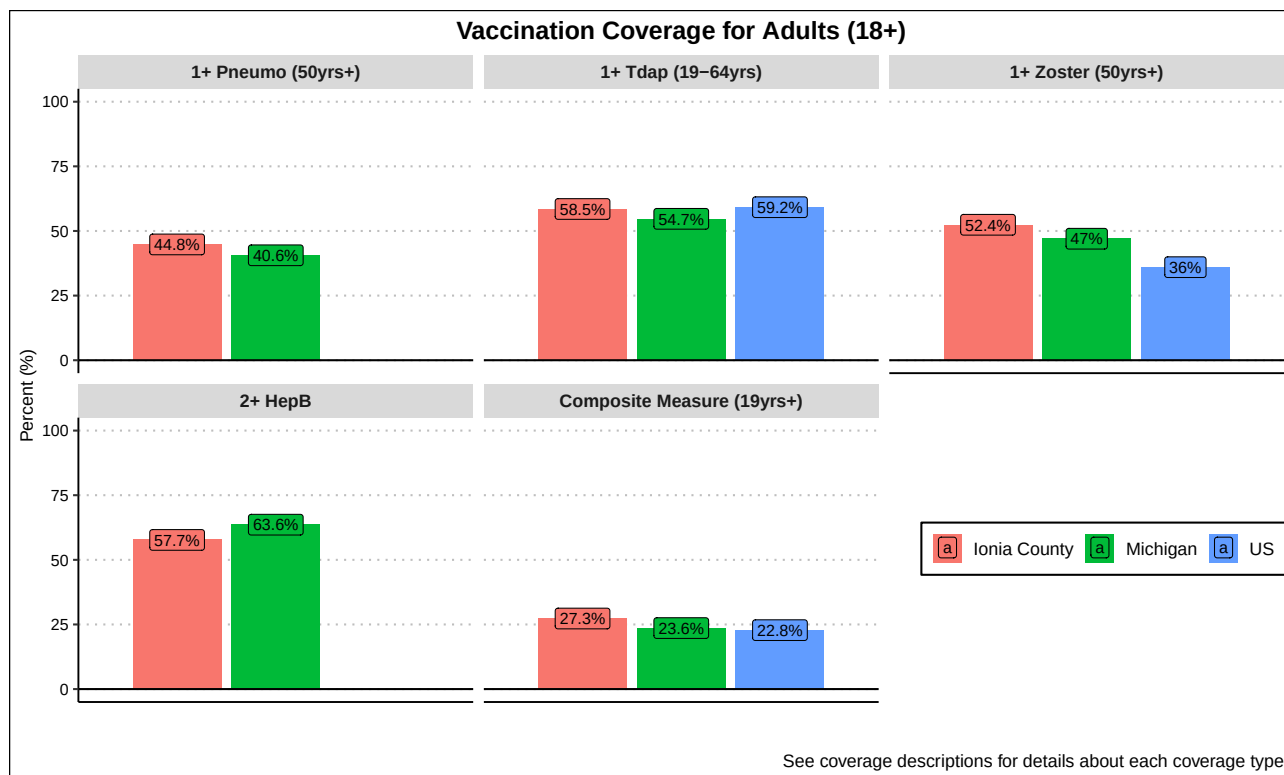


Figure 51: Vaccination Coverage for Adults in Ionia County

See: Coverage descriptions for Adults 18+ Years Old.

In Ionia County, coverage for the Tdap vaccine for adults sharply decreased from 2022 Q4 to 2023 Q1 (71.6 % to 56.2%), and has remained around (56-58%) since then. In Ionia County, coverage for the Zoster/Shingles vaccines for adults has been increasing since 2018 Q4, when coverage was 21% up to 52.4% in 2026 Q1. In Ionia County, the Pneumococcal Polysaccharide vaccine for adults mainly increased from 2014 Q1 until 2021 Q3 (32.3% to 70.4%) until coverage data is no longer available.

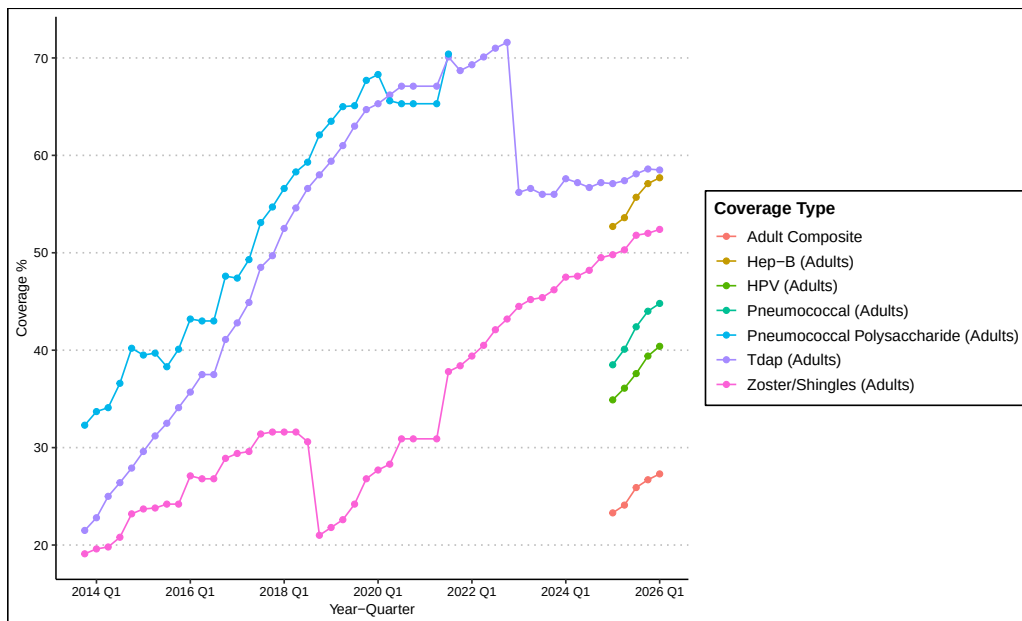


Figure 52: Vaccination Coverage Trends for Adults in Ionia County

6.3.4 Influenza

Ionia County has similar Influenza vaccination to the State of Michigan, while both tend to be lower than the U.S. national Influenza coverage.

Flu vaccination coverage ranges between 12-25% in Ionia County.

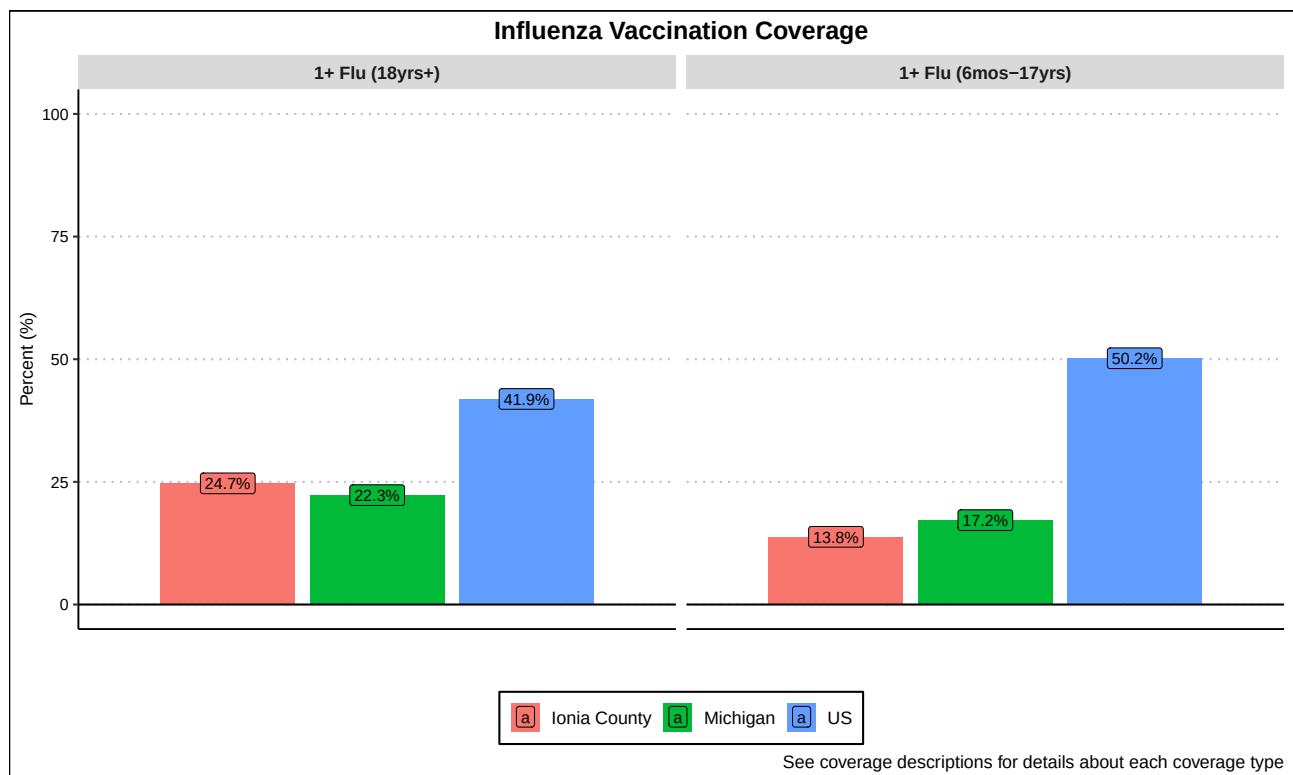


Figure 53: Influenza Vaccination Coverage in Ionia County

See: Coverage descriptions for Influenza Vaccines.

6.3.5 COVID-19

Ionia County has similar COVID-19 vaccination coverage to the State of Michigan, while both tend to be considerably lower than COVID-19 vaccination coverage nationally across the U.S..

COVID-19 complete coverage (6 months to 17 years old) in Ionia County is around half of the coverage of the state (1.3% vs. 2.6%), and less than a seventh of the national coverage (1.3% vs. 9.4%)

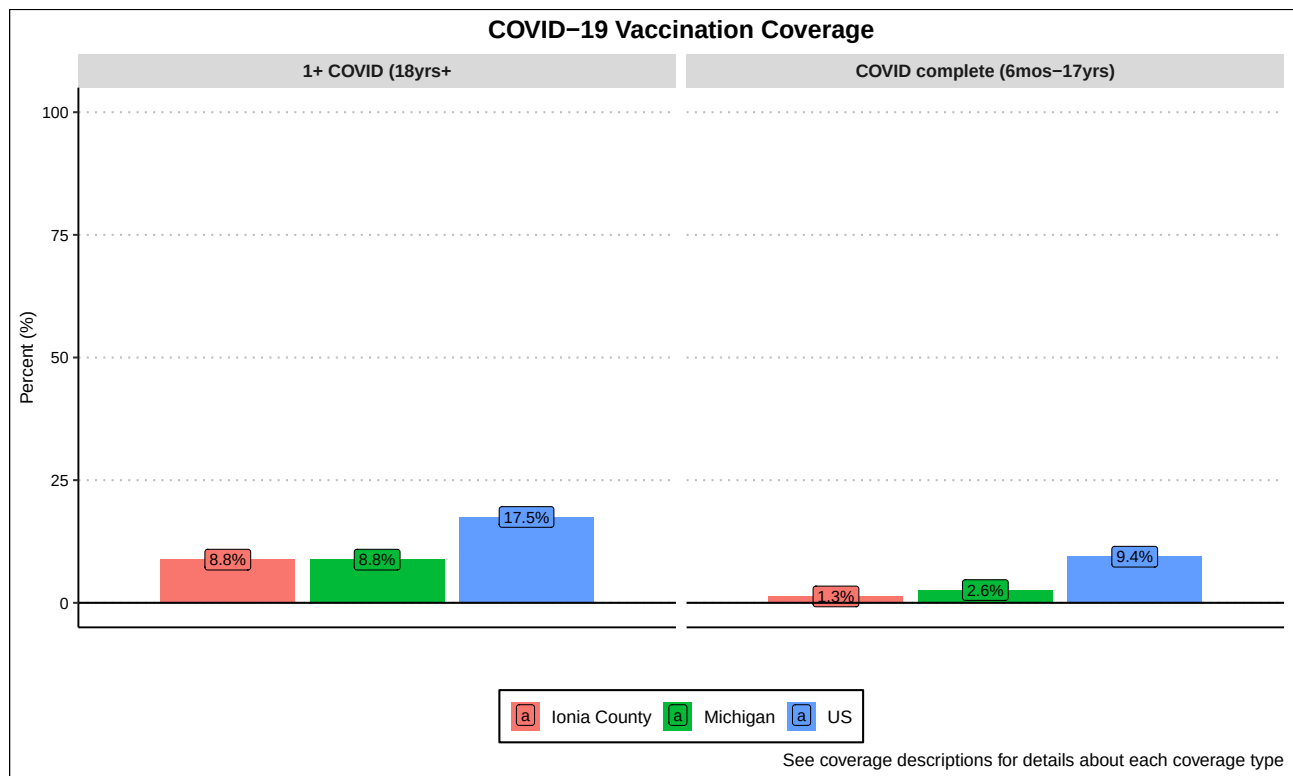


Figure 54: COVID-19 Vaccination Coverage in Ionia County

See: Coverage descriptions for COVID-19 Vaccines.

6.3.6 Coverage Descriptions

Coverages for Children 19-35 Months Old:

- 1+ MMR: 1 or more doses of the Measles, Mumps, and Rubella vaccine
- 2+ Hep A: 2 or more doses of the Hepatitis A vaccine
- 4+ DTaP: 4 or more doses of the diphtheria, tetanus, and pertussis vaccine
- 4313314: 4 or more doses of DTaP/DTP/DT (diphtheria, tetanus, and pertussis), 3 or more doses of Polio, 1 or more dose of MMR (Measles, Mumps, and Rubella), 3 or more doses of Hib (*Haemophilus influenzae* type b), 3 or more doses of HepB (Hepatitis B), 1 or more dose of Varicella, 4 or more doses of PCV (pneumococcal conjugate)
- 43133142: 4 or more doses of DTaP/DTP/DT (diphtheria, tetanus, and pertussis), 3 or more doses of Polio, 1 or more dose of MMR (Measles, Mumps, and Rubella), 3 or more doses of Hib (*Haemophilus influenzae* type b), 3 or more doses of Hep B (Hepatitis B), 1 or more dose of Varicella, 4 or more doses of PCV (pneumococcal conjugate), 2 or more doses of HepA (Hepatitis A)
- Birth Dose Hep B: Birth dose of Hepatitis B
- Medicaid coverage (4313314): coverage of the 4313314 vaccine combination for those on Medicaid
- PCV Complete: Recommended number of pneumococcal conjugate vaccine doses
- Rota. Complete (8-24 months): recommended number of rotavirus vaccine doses
- WIC coverage (4313314) coverage of the 4313314 vaccine combination for those on the Women, Infants, and Children Program

Coverages for Children 13-17 Years Old:

- 1+ MenACWY: 1 or more doses of meningococcal conjugate vaccine
- 1+ MenB (16-18 years old): 1 or more doses of meningococcal B vaccine for those 16-18 years old
- 1+ Tdap: 1 or more doses of the tetanus, diphtheria, and pertussis vaccine administered on or after age 11
- 132321 coverage: 1 or more doses of Tdap (tetanus, diphtheria, and pertussis), 3 or more doses of Polio, 2 or more doses of MMR (Measles, Mumps, and Rubella), 3 or more doses of HepB (Hepatitis B), 2 or more doses of varicella vaccine, 1 or more dose of MenACWY (meningococcal conjugate)
- 1323213 coverage: 1 or more doses of Tdap (tetanus, diphtheria, and pertussis), 3 or more doses of Polio, 2 or more doses of MMR (Measles, Mumps, and Rubella), 3 or more doses of HepB (Hepatitis B), 2 or more doses of varicella vaccine, 1 or more dose of MenACWY (meningococcal conjugate), HPV (Human Papillomavirus) complete (with 2 or 3 doses (Males & Females))
- 2+ MenACWY (17 years old): 2 or more doses of the meningococcal conjugate vaccine at 17 years old
- 2+ MMR: 2 or more doses of the Measles, Mumps, and Rubella vaccine
- HPV complete: the recommended number of doses of the Human Papillomavirus vaccine

Coverages for Adults:

- 1+ Pneumo (60+ years old): 1 or more doses of the pneumococcal vaccine
- 1+ Tdap (19-64 years old): 1 or more doses of the tetanus, diphtheria, and pertussis vaccine
- 1+ Zoster: 1 or more doses of the zoster (shingles) vaccine
- 2+ HepB: 2 or more doses of the Hepatitis B vaccine
- Composite Measure (19+ years old): The adult composite measure assesses adults 19 years of age and older for completion of their age-based recommended vaccines

Coverage for Influenza Vaccination Coverage:

- 1+ Flu (18+ years old): 1 or more doses of the influenza vaccine in the specified season for adults 18+ years old
- 1+ Flu (6 months to 17 years old): 1 or more doses of the influenza vaccine in the specified season for children 6 months to 17 years old
- Flu Complete (6 months to 8 years old): The recommended number of doses of the influenza vaccine in the specified season for children 6 months to 8 years old.

Coverage for COVID-19 Vaccination Coverage:

- 1+ COVID (18+ years old): 1 or more doses of the COVID-19 vaccine in the specified season for adults 18+ years old
- COVID complete (6 months to 17 years old): The recommended number of doses of the COVID-19 vaccine in the specified season for children 6 months to 17 years old.

6.4 Health Conditions

The data on health conditions is sourced from the Population Level Analysis and Community Estimates (PLACES) at the CDC, and goes to the census tract level.

6.4.1 Arthritis

Compared to the state average, Ionia County has a lower rate of arthritis (29.7% vs. 30%). Only one census tract, 301.01 (northeast Ionia County), has a higher rate than the state average (33.7%). Tract 321 (West Ionia City) has the lowest rate of arthritis in the county (24%).

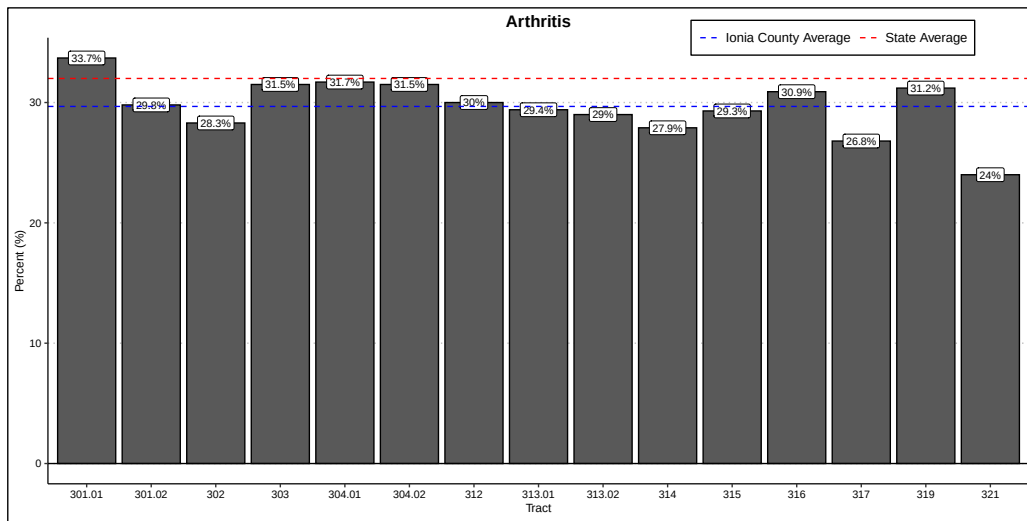


Figure 55: Arthritis by Census Tract in Ionia County

6.4.2 Chronic Obstructive Pulmonary Disease

Ionia County has a marginally higher rate of COPD than the State of Michigan (8.5% vs. 7.7%). The highest rates of COPD are in tracts 301.01 (northeast Ionia County) (11.4%) and 303 (West Belding) (11.2%). Tract 314 (Portland City & Township) has the lowest rate of COPD (6%).

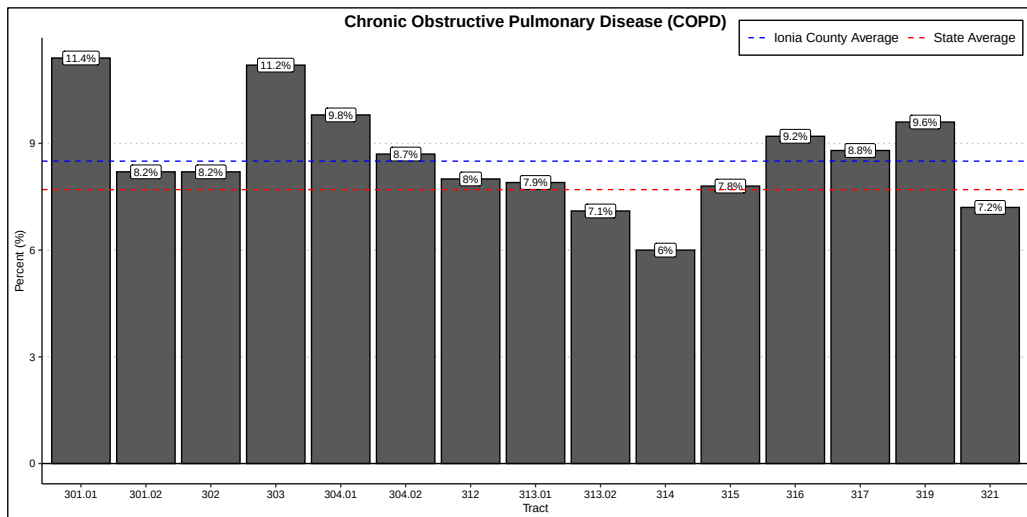


Figure 56: COPD by Census Tract in Ionia County

6.4.3 Coronary Heart Disease

Ionia County has a 1.6% higher coronary heart disease rate than the State of Michigan (6.9% vs. 5.3%), and each census tract individually is higher than the state average. Tract 301.01 (Northeast Ionia County) has the highest rate of coronary heart disease (8.8%), while tract 314 (Portland city % Township) has the lowest rate of coronary heart disease (5.7%).

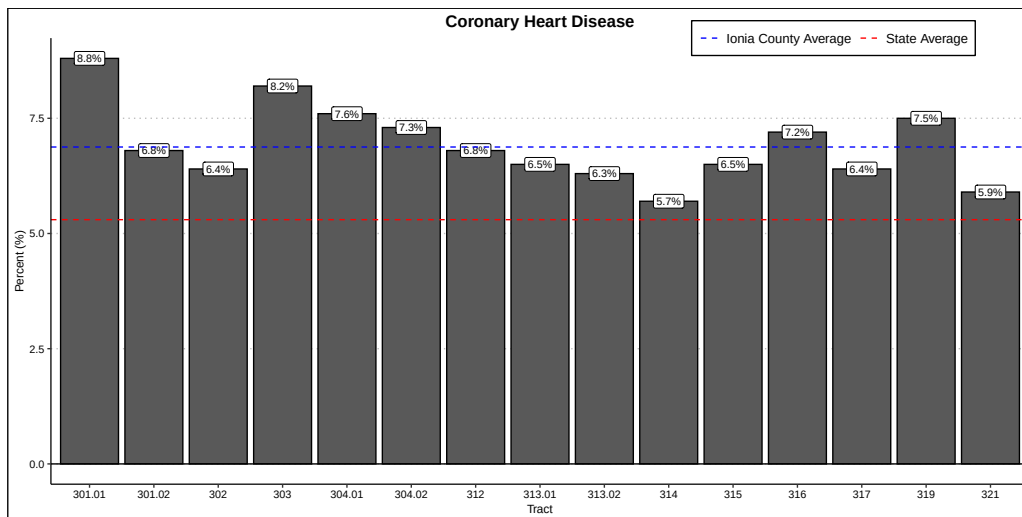


Figure 57: Coronary Heart Disease by Census Tract in Ionia County

6.4.4 Asthma

Ionia County has a similar rate of asthma compared to the State of Michigan ($\sim 11.5\%$). All census tracts have asthma rates between 10.7% - 12.3% .

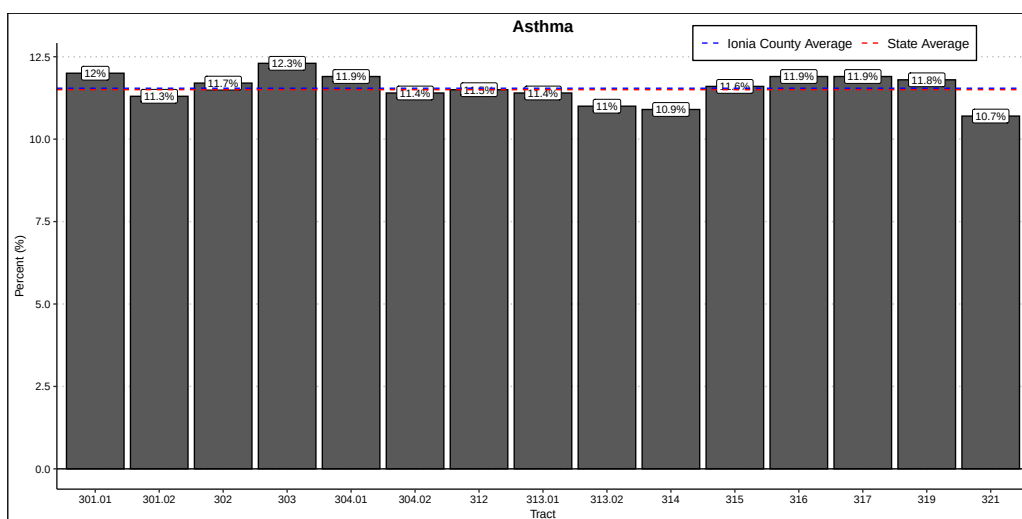


Figure 58: Asthma by Census Tract in Ionia County

6.4.5 High Blood Pressure

High blood pressure rates are similar for Ionia County and the State of Michigan (36.1% vs. 37%), with a small amount of variability in each of the census tracts making up Ionia County. Tract 301.01 (Northeast Ionia County) has the highest rate of high blood pressure (40.8%) and tract 314 (Portland city & Portland Township) has the lowest rate of high blood pressure (32.8%).

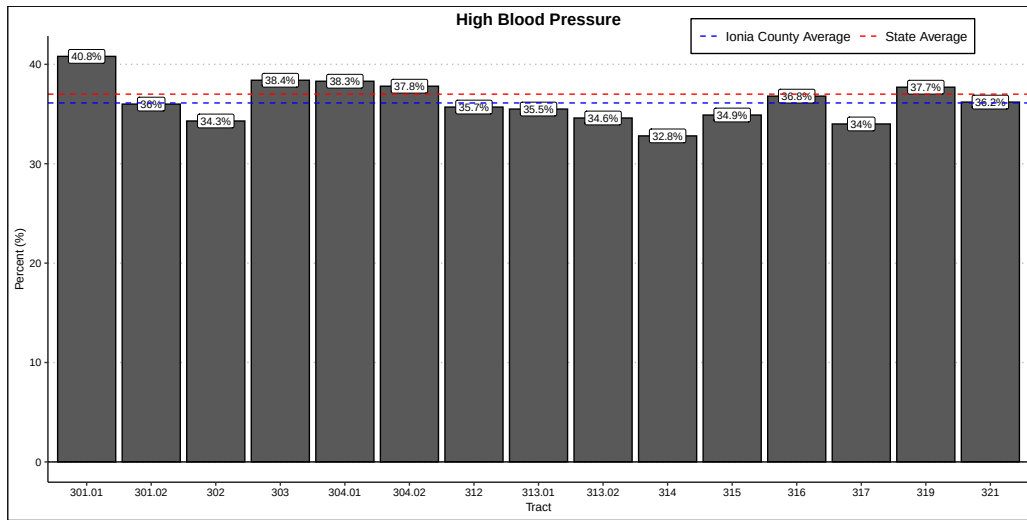


Figure 59: High Blood Pressure by Census Tract in Ionia County

6.4.6 High Cholesterol

High cholesterol pressure rates are similar for Ionia County and the State of Michigan (34.5% vs. 36%), with a small amount of variability in each of the census tracts making up Ionia County. Tract 301.01 (Northeast Ionia County) has the highest rate of high cholesterol (36.9%) while tract 321 (West Ionia City) has the lowest rate of high cholesterol (30.4%).

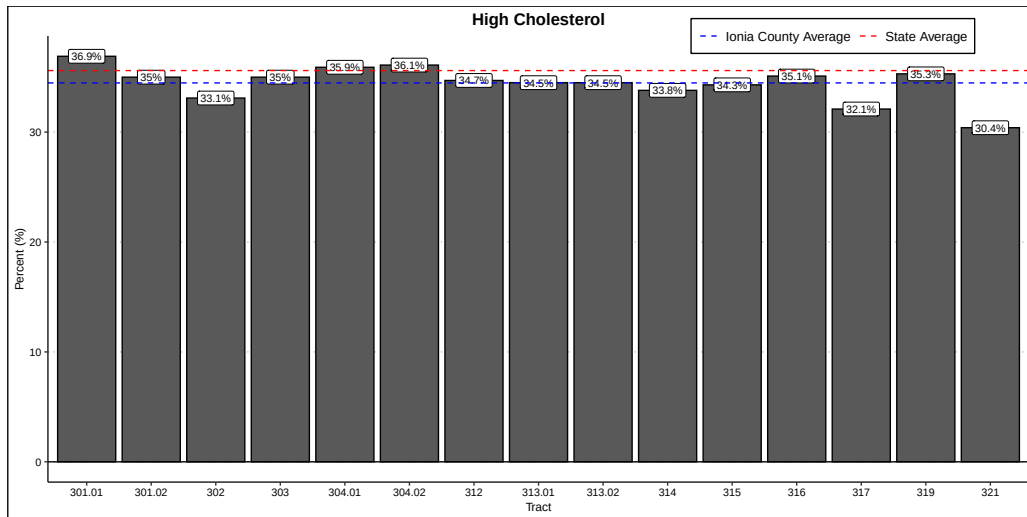


Figure 60: High Cholesterol by Census Tract in Ionia County

6.4.7 Stroke

The average stroke rates for Ionia County and the State of Michigan are similar (3.5% vs. 3.6%). Tracts 301.01 (Northeast Ionia County) and 303 (West Belding) have the highest rates of stroke (4.5% and 4.3%, respectively). Tract 314 (Portland City & Portland Township) has the lowest rate of stroke (2.7%).

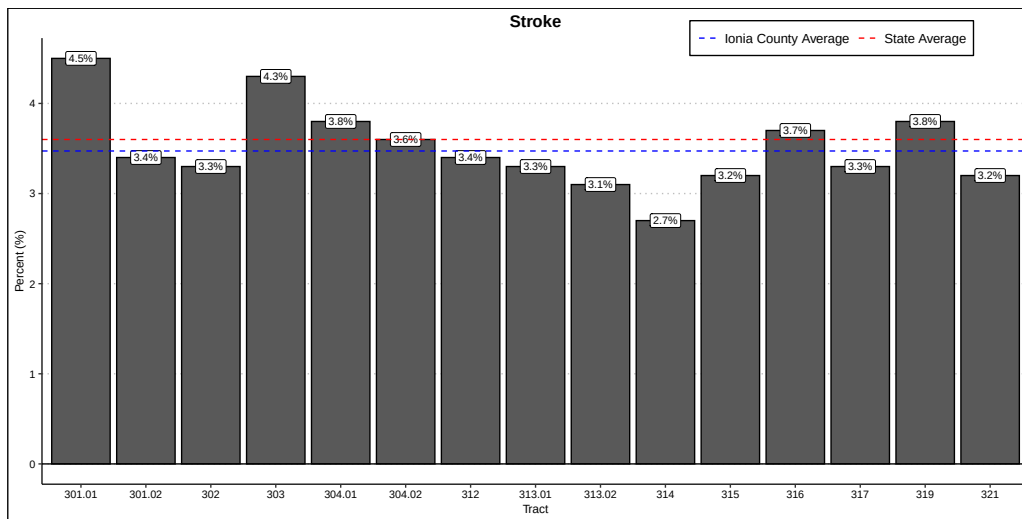


Figure 61: Stroke by Census Tract in Ionia County

6.4.8 Binge Drinking

Ionia County has a slightly lower binge drinking percentage than the State of Michigan (17.2% vs. 18.7%), with only one census tract, 321 (West Ionia city) having a rate of binge drinking higher than the state average (18.9%). Tract 301.01 (Northeast Ionia County) has the lowest rate of binge drinking (15.7%).

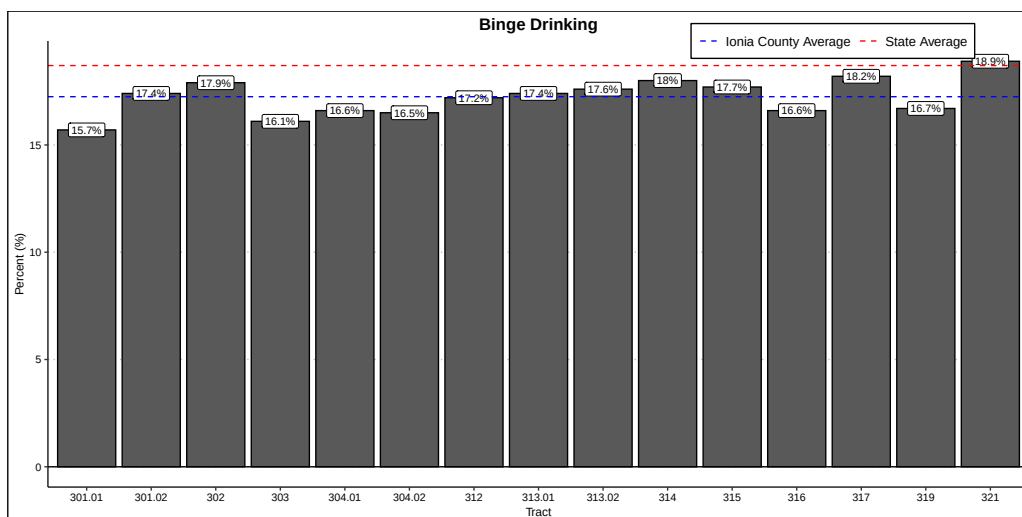


Figure 62: Binge Drinking by Census Tract in Ionia County

6.5 Pregnancy & Infancy

Data on pregnancy and infancy sourced from the Michigan Department of Health and Human Services Division for Vital Records and Health Statistics, specifically the 2024 Geocoded Michigan Birth Certificate Registry.

6.5.1 Maternal Health

Compared to the State of Michigan, Ionia County has:

- A slightly higher percent of pregnant women being under 20 years old (5.1% vs. 3.3%)
- A similar percent of births that are first births (37% vs. 39%)
- The same percent of births that are fourth and higher order births (13.3%)
- A slightly lower percent of pregnant women with less than 12 years of education (8.7% vs. 9.7%)
- A similar rate of pregnant women who are unmarried (38% vs. 37.7%)

- A lower percent of Group-B Strep present during pregnancy (13.5% vs. 16.7%)
- A higher percent of maternal HIV testing performed (95.3% vs. 91.7%).
- A similar percent of prenatal care during the first trimester (76.3% vs. 75.7%)
- And, the same percent of woman delivering a live birth who had a healthy weight prior to pregnancy (36.6%)



Figure 63: Maternal Health Statistics in Ionia County

6.5.2 Pregnancy Weight Gain

Compared to the State of Michigan, Ionia County has:

- A similar percent of women who gain less than 16 pounds while pregnant (16% vs. 16.5%)
- A slightly lower percent of women giving birth to a single baby whose weight gained was low (17% vs. 19.5%)
- A slightly lower percent of women giving birth to a single baby whose weight gain was recommended (23.7% vs. 26.4%)
- A higher percent of women giving birth to a single baby whose weight gain was excessive (55.2% vs. 50.5%)

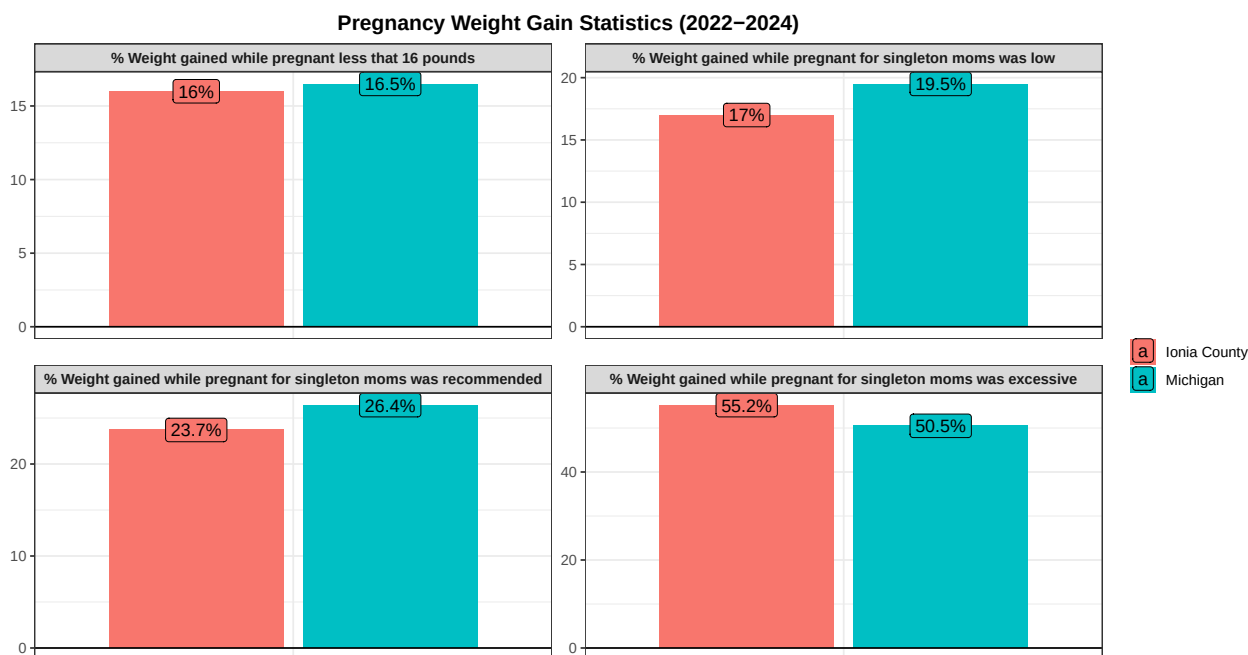


Figure 64: Pregnancy Weight Gain Statistics in Ionia County

Terminology via MDHHS:

- Weight gain is considered low or excessive if it falls below or above the IOM recommended range of weight gain. For a medically normal prepregnancy BMI, a weight gain of 25-35 lbs is recommended. See “Weight Gain During Pregnancy: Reexamining the Guidelines” for details.

6.5.3 Pregnancy Smoking (Cigarettes)

Ionia County has a slightly higher rate of smoking (cigarettes) during pregnancy (6% vs. 5.3%) than the State of Michigan.

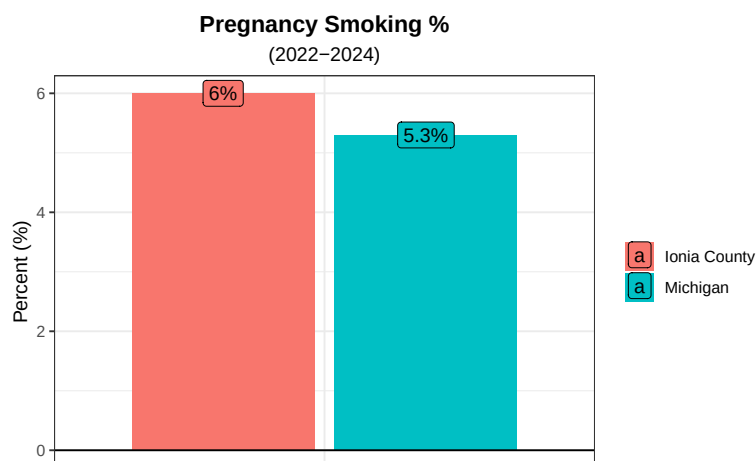


Figure 65: Pregnancy Smoking (cigarettes) Statistics in Ionia County

Terminology via MDHHS:

- Mothers who smoked while pregnant where those who had a history of smoking, but never quit, plus those who quit at some point between estimated conception and birth date.

6.5.4 Breast Feeding

Compared to the State of Michigan, Ionia County has:

- A lower percent of women whose breast feeding was not planned (5.2% vs. 11.6%)
- A higher percent of women whose breast feeding was initiated (62.1% vs. 53.7%)
- A similar percent of women whose breast feeding was planned (31.3% vs. 32.7%)
- A higher percent of women whose breast feeding was planned or initiated (93.5% vs. 86.4%)

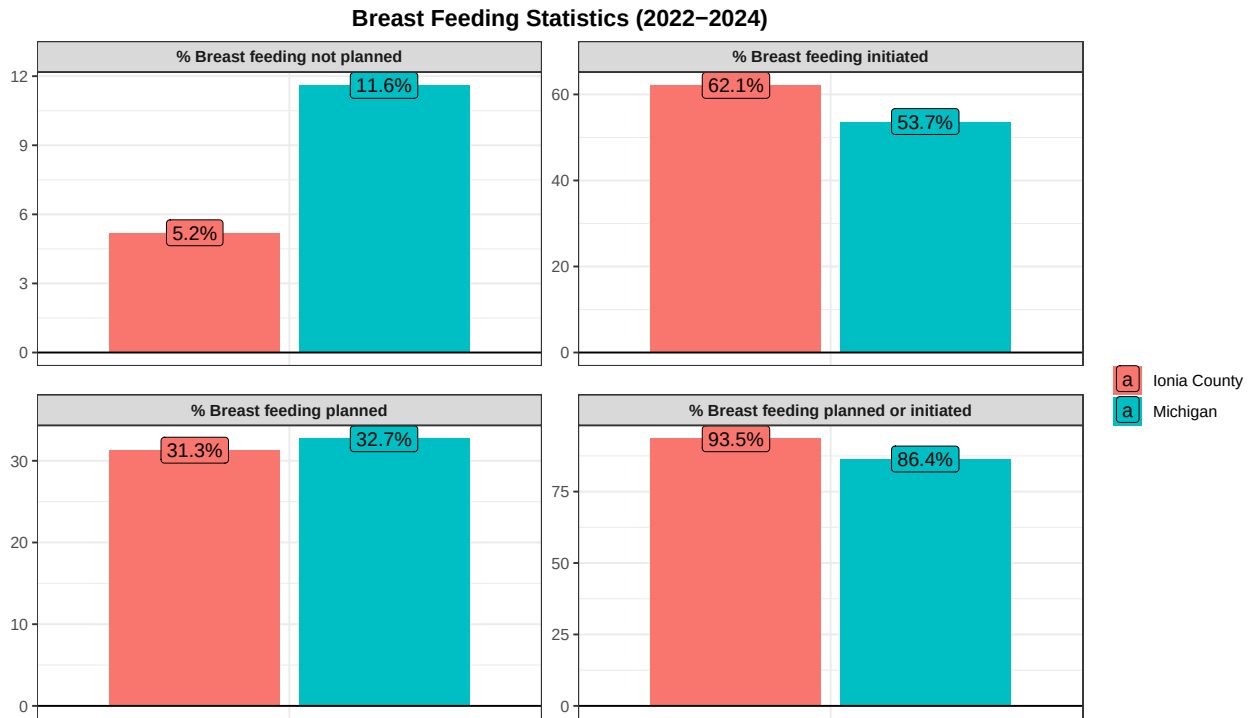


Figure 66: Breastfeeding Statistics in Ionia County

6.5.5 Cesarean Sections

Compared to the State of Michigan, Ionia County has:

- A lower percent of births of Cesarean delivery (27.8% vs. 32.4%),
- A lower percent of women with low-risk live births with Cesarean delivery (22.8% vs. 28.2%)
- A lower percent of women with low-risk live births that were first birth Cesarean sections (11.6% vs. 15.9%)
- A lower percent of Cesarean births among low-risk women with prior Cesarean births (78.2% vs. 81.9%)

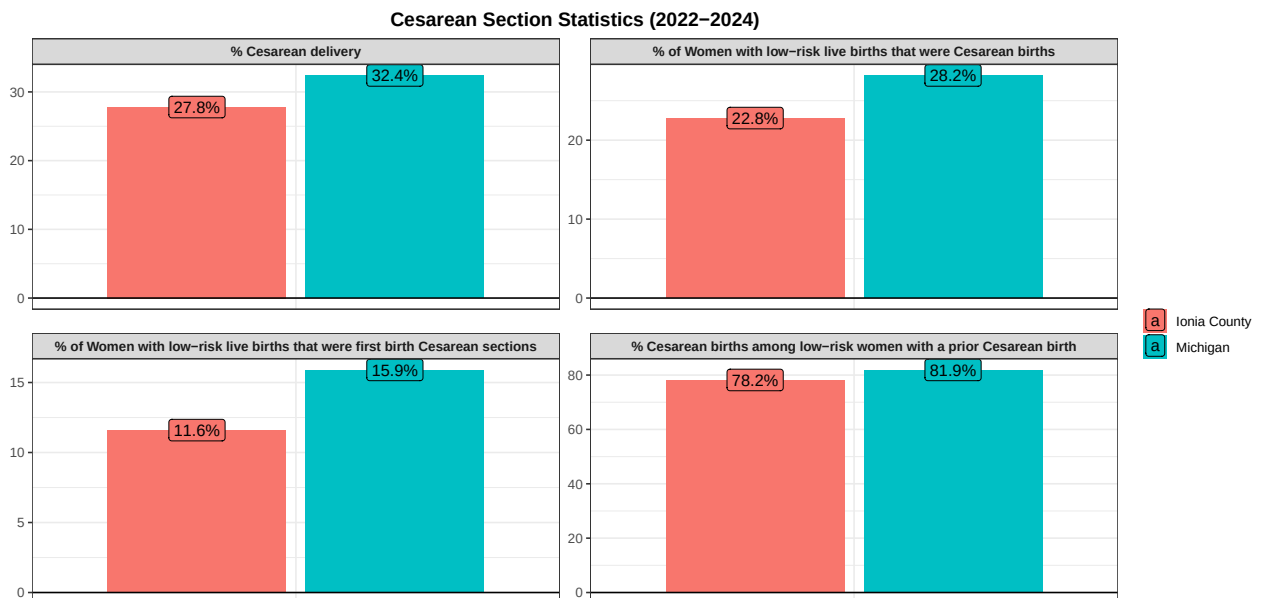


Figure 67: Cesarean Section Statistics in Ionia County

Terminology via MDHHS:

- Low-risk Cesarean births are singleton, full-term (37 weeks gestation or older), with vertex presentation that were delivered using a C-section.

6.5.6 Infants

Compared to the State of Michigan, Ionia County has:

- A similar percent of infants with low birthweight (8.8% vs. 8.9%)
- A similar percent of infants with very low birthweight (1.1% vs. 1.4%)
- A slightly lower percent of very preterm births (1.3% vs. 2.4%)
- A similar percent of live births with 32 to 33 weeks of gestation (2% vs. 1.3%)
- A slightly lower percent of late preterm births (6.4% vs. 7.7%)
- A slightly lower percent of preterm births (9.6% vs. 11.4%).
- A similar percent of births with an Apgar score < 7 in the first 6 minutes of life
- A similar percent of births of a single child (singleton births)
- a slightly lower male/female ratio than the State of Michigan. For every 100 women, 101.4 men are born in Ionia County. For every 100 women, 104.5 men are born in the State of Michigan.

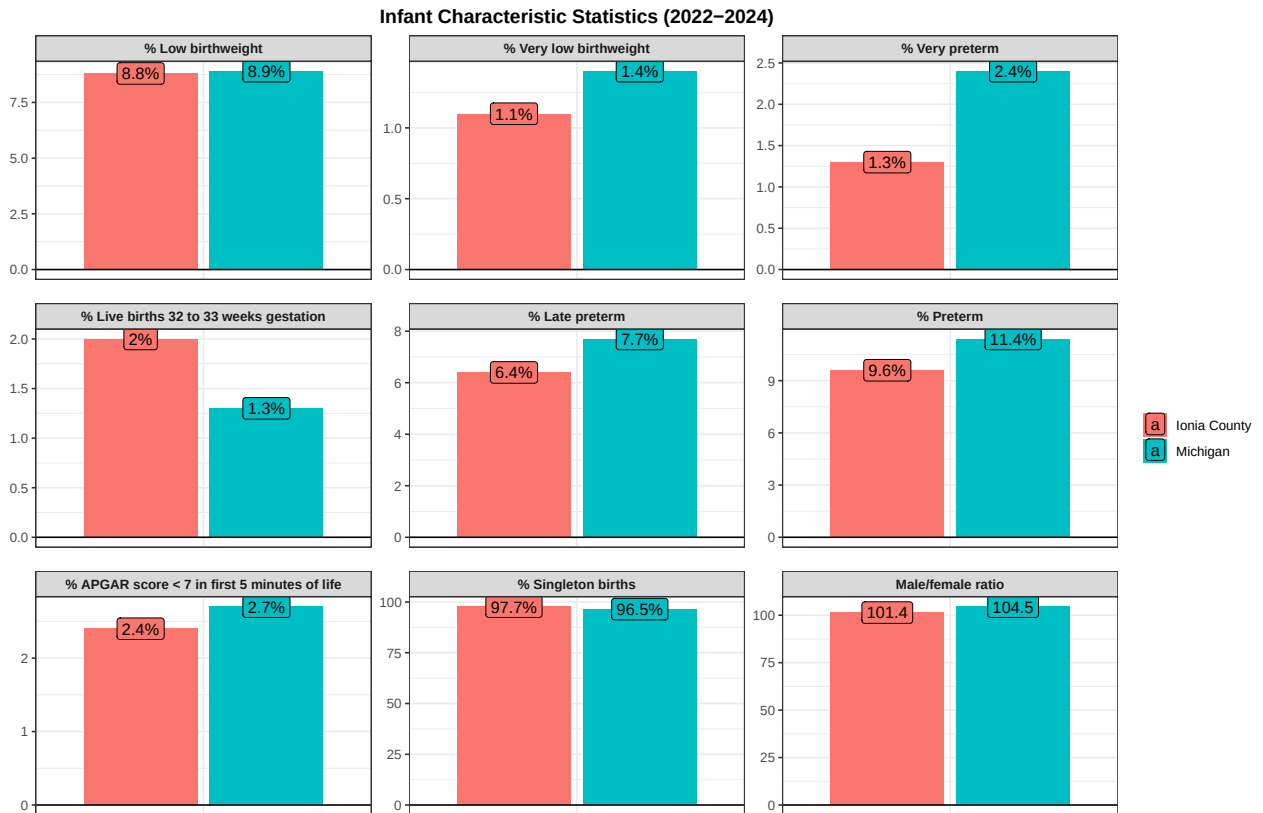


Figure 68: Infancy Statistics in Ionia County

Terminology via MDHHS:

- A healthy weight for a woman prior to pregnancy is defined as a BMI of 18.5 to 24.9.
- Low weight births are less than 2,500 grams (~ 5.5 lbs).
- Very low weight births are less than 1,500 grams (~ 3.3 lbs).
- Very preterm births are infants born prior to 32 completed weeks of gestation.
- Late preterm births are infants born between 34 to 36 completed weeks of gestation.
- Preterm births are infants born prior to 37 completed weeks of gestation.
- The Apgar score is a metric used to assess newborn health immediately after birth (NIH)
- Singleton birth is a single birth delivery.
- Male/female ratio is the number of male resident live births divided by the female resident live births $\times 100$.

6.5.7 Women, Infants, and Children Program

Ionia County has a lower percent of pregnant women on WIC food during pregnancy than the State of Michigan (23.8% vs. 30.1%).

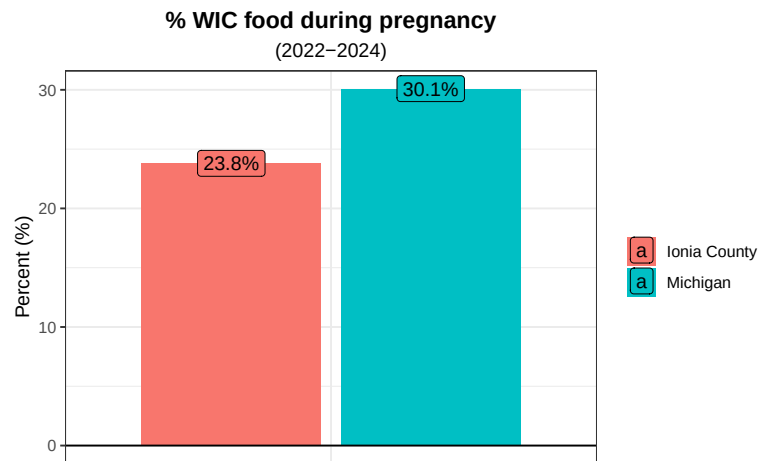


Figure 69: WIC Statistics in Ionia County

6.6 STI & HIV Prevalence

6.6.1 STI

STI & HIV data sourced from Michigan Department of Health and Human Services. Compared to the State of Michigan, Ionia County has (per 100,000 people):

- A lower rate of chlamydia (191.7 vs. 382.5)
- A lower rate of gonorrhea (46.8 vs. 134.4)
- A lower rate of latent syphilis (3 vs. 21.5)
- A lower rate of primary and secondary (P&S) syphilis (3 vs. 6.8)
- A lower percent of chlamydia re-infection (2% vs. 9%)
- And, a lower percent of gonorrhea re-infection (6% vs. 8%)

6.6.2 HIV



Figure 70: STI Statistics in Ionia County

Compared to the State of Michigan, Ionia County has:

- A lower rate of HIV (per 100,000) (58.9 vs. 186.9)
- A lower percent of Stage 3 HIV (AIDS) among persons with HIV (5.1% vs. 6.8%)
- A similar prevalence of gonorrhea/chlamydia among persons with HIV (5.1% vs. 5.2%)
- A slightly lower prevalence of syphilis among persons with HIV (2.6% vs. 3.1%)
- And, a lower HIV transmission rate (new diagnoses/reported prevalence $\times$ 100) (2.6 vs. 3.6)

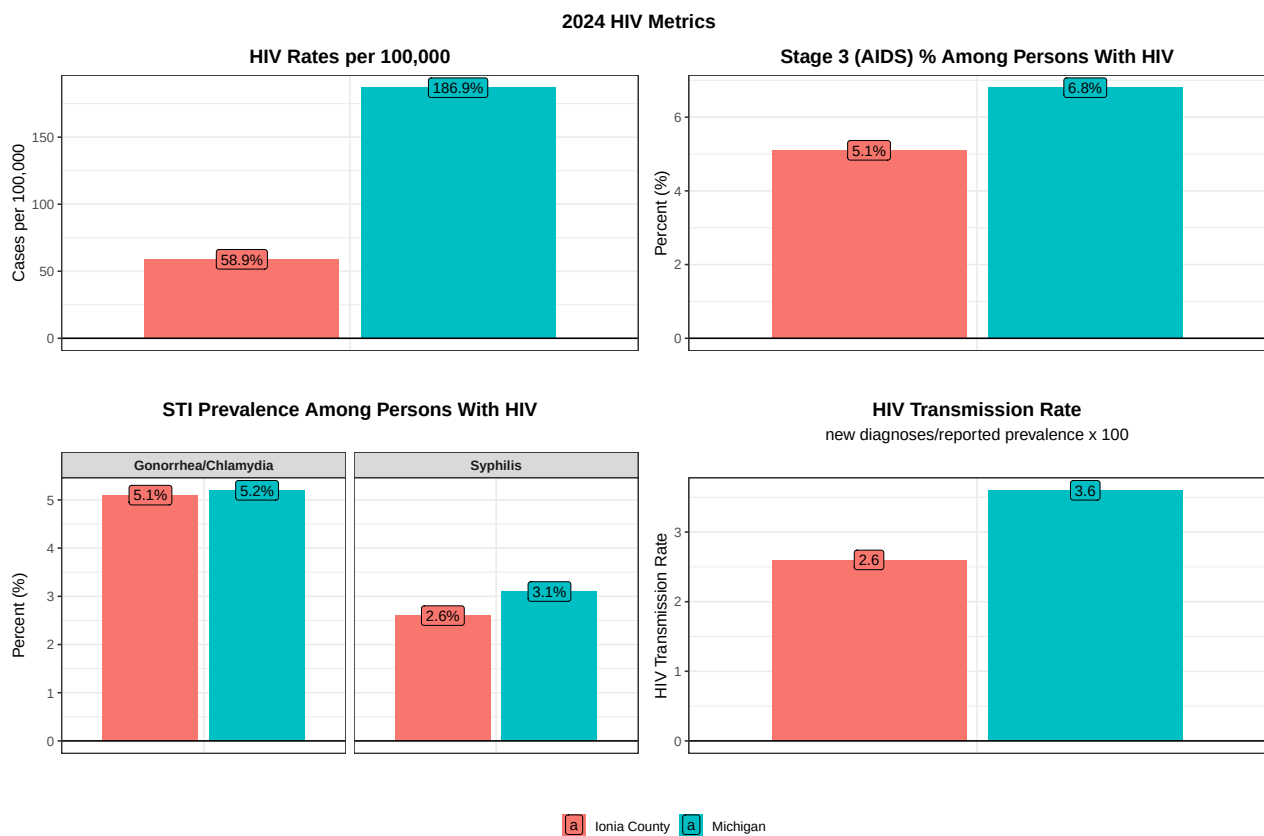


Figure 71: HIV Statistics in Ionia County

6.7 Leading Causes of Death

The leading causes of death in Ionia County, the State of Michigan, and the USA are similar, with the two greatest causes of death being heart disease and cancer. The age-adjusted rate of heart disease death in Ionia County is less than the age-adjusted rate in Michigan, but still higher than the national age-adjusted rate. For cancer, Ionia County has a higher age-adjusted rate of cancer deaths than the State of Michigan and the United States. Ionia County also has a higher age-adjusted rate of deaths from stroke and from suicide than the State of Michigan and United States.

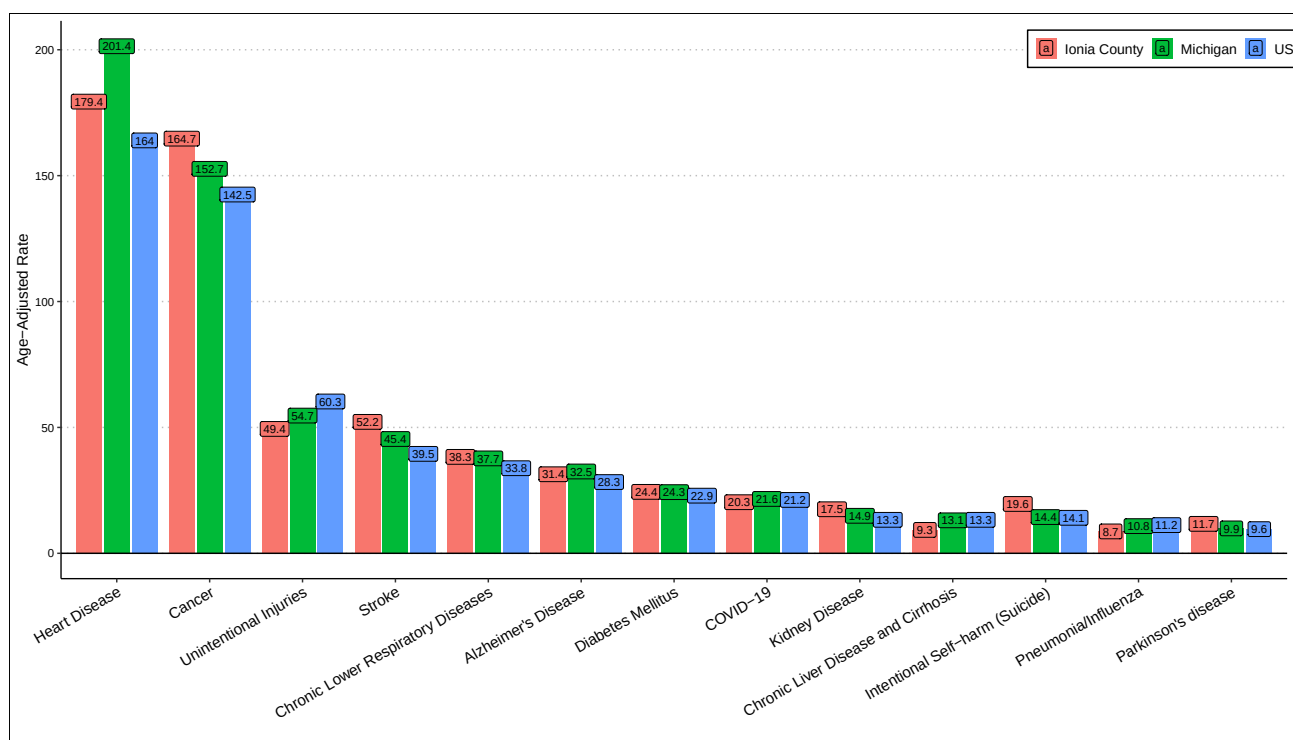


Figure 72: Ionia County vs. Michigan vs. USA: Leading Causes of Death

Death rates in Ionia County are lower for drug overdose and firearm but higher for motor vehicle and drunk driving deaths compared to the State of Michigan:

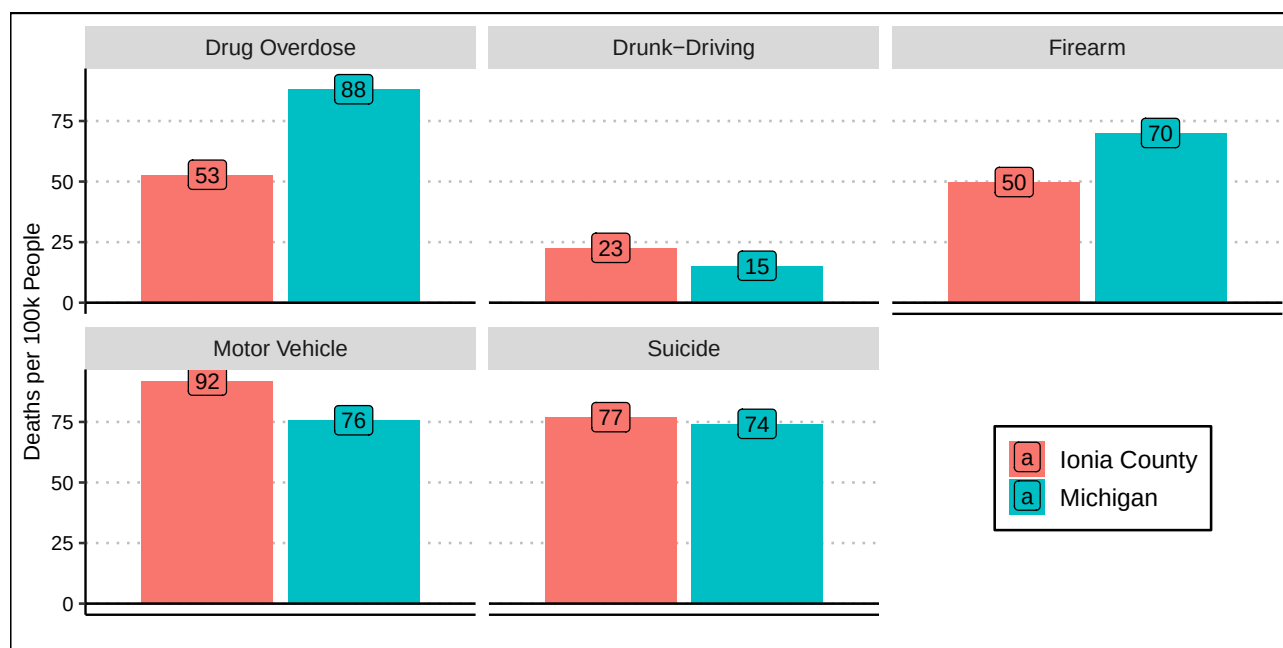


Figure 73: Ionia County vs. Michigan: Specific Death Rates

6.8 Substance Use Vulnerability Index

This section contains data from the 2022 County & Zip Code Tabulation Area (ZCTA) Level Substance Use Vulnerability Index Report sourced from the Michigan Department of Health & Human Services. Census Bureau created ZCTAs align with actual ZIP codes in Michigan 86% of the time. For ease of understanding, ZCTAs have been translated to the most recognizable geographic name they represent.

The stated purpose of the Michigan SUVI is to identify the extent to which certain communities have been affected by substance use by identifying "Substance Use Burden," "Substance Use Resources," and

”**Social Vulnerability.**” Thus, the variables and metrics highlighted were specifically chosen to best quantify those factors.

Details on the ZIP Code tabulations and calculated metrics can be found in the methodology for the Michigan SUVI.

6.8.1 Opioid Prescription

Ionia County has a lower rate of opioid prescriptions per person compared to the State of Michigan (34.5 vs. 42). Ionia County has the 57th highest opioid prescription rate per person out of 83 counties in Michigan.

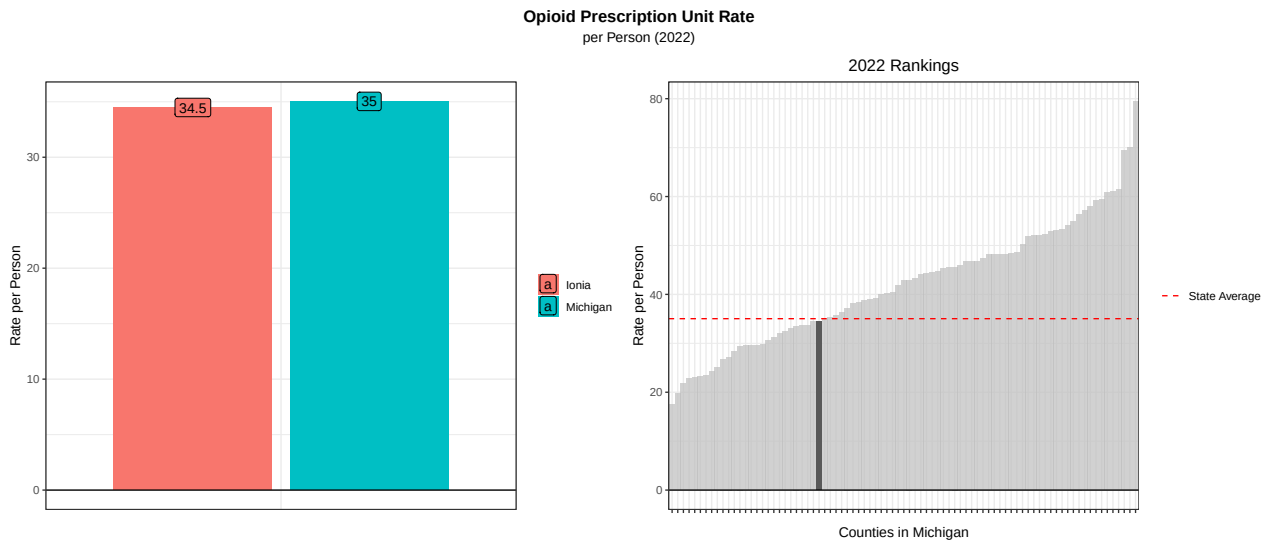


Figure 74: Opioid Prescription Rate in Ionia County

Four ZIP codes in Ionia County, Belding, Lyons, Orleans, and Hubbardston have higher opioid prescription unit rates per person than the state average.

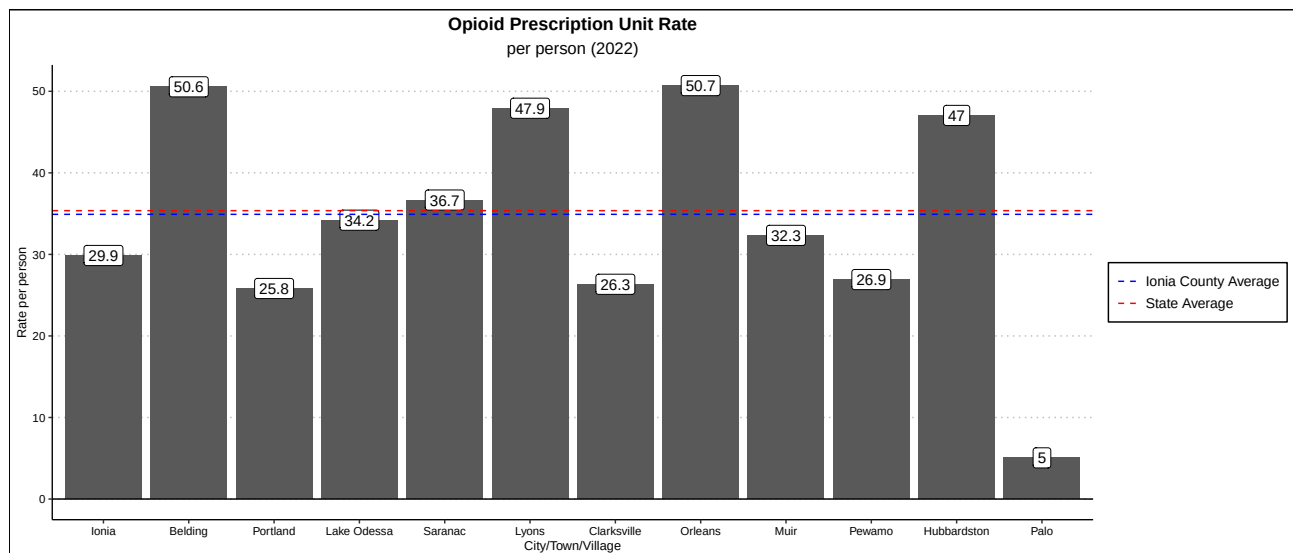


Figure 75: Opioid Prescription Rate in Ionia County ZIP Codes

6.8.2 Buprenorphine Prescription

Ionia County has a lower unit rate of buprenorphine (an opioid medication) prescriptions per 1,000 than the State of Michigan (2479.9 vs. 3086.2). Ionia County has the 53rd highest buprenorphine prescription unit rate out of 83 counties in Michigan.

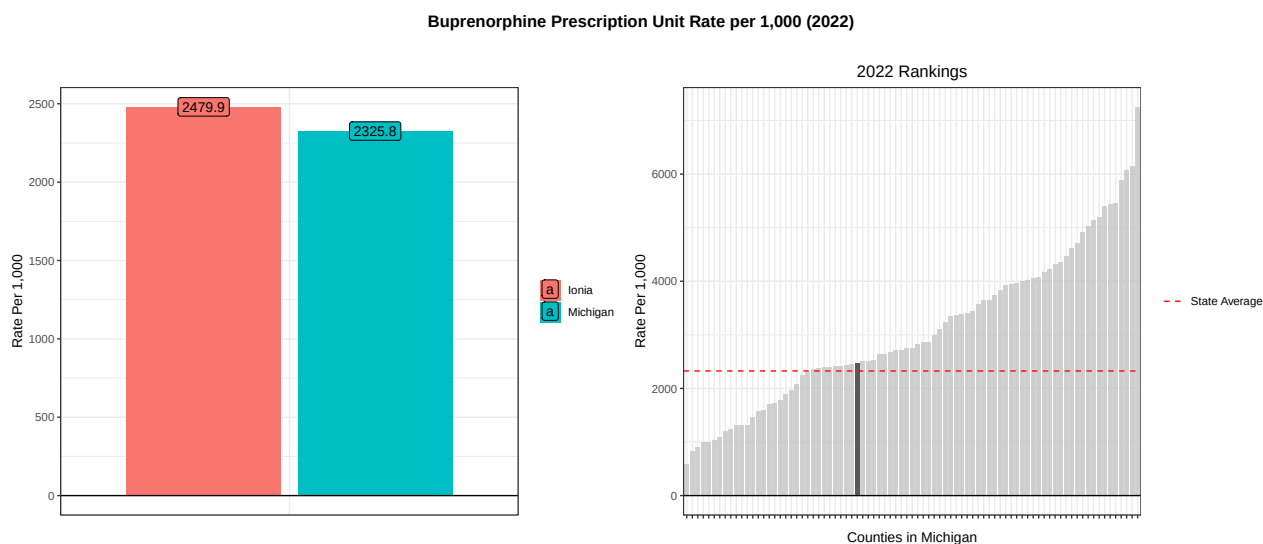


Figure 76: Buprenorphine Prescription Rate in Ionia County

Five ZIP codes, Ionia, Belding, Lyons, Orleans, and Muir have higher buprenorphine prescription unit rate per 1,000 than the state average.

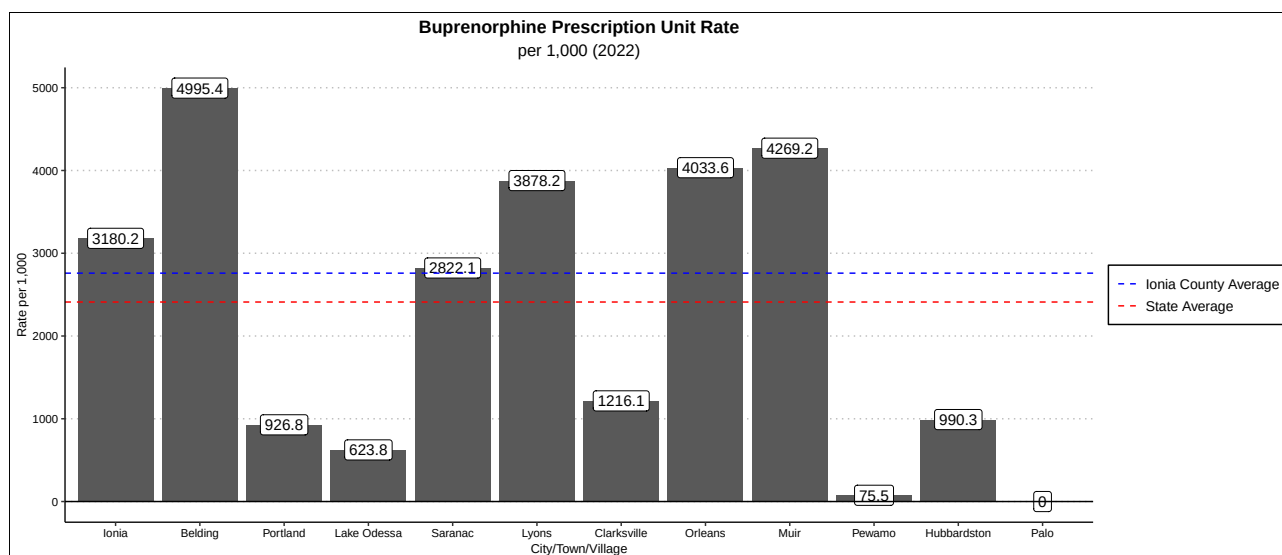


Figure 77: Buprenorphine Prescription Rate in Ionia County ZIP Codes

6.8.3 Non-Fatal Overdose

Ionia County has a lower rate of 3-year non-fatal overdose per 100,000 than the State of Michigan (233 vs. 270.2). Ionia County has the 60th highest 3-year non-fatal overdose rate per 100,000 out of 83 counties in Michigan.

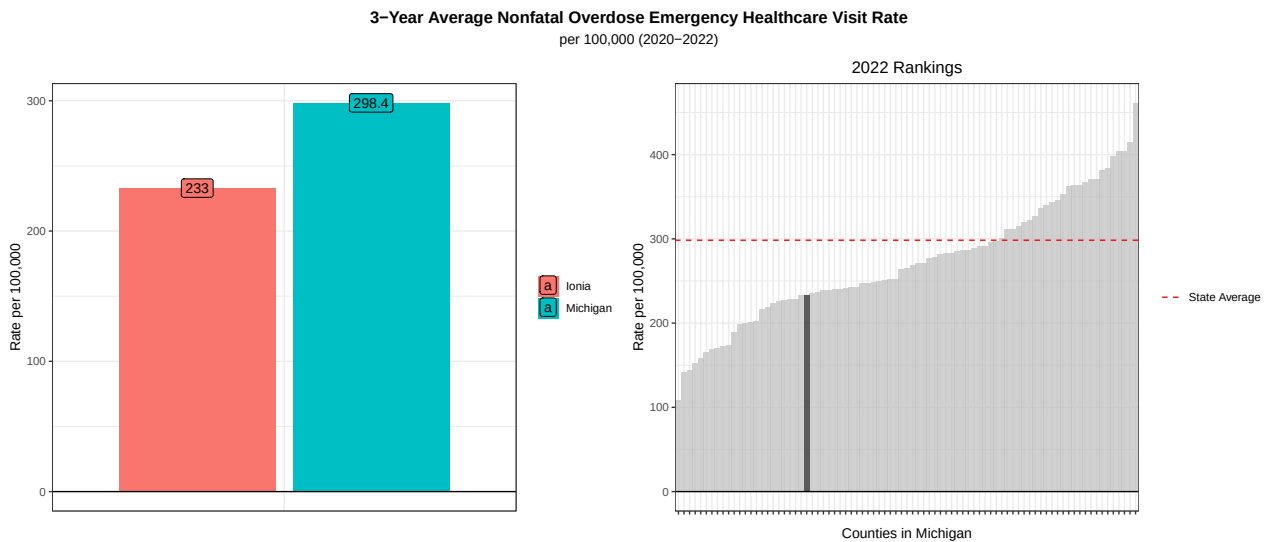


Figure 78: 3-Year Non-Fatal Rate in Ionia County

Three ZIP codes, Ionia, Muir, and Hubbardston have a higher 3-year non-fatal overdose rate per 100,000 than the state average.

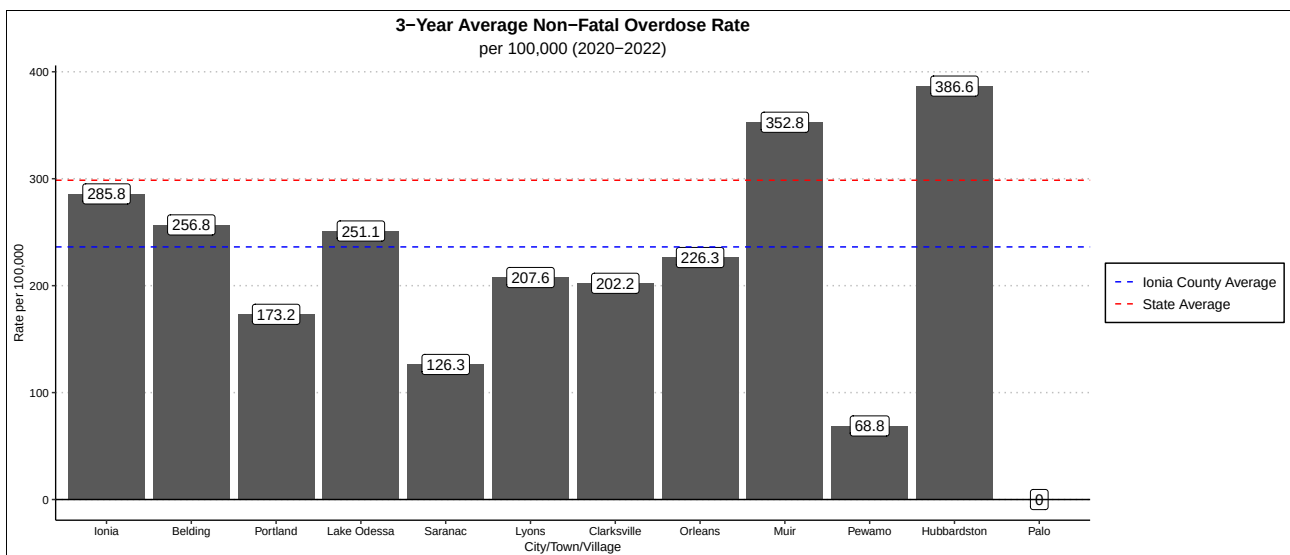


Figure 79: Nonfatal Overdose Rate in Ionia County ZIP Codes

6.8.4 Fatal Overdose

Ionia County has a lower rate of 5-year non-fatal overdose rate per 100,000 than the State of Michigan (15.4 vs. 18.8). Ionia County has the 51st highest fatal overdose rate per 100,000 out of 83 counties in Michigan.

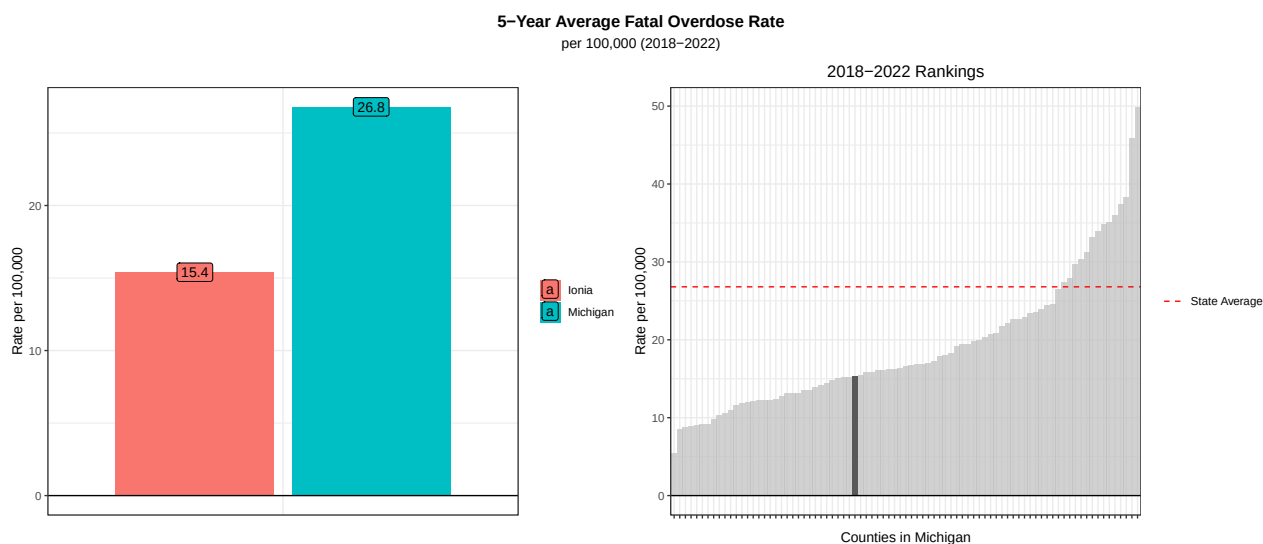


Figure 80: 5-Year Fatal Overdose Rate in Ionia County

Four ZIP codes, Ionia, Lyons, Orleans, and Muir have higher 5-year fatal overdose rate per 100,000 than the state average.

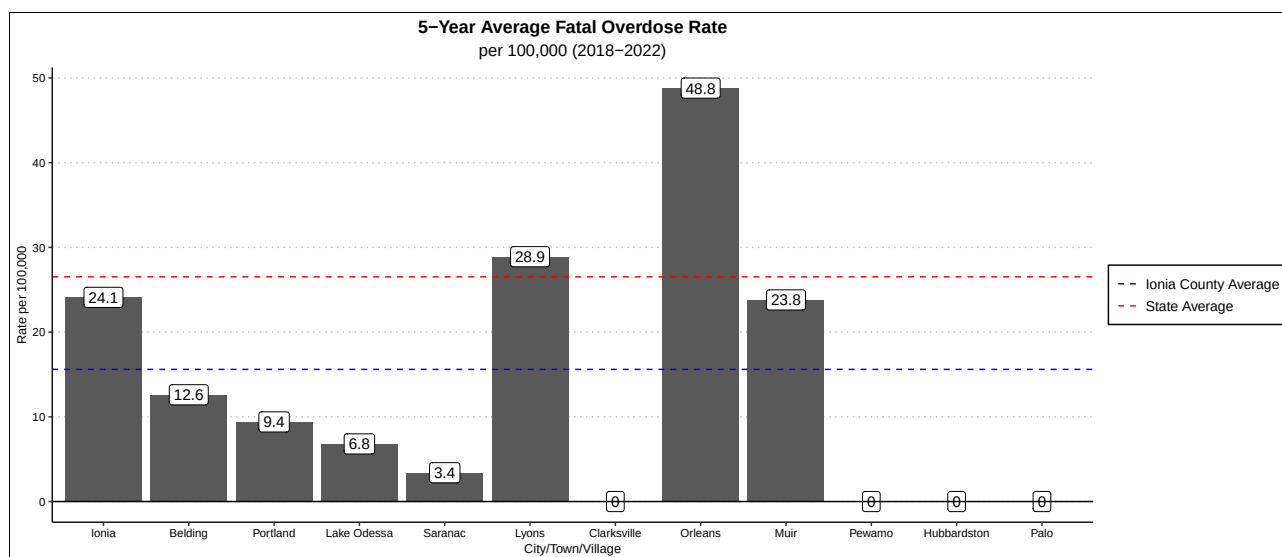


Figure 81: Fatal Overdose Rate in Ionia County ZIP Codes

6.8.5 Substance Use Disorder Treatment Centers

Ionia County has a higher percentage of its population within a 30 minute drive of a Substance Use Disorder (SUD) Treatment Center than the State of Michigan (100% vs. 85.7%). Ionia County is one of 23 counties with 100% of its population within a 30 minute drive of a SUD treatment center.

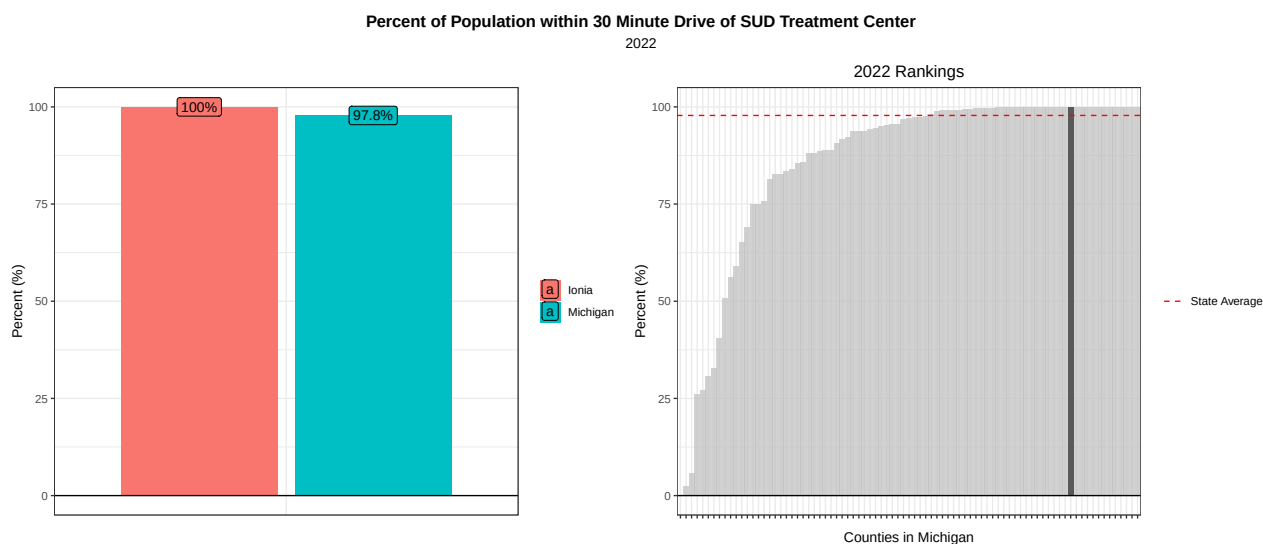


Figure 82: Population Near SUD Treatment Center in Ionia County

There is one SUD treatment center within Ionia County: Samaritas Belding in Belding City (indicated by the pink circle).

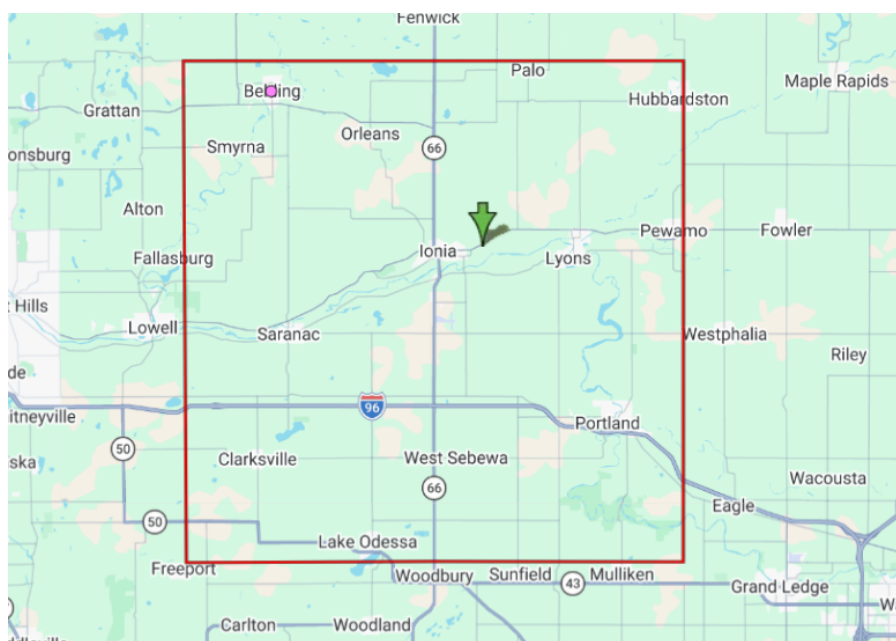


Figure 83: Substance Use Treatment Center Map

(Map courtesy of findtreatment.gov)

6.8.6 Syringe Service Programs

Ionia County has a lower percentage of its population within a 15 minute drive of a syringe service program (SSP) than the State of Michigan (0% vs. 27.6%). Ionia County is one of 21 counties with 0% of its population within a 15 minute drive of an SSP.

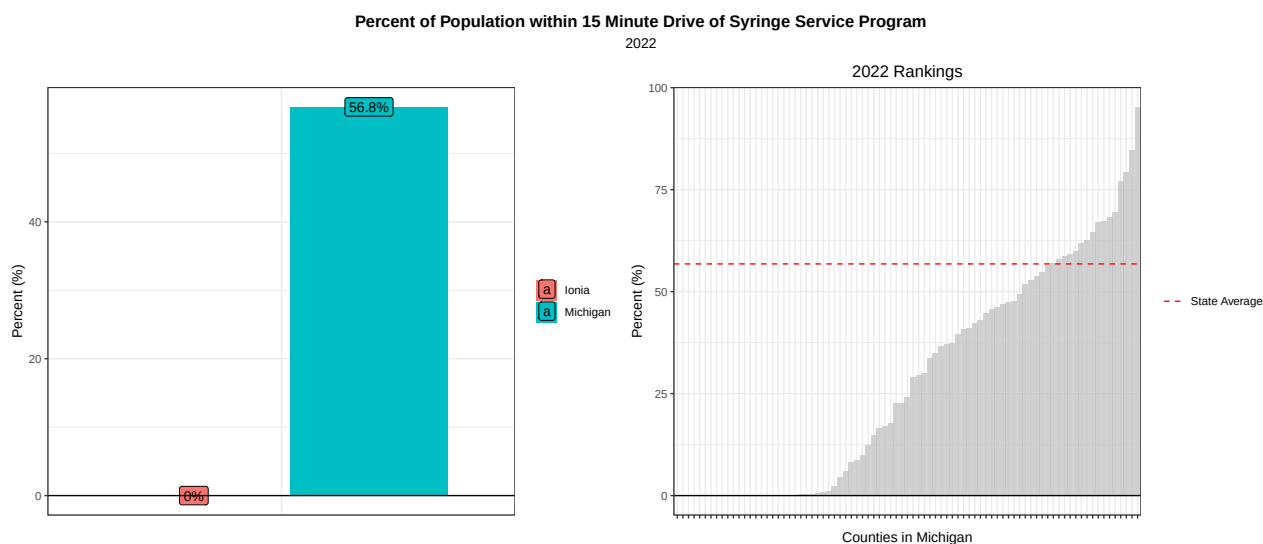


Figure 84: Population Near Syringe Service Program in Ionia County

There are no reported SSPs in Ionia County, and the nearest reported SSP to Ionia County Residents could either be in Greenville, Lansing, or Grand Rapids.

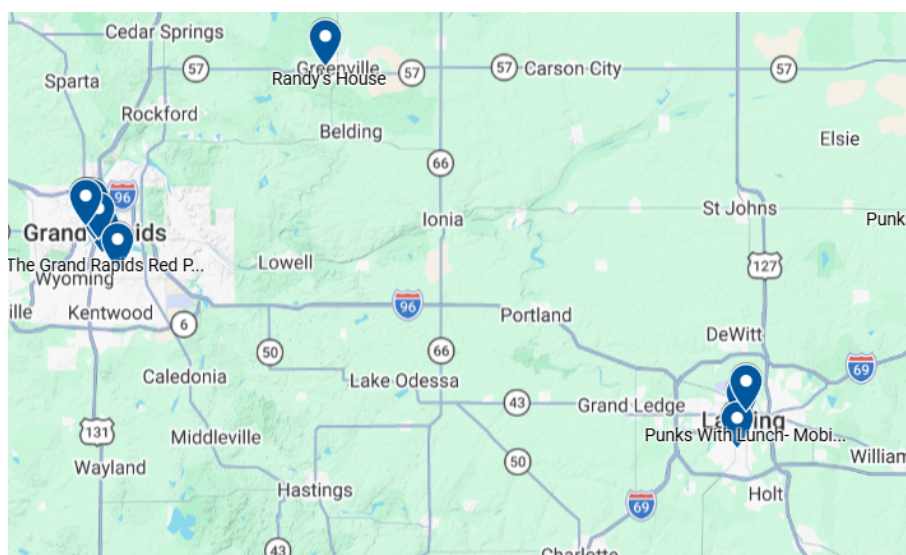


Figure 85: Syringe Service Program Map

6.8.7 Narcan Vending Machines

While not included in the Substance Use Vulnerability Report, the map below shows the availability of Narcan: an opioid antagonist medication that can be administered to restore breathing after an opioid overdose. There are no *reported* Narcan vending machines in Ionia County, and the nearest *reported* Narcan vending machine to somebody in Ionia County could be in Greenville, Lowell, or Lansing.

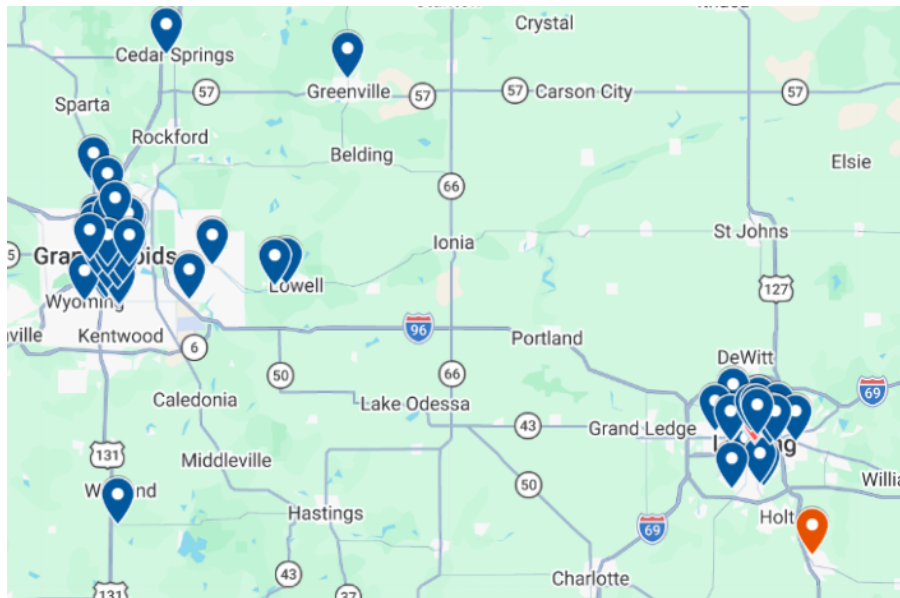


Figure 86: Narcan Vending Machine Map

Remark: It should be noted there is currently Narcan available at the Ionia County Health Department building, so it can be assumed this map is not all encompassing.

6.8.8 Drug Arrests

Ionia County has a lower rate of drug arrests per 100,000 than the State of Michigan (79.3 vs. 158.5). Ionia County has the 65th highest rate of drug related arrests per 100,000 out of 83 counties.

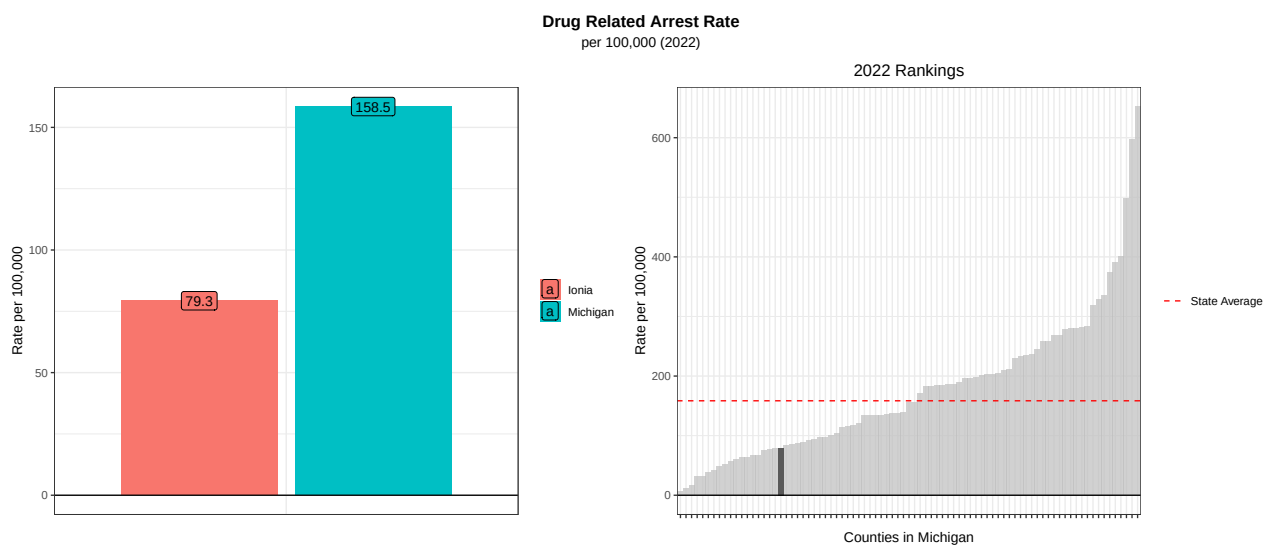


Figure 87: Drug Related Arrest Rate in Ionia County

Remark: drug arrest data is not available on the ZIP code level because of missing data issues.

6.8.9 Substance Use Vulnerability Index Score

Shown below is the methodology for calculating the score for the substance use index:

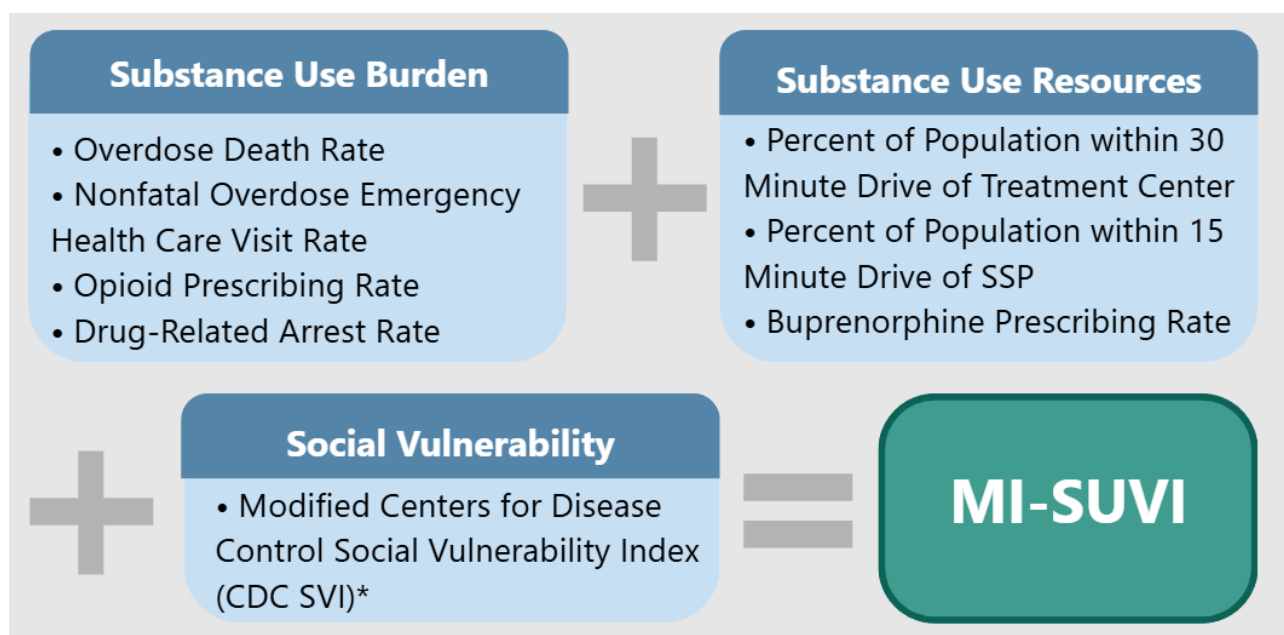


Figure 88: Substance Use Vulnerability Index Methods

Ionia County has a lower substance use vulnerability index, meaning lower vulnerability, than the state average (-0.4 standard deviations vs. -0.1 standard deviations). Ionia County has the 54-56th (tied) highest substance use vulnerability index score out of 83 counties.

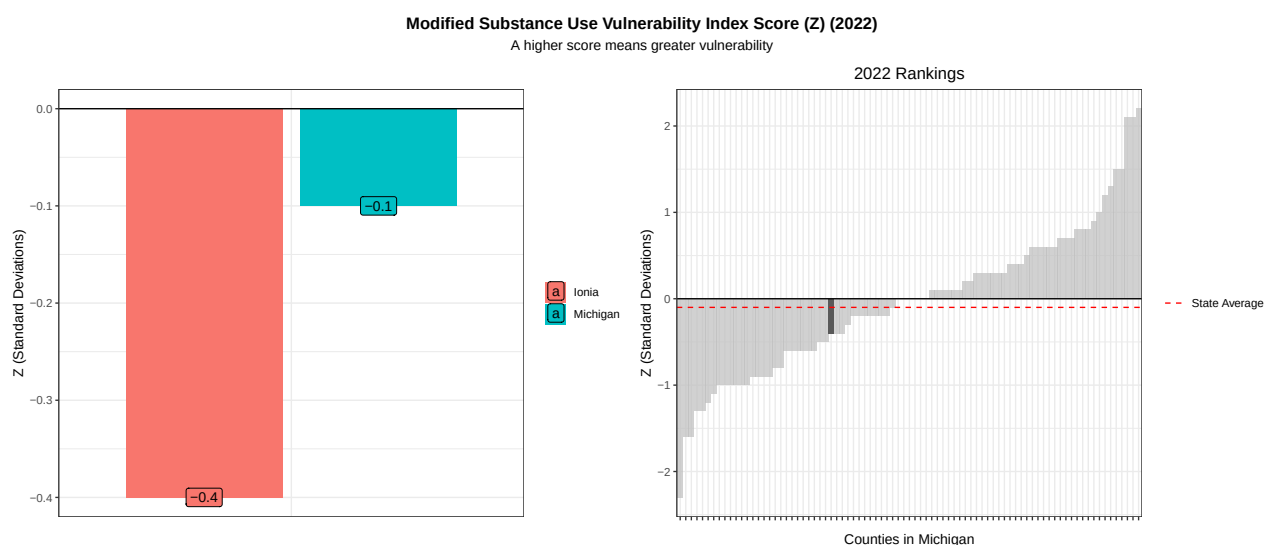


Figure 89: Substance Use Vulnerability Index in Ionia County

Three ZIP codes, Ionia, Orleans, and Hubbardston have a higher substance use vulnerability index, meaning more greater vulnerability to substance abuse, than the state average.

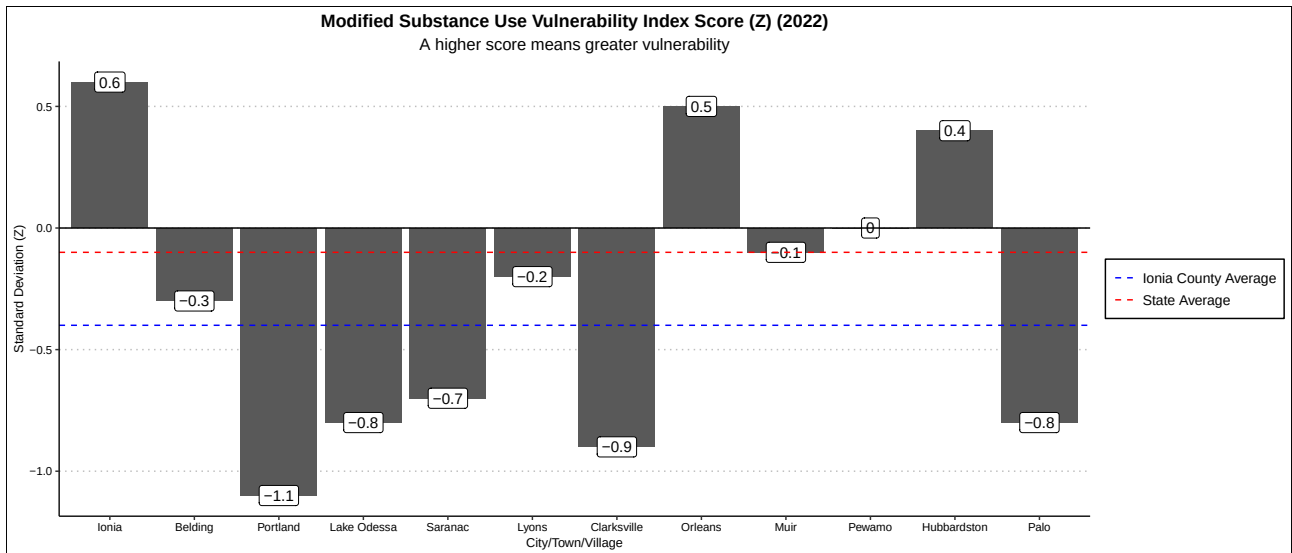


Figure 90: Substance Use Vulnerability Index in Ionia County ZIP Codes

7 Environmental Health, Pollution, & Climate

All data on environmental health and climate was sourced from the Centers for Disease Control and Prevention and Agency for Toxic Substances Disease Registry: Environmental Justice Index (2024).

Factors and metrics of particularly interest, combined with reliability of data are shown alongside aggregated metrics concerning environmental health, pollution, and climate impacts.

7.1 Greenspaces & Walkability

All census tracts in Ionia County are below the national average for proportion of the tract within 1 mile of a park or greenspace. While some tracts are at 100%, such as 303 (West Belding), 317 (East Ionia City), and 319 (Orange & Berlin Townships), the national average is higher than the rounded 100% figure that those tracts have. In census tract 313.01 (Boston Township), only 1% of the tract lives within 1 mile of a park or greenspace.

All census tracts in Ionia County are below the national average for National Walkability Index, a metric defined by the EPA, which considers factors such as land usage, proximity to transit stops, employment mix, and intersection proximity. The tracts with the greatest walkability scores are 302 (East Belding) and 317 (East Ionia City). The lowest walkability scores are in 304.02 (Keene Township and most of Otisco Township) and 321 (West Ionia City).

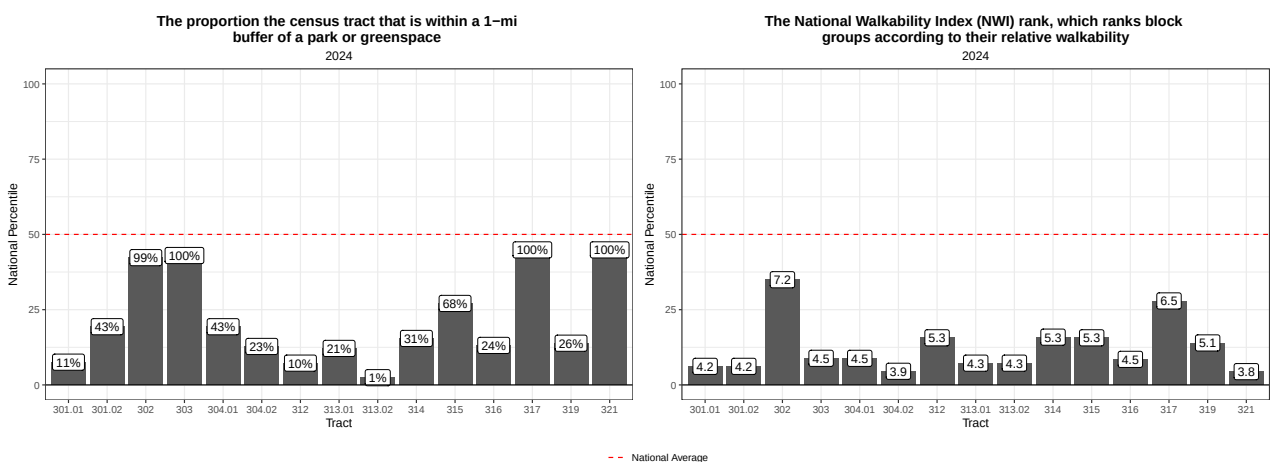


Figure 91: Walkability and Green Space metrics in Ionia County Census Tracts

7.2 Pollution

7.2.1 Air Pollution

Almost all census tracts in Ionia County are above the national average for days above the regulatory standard for Ozone. The highest tracts for Ozone exposure are 313.01 and 313.02 (South Boston and Campbell Townships). Census tract 315 (Danby Township) has the lowest and only below average exposure nationally.

Ionia County census tracts all have below average exposure to ambient concentrations of diesel particulate matter.

Five census tracts in Ionia County have above average mean number of days above the regulatory standard for Particulate Matter in the air: 312 (Odessa & Sebewa Townships), 315 (Danby Townships), 317 (East Ionia City), 319 (Berlin, Orange, & part of Easton Township), and 321 (West Ionia City).

All Ionia County census tracts except 319 (Berlin, Orange, & part of Easton Township) are below the national average for the likelihood of developing cancer from air toxics over the course of a lifetime, assuming continuous exposure.

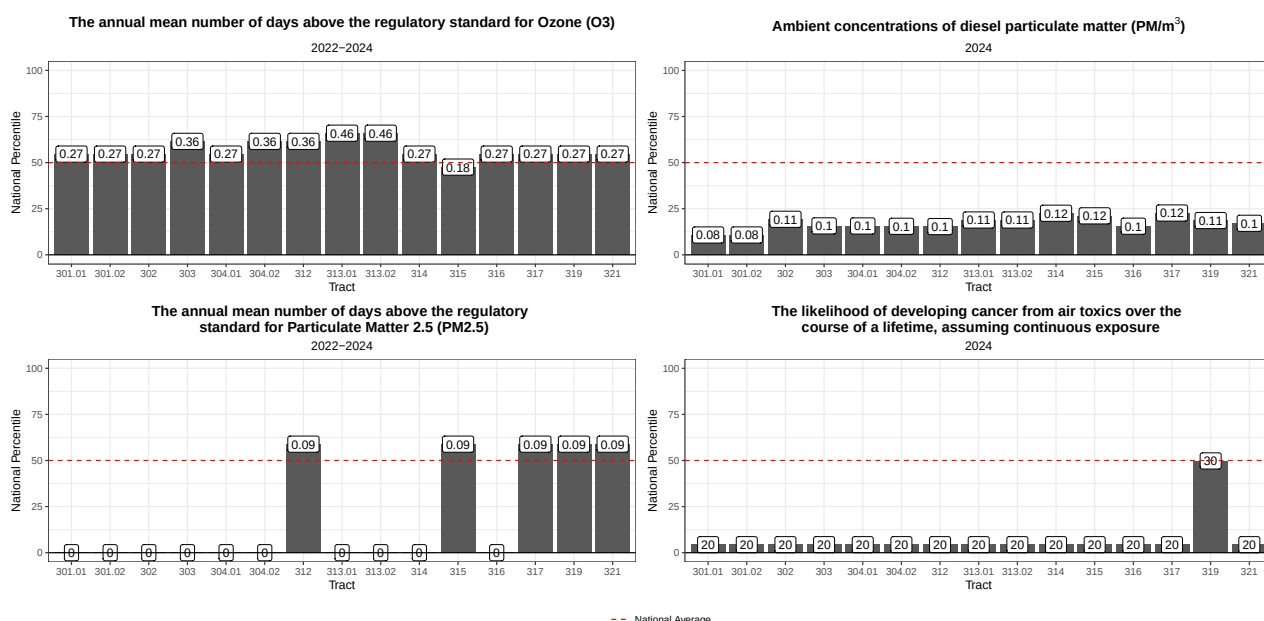


Figure 92: Air Pollution Metrics in Ionia County Census Tracts

7.2.2 Waste Pollution

Six Ionia County census tracts, 302 (East Belding), 303 (West Belding), 304.01 (North Otisco Township), 316 (North Easton & Ionia Township), 317 (East Ionia City), and 319 (Berlin, Orange, & part of Easton Township), are above the national average for the proportion of the population within 1 mile of an EPA National Priority List site, a site with “known releases or threatened releases of hazardous substances, pollutants, or contaminants” (EPA).

Three Ionia County census tracts, 302 (East Belding), 303 (West Belding), and 317 (East Ionia City) are above the national average for proportion of the population within a mile of an EPA Toxic Release Inventory Site.

Many Ionia County census tracts are above the national average for the proportion of the population within a mile of an EPA Treatment, Storage, and Disposal Facility. Two tracts, 302 (East Belding) and 317 (East Ionia City), are above 80% for proportion of the population within a mile of an EPA Treatment, Storage, and Disposal Facility.

Several Ionia County census tracts are above the national average for the proportion of the population within a mile of an EPA Risk Management Plan site. Four tracts, 301.02 (Lyons and southern North Plains Township), 304.01 (North Otisco Township), 312 (Odessa and Sebewa Townships), and 315 (Danby Township) are above 5% for the proportion of the population within a mile of an EPA Risk Management Plan site.

Remark: 0% or 100% labels shown in the figure above convey the most precise statistical metric provided by the CDC, but more precise metrics used for national percentile calculations can provide visual discrepancies.

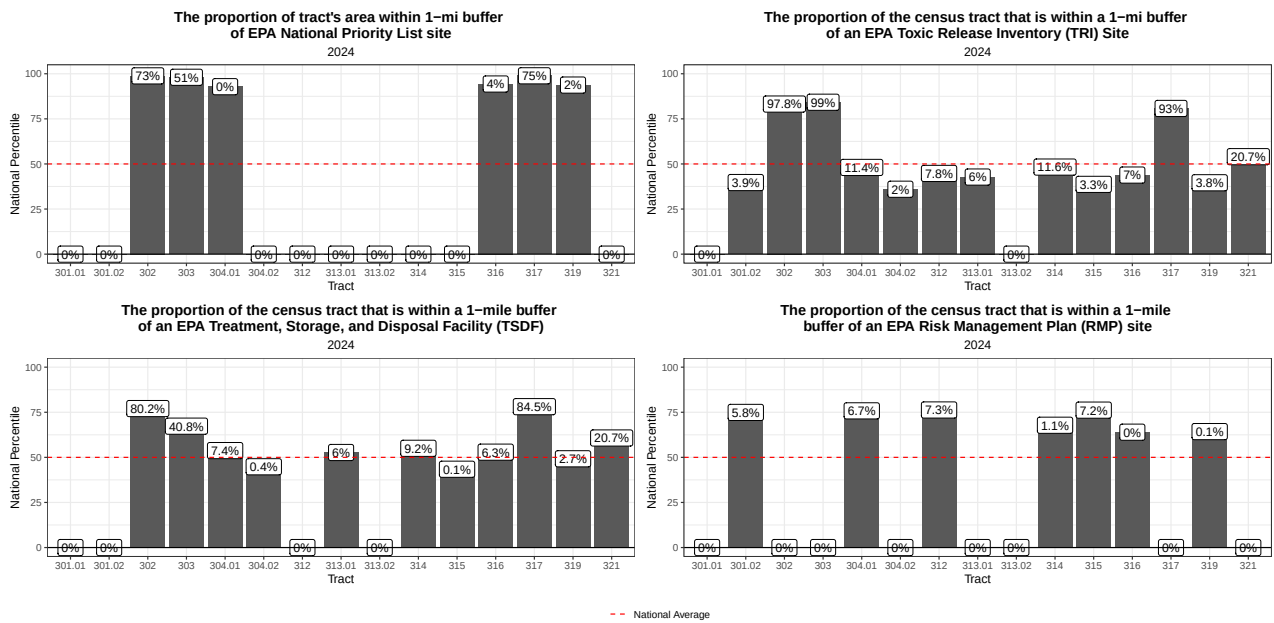


Figure 93: Waste Pollution Metrics in Ionina County Census Tracts

7.2.3 Water Pollution

All Ionina County Census tracts are above the national average for the percentage of the census tract that intersects an impaired or impacted watershed at the HUC-12 level (the smallest level of the U.S. watershed hierarchy, 40 mi² on average), usually due to pollution levels (EPA).

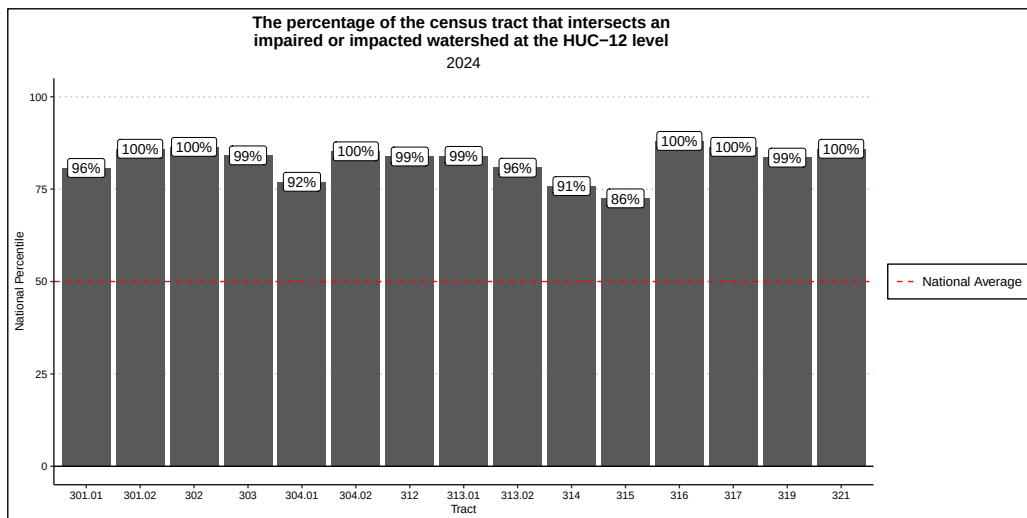


Figure 94: Impacted Watershed Metrics in Ionina County Census Tracts

7.2.4 Potential Exposure to Lead

Three Ionina County census tracts are well above the national average for percentage of houses built before 1980, increasing potential exposure to lead: 302 (East Belding), 312 (Odessa and Sebewa Townships), and 317 (East Ionina City).

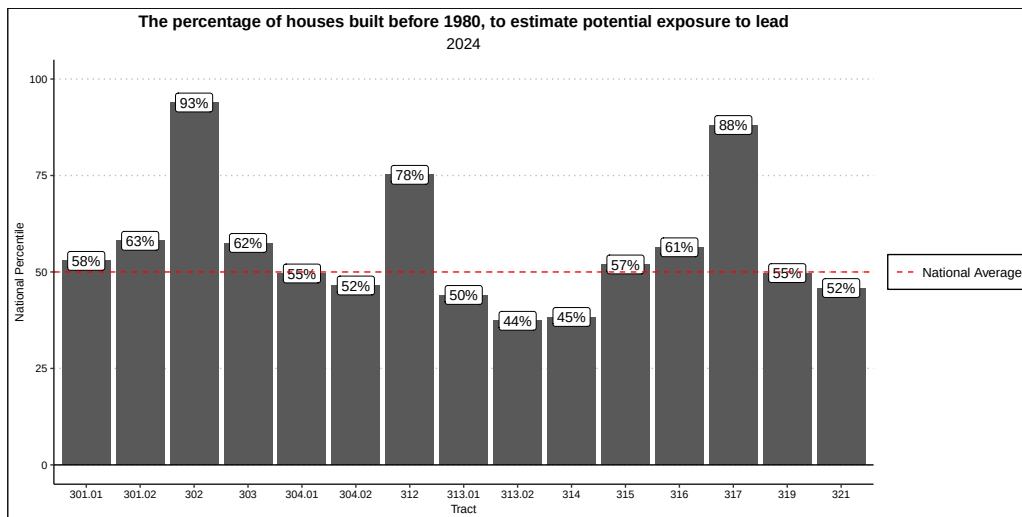


Figure 95: Older Houses and Potential Lead Exposure in Ionia County Census Tracts

7.2.5 Environmental Burden

Most of the Census Tracts in Ionia County are above the national average for the Environmental Burden Metric. The Environmental Burden Metric combines metrics surrounding “air pollution, potentially hazardous and toxic sites, the built environment, pollution related to transportation, and water pollution.” (EPA).

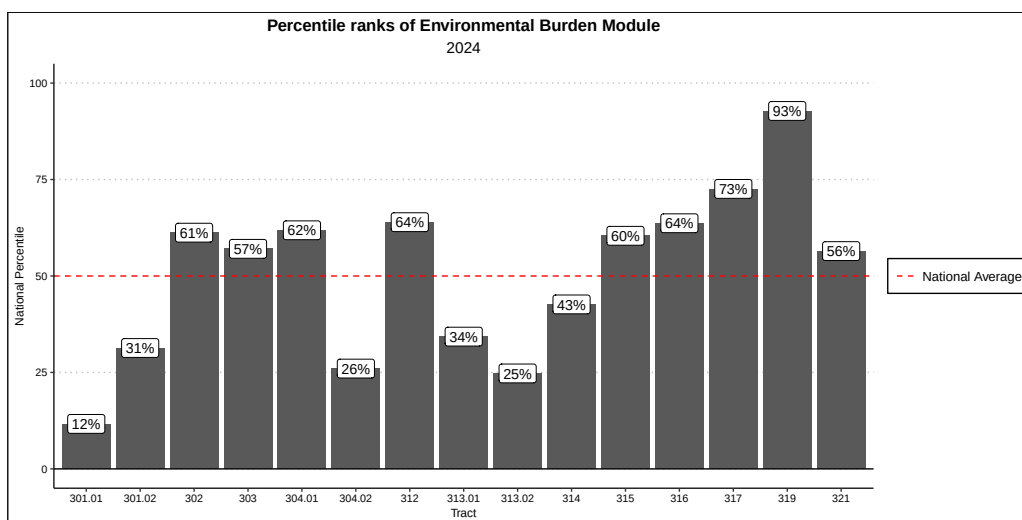


Figure 96: Environmental Burden Metrics in Ionia County Census Tracts

7.3 Climate

All Ionia County census tracts are above the national average for average annualized frequency of smoky days from wildfire smoke, most around 15 days annually.

All Ionia County census tracts are above the national average for strong wind events, all between 3 and 5 annually.

Five Ionia County census tracts are above the national average for the Climate Burden metric: 301.02 (Lyons and southern North Plains Township), 304.01 (North Otisco Township), 313.02 (South Boston and Campbell Townships), 314 (Portland Township), and 315 (Danby Township). The Climate Burden Metric “contains environmental indicators that are or have been influenced by climate change, such as extreme heat and wildfires” (EPA). It should be noted that lower values for other census tracts could be due to unavailability of data for other metrics relating to climate burden.

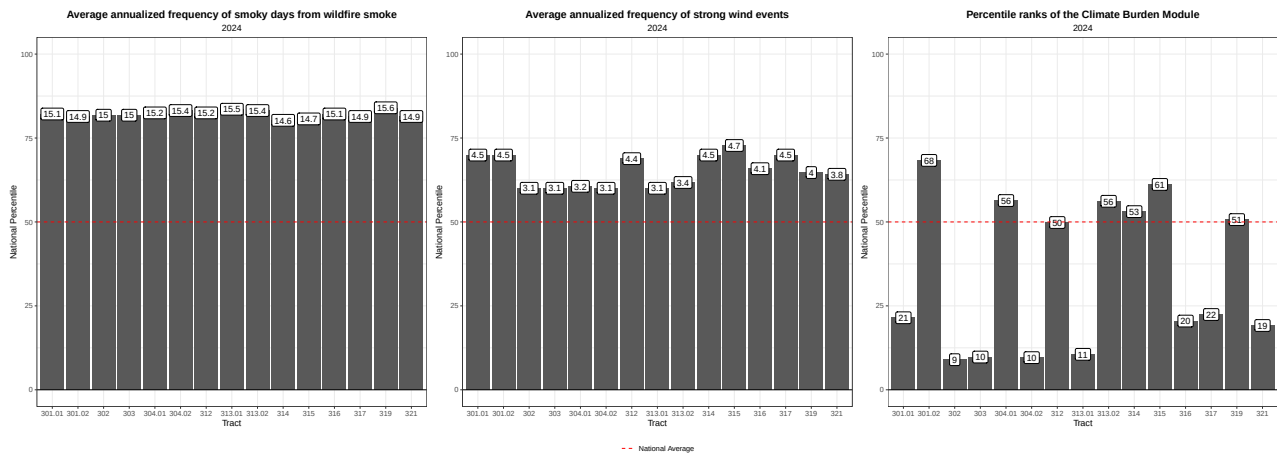


Figure 97: Wind, Wildfires, and Climate Burden Metrics in Ionia County Census Tracts

8 HRSA Unmet Needs Report

The Health Resources and Services Administration (HRSA) compiles many different metrics to assess unmet needs in preventative care across all United States ZIP codes. Below is a look at some of the metrics that make up the score and the overall unmet needs score of each ZIP code in Ionia County.

8.1 Limited Healthy Food

Limited healthy food is defined as the "Proportion of population below 200% of the [federal poverty level] who do not live close to a grocery store." For a rural county such as Ionia, that means within 10 miles.

Five ZIP codes are above the national average for percent of the population with limited healthy food available: Ionia (27%), Portland (15%), Saranac (24%), Lyons (15%), and Muir (27%)

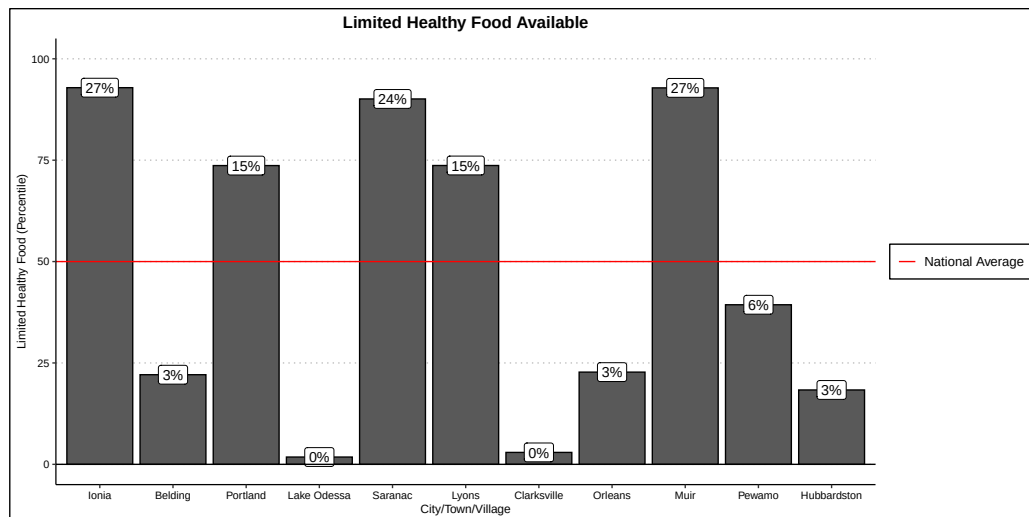


Figure 98: Limited Healthy Food % by Unique ZIP code

8.2 Housing Stress

Housing stress is defined as the "Proportion of households with one or more housing unit problems" And, "Housing unit problems include the following:

1. Housing Expense: monthly housing costs, including utilities, exceed 30% of income
2. Crowding: more household members than rooms
3. Incomplete Plumbing: home lacks necessary bathroom facilities
4. Incomplete Kitchen: home lacks essential kitchen facilities."

Two ZIP codes are above the national average for percent of the population facing housing stress: Ionia (26%) & Palo (25%)

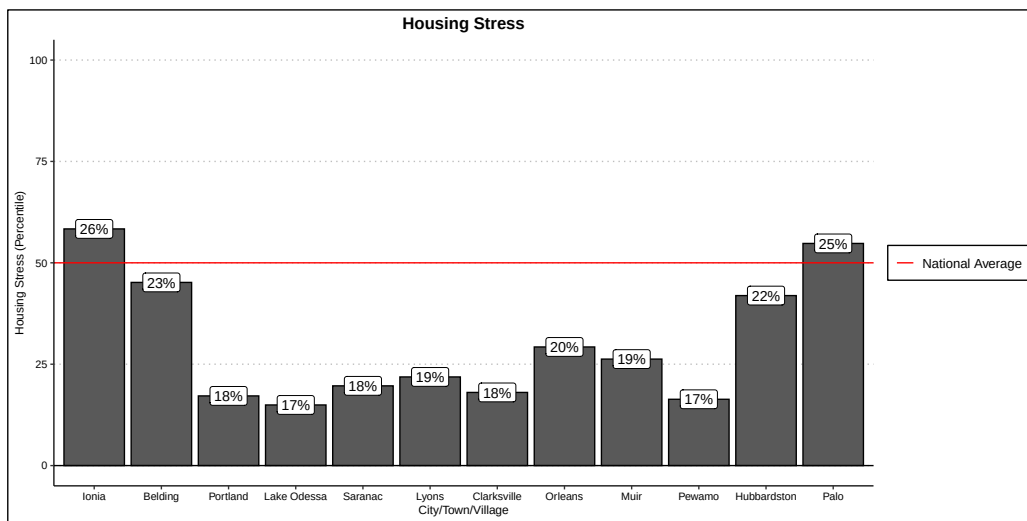


Figure 99: Housing Stress % by Unique ZIP code

8.3 No Vehicle Access

No vehicle access is defined as the "Proportion of households with no vehicles available for personal use."

Only two ZIP codes are below the national average for percent of population with no vehicle access: Lyons (2%) & Clarksville (1%). The ZIP Code with the highest percent of the population with no vehicle access is Hubbardston (11%).

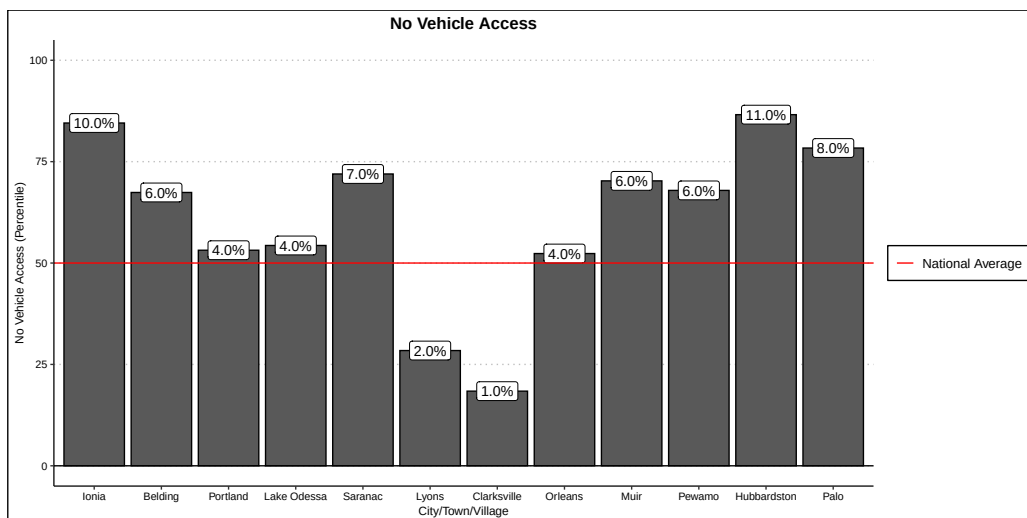


Figure 100: No Vehicle Access % by Unique ZIP code

8.4 Poor Physical Health

Poor physical health is defined as the "Percentage of adults 18 years and older who report that their physical health was not good for 14 or more days during the past 30 days."

Four ZIP codes are above the national average for percent of the population with reported poor physical health: Belding (15.1%), Orleans (15.9%), Hubbardston (16.5%), and Palo (17.3%). Clarksville has the lowest percentage of the population reporting poor physical health (12.7%).

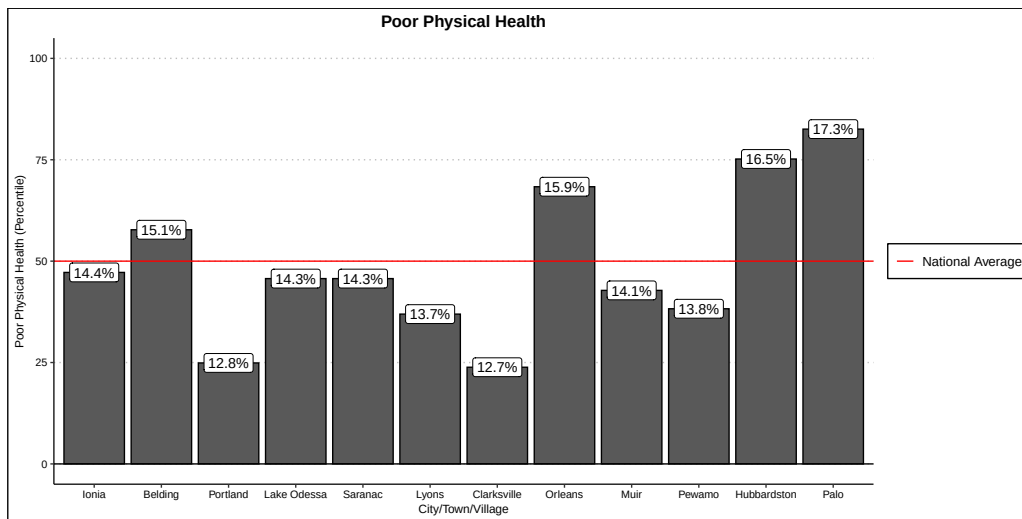


Figure 101: Poor Physical Health % by Unique ZIP code

8.5 Poor Mental Health

Poor Mental Health is defined as the "Percentage of adults 18 years and older who report that their mental health was not good for 14 or more days during the past 30 days"

Every ZIP code in Ionia County is above the national average for percent of the population reporting poor mental health. The ZIP code with the highest percentage of the population reporting poor mental health is Palo (20.3%). The lowest percentage is Clarksville (17.2%).

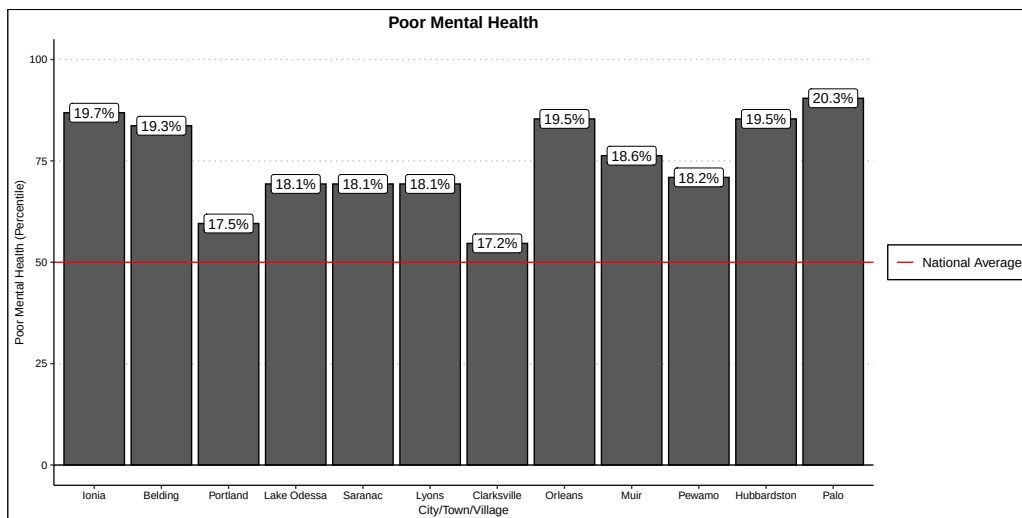


Figure 102: Poor Mental Health % by Unique ZIP code

8.6 Obesity

Obesity is defined as the "Percentage of adults 18 years and older with body mass index $\geq 30 \text{ kg/m}^2$, based upon self-reported height and weight."

Every ZIP code in Ionia County is above the national average for percentage of the population with clinical obesity. Ionia has the highest percentage of the population with clinical obesity (43%). Portland and Clarksville are tied for the lowest percentage of the population with clinical obesity (37.4%).

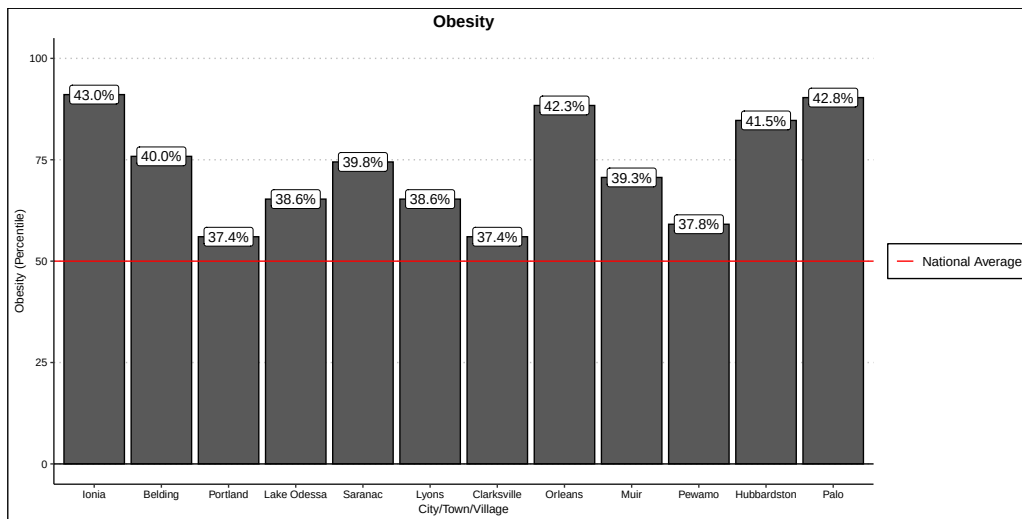


Figure 103: Obesity % by Unique ZIP code

8.7 Diabetes

Diabetes is defined as the "Percentage of adults 18 years and older who report having been diagnosed with diabetes."

Two ZIP codes are above the national average for percentage of the population with diabetes: Hubbardston (13.2%) & Palo (14.2%). Portland and Clarksville are tied for the lowest percentage of the population with diabetes (9.9%).

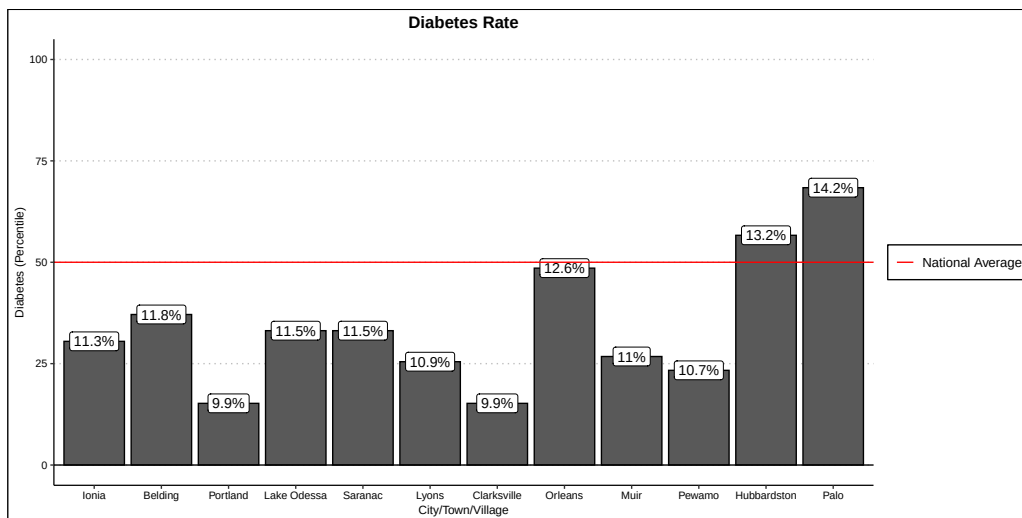


Figure 104: Diabetes % by Unique ZIP code

8.8 Smoking

Smoking is defined as the "Percentage of adults 18 years and older who are current smokers." Specifically, this refers to cigarette smoking.

Seven Zip codes are above the national average for percentage of the population who are current smokers: Ionia (20%), Belding (18%), Saranac (16%), Orleans (20%), Muir (16%), Hubbardston (20%), and Palo (20%). Three ZIP codes are below the national average for smoking and all have a reported smoking rate of 14%: Portland, Clarksville, and Pewamo.

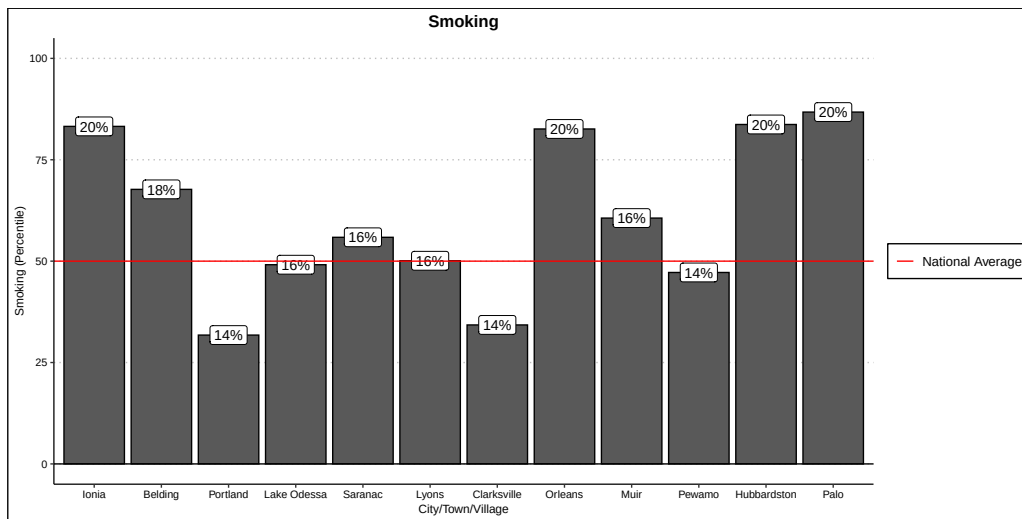


Figure 105: Smoking % by Unique ZIP code

8.9 All-Cause Mortality

All-Cause Mortality is defined as the "Rate of deaths from all causes per 100,000 population (age-adjusted)."

Three ZIP codes are above the national average for all-cause mortality rate per 100,00 people: Lake Odessa (777), Saranac (801), and Lyons (765). The lowest all-cause mortality rates are in Clarksville (693) and Pewamo (692).

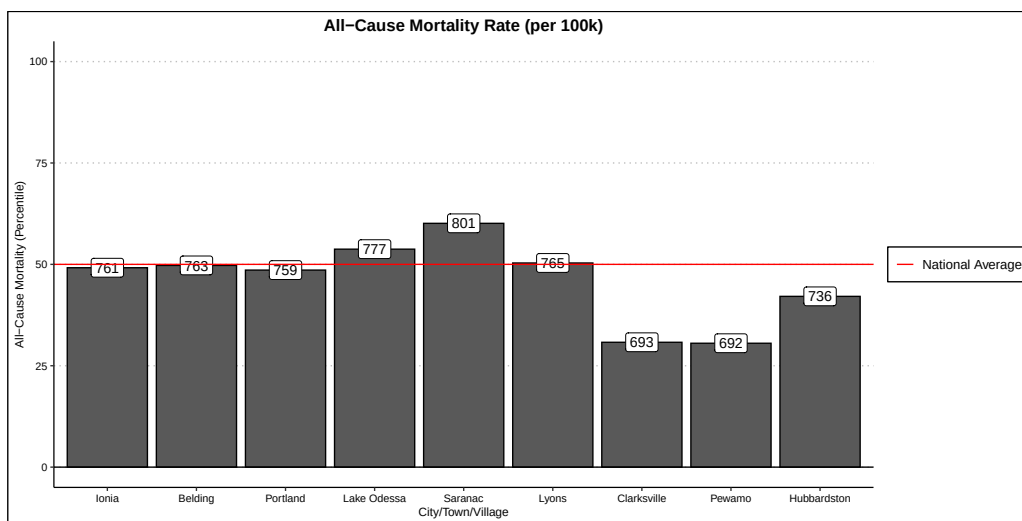


Figure 106: Mortality Rate by Unique ZIP code

8.10 Life Expectancy

Life Expectancy is defined as the "Expected number of years of life at birth."

All ZIP codes are around the 50th percentile nationally for life expectancy. The highest life expectancy is in Pewamo (78.8 years old). The lowest life expectancy is tied between Ionia, Saranac, Lyons, Clarksville, Orleans, and Muir, who all have an average life expectancy of 78.2 years old.

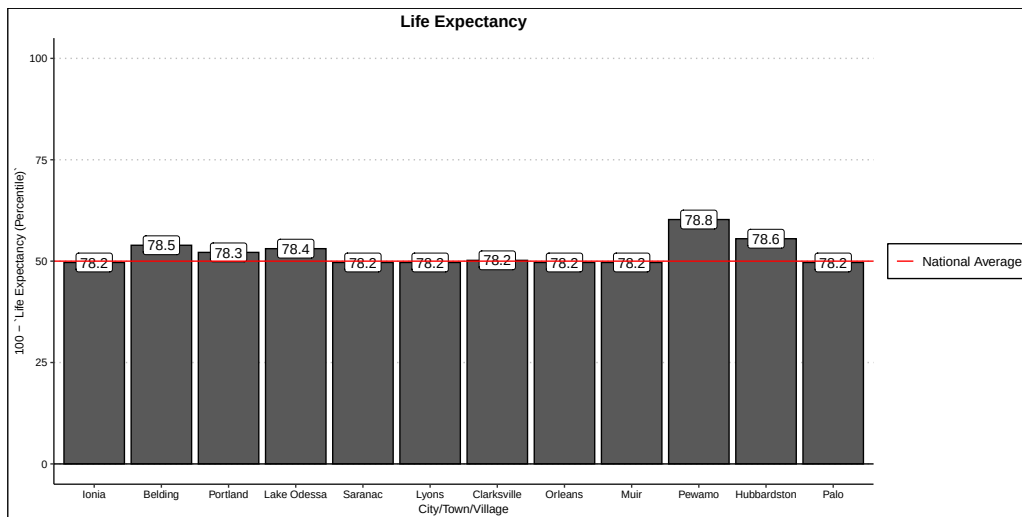


Figure 107: Life Expectancy by Unique ZIP code

8.11 Unmet Needs Score

Below is a summary of the methodology used to calculate the unmet needs score:

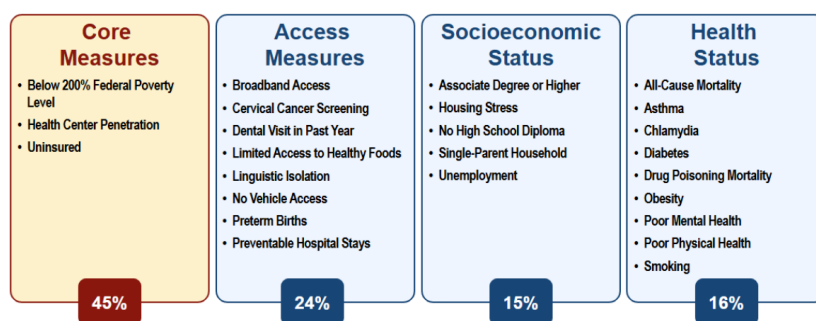


Figure 108: Unmet Needs Score Methods

All but 4 of the ZIP codes in Ionia County are above the national average for unmet needs: Portland (46.1/100), Lake Odessa (46/100), Clarksville (33.8/100), and Pewamo (48.6). The highest unmet needs scores are in Ionia (76.8/100), Hubbardston (77.3/100), and Palo (81.1/100).

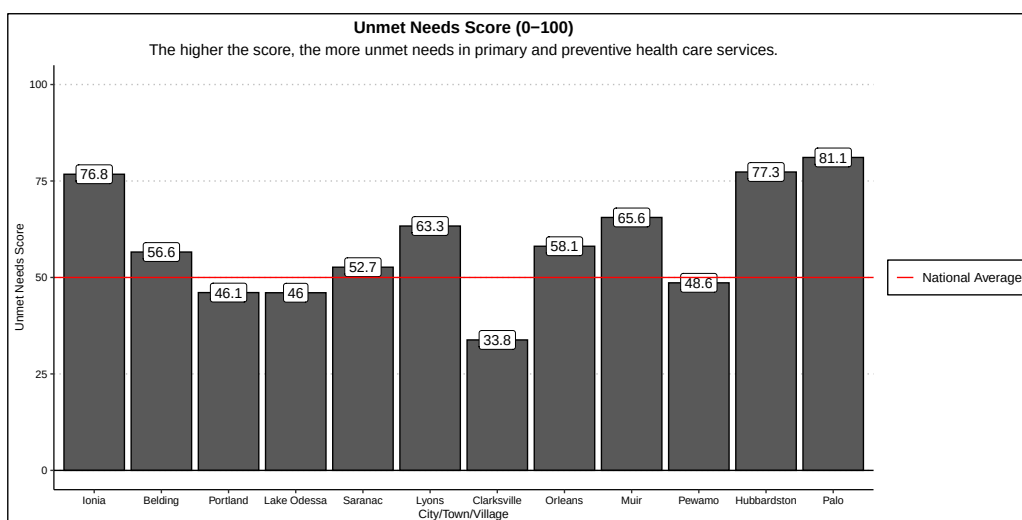


Figure 109: Unmet Needs Score by Unique ZIP code

9 High School Student Behavior Survey

The data on the surveying of high schoolers in this section comes from the Michigan Profile for Healthy Youth (MiPHY) online survey conducted bi-annually by the Michigan Department of Education. Ionia County participated in the years of 2012, 2014, and 2020. Due to the anonymity of the survey, "The MiPHY data reflect student survey responses from high schools that voluntarily participated in the county, thus may not be representative of all schools in the county." (MDHHS) Statewide averages were taken from the High School Youth Risk Behavior Survey (YRBS) conducted by the CDC. Certain questions are worded slightly differently and therefore cannot be compared directly, so statewide averages are not available for every subject.

9.1 Demographics and Information

9.1.1 2011-2012 Cycle

Shown below is the school, demographics, and sample size report from high schoolers surveyed in Ionia County from the 2011-2012 cycle:

	MiPHY District Participation		
	#	Total	%
	Participating Districts ¹	Districts in County	Participating Districts
Public	4	10	40.0%
Public Districts with Private School	3	5	60.0%
Public School Academy	0	2	0.0%
Total²	4	12	33.3%

¹ A district is counted as participating if at least one building takes the survey.

² Public districts may be represented in the "Public" and "Public District with Private School" categories, but are only counted once in the total.

	MiPHY Building Participation		
	#	Total	%
	Participating Buildings	Buildings in County ³	Participating Buildings
Public	3	12	25.0%
Private	0	1	0.0%
Public School Academy	0	0	0.0%
Total	3	13	23.1%

³ Buildings that include 9th and 11th grades in the county as identified in the Educational Entity Master.

	N	Missing	%		N	Missing	%		N	Missing	%
County (9 & 11th grade)	663	35	100.0%	Gender				Race / Ethnicity			
				Males	321	19	48.4%	American Indian	8	0	1.2%
Grade				Females	331	15	49.9%	Arab/Chaldean	4	0	0.6%
9th Grade	364	7	54.9%					Asian	1	2	0.2%
11th Grade	299	1	45.1%	Academics				Black	3	0	0.5%
				As/Bs	420	17	63.3%	Hispanic/Latino	43	3	6.5%
				Ds/Fs	58	5	8.7%	White	586	27	88.4%

Figure 110: Information on Schools and Students in 2012 Ionia County MiPHY Survey

9.1.2 2013-2014 Cycle

Shown below is the school, demographics, and sample size report from high schoolers surveyed in Ionia County from the 2013-2014 cycle:

Michigan Profile for Healthy Youth District Participation			
	#	Total	%
	Participating Districts ¹	Districts in County	Participating Districts
Public	3	11	27.3%
Public Districts with Private School	1	1	100.0%
Public School Academy	0	2	0.0%
Total²	3	13	23.1%

¹ A district is counted as participating if at least one building takes the survey.

² Public districts may be represented in the "Public" and "Public District with Private School" categories, but are only counted once in the total.

Michigan Profile for Healthy Youth Building Participation			
	#	Total	%
	Participating Buildings	Buildings in County ³	Participating Buildings
Public	2	12	16.7%
Private	0	1	0.0%
Public School Academy	0	0	0.0%
Total	2	12	16.7%

³ Buildings that include 9th and 11th grades in the county as identified in the Educational Entity Master.

	N	Missing	%		N	Missing	%		N	Missing	%
County (9 & 11th grade)	263	4	100.0%	Gender				Race / Ethnicity			
				Males	145	2	55.1%	American Indian	5	1	1.9%
Grade				Females	117	1	44.5%	Arab/Chaldean	1	1	0.4%
9th Grade	124	1	47.1%	Academics				Asian	0	0	0.0%
11th Grade	139	0	52.9%	As/Bs	169	3	64.3%	Black	4	0	1.5%
				Ds/Fs	24	0	9.1%	Hispanic/Latino	22	0	8.4%
								White	226	2	85.9%

Figure 111: Information on Schools and Students in 2014 Ionia County MiPHY Survey

9.1.3 2019-2020 Cycle

Michigan Profile for Healthy Youth District Participation			
	#	Total	%
	Participating Districts ¹	Districts in County	Participating Districts
Public	2	11	18.2%
Public Districts with Private School	1	1	100.0%
Public School Academy	0	2	0.0%
Total²	2	13	15.4%

¹ A district is counted as participating if at least one building takes the survey.

² Public districts may be represented in the "Public" and "Public District with Private School" categories, but are only counted once in the total.

Michigan Profile for Healthy Youth Building Participation			
	#	Total	%
	Participating Buildings	Buildings in County ³	Participating Buildings
Public	2	12	16.7%
Private	0	1	0.0%
Public School Academy	0	0	0.0%
Total	2	12	16.7%

³ Buildings that include 9th and 11th grades in the county as identified in the Educational Entity Master.

	N	Missing	%		N	Missing	%		N	Missing	%
County (9 & 11th grade)	335	13	100.0%	Gender				Race / Ethnicity			
				Males	165	6	49.3%	American Indian	4	0	1.2%
Grade				Females	169	7	50.4%	Arab/Chaldean	2	0	0.6%
9th Grade	168	1	50.1%	Academics				Asian	1	0	0.3%
11th Grade	167	0	49.9%	As/Bs	240	7	71.6%	Black	3	0	0.9%
				Ds/Fs	20	4	6.0%	Hispanic/Latino	27	7	8.1%
								White	287	5	85.7%

Figure 112: Information on Schools and Students in 2020 Ionia County MiPHY Survey

9.2 Tobacco & Substance Use

9.2.1 Cigarettes

In the 2012 survey, cigarette use among high schoolers in Ionia County was about the same as the State of Michigan (13.9% vs. 14%). In the 2014 survey, cigarette use among high schoolers in Ionia County was higher than the state average (16.8% vs. 11.8%). In 2020, survey response for cigarette use is about the same for Ionia County and the State of Michigan (4.1% vs. 4.5%).

Compared to other counties, Ionia County has the 17-18th (tied) highest rate of cigarette use in high schoolers in 46 counties with available data.

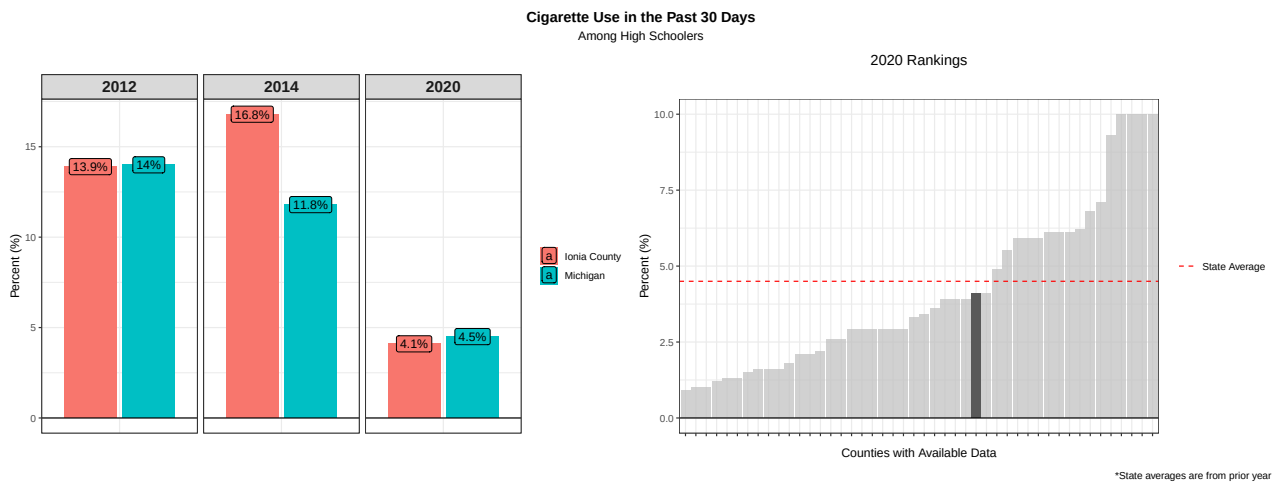


Figure 113: Cigarette Use Among High schoolers in Ionia County

9.2.2 E-Cigarettes (Vapes)

In the 2020 survey, e-cigarette (vape) use among high schoolers in Ionia County was about the same as the State of Michigan (21% vs. 20.8%).

Compared to other counties, Ionia County has the 19th highest rate of e-cigarette (vape) use in high schoolers in 46 counties with available data.

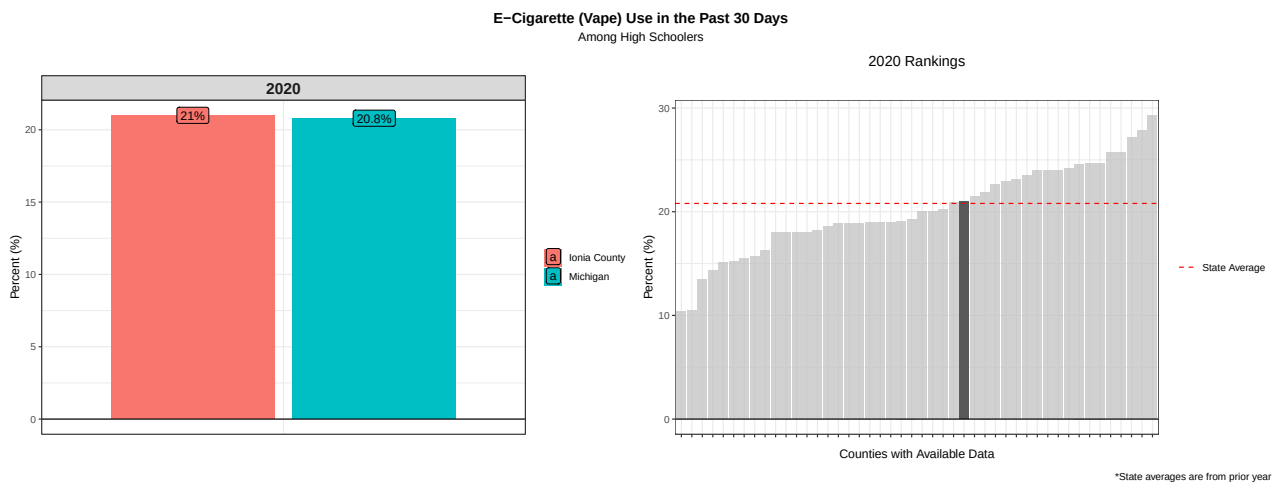


Figure 114: E-Cigarette (Vape) Use Among Highschoolers in Ionia County

9.2.3 Alcohol

In the 2012 survey, alcohol use among high schoolers in Ionia County was higher than the State of Michigan (25.2% vs. 35.5%). In the 2012 survey, the difference between Ionia County and the State of Michigan was even higher (35.5% vs. 14.0%). In the 2020 survey, rates of alcohol use among high schoolers in Ionia County was only slightly higher than the State of Michigan (15.2% vs. 13.6%).

Compared to other counties, Ionia County has the 28-29th (tied) highest rate of alcohol use in high schoolers in 46 counties with available data.

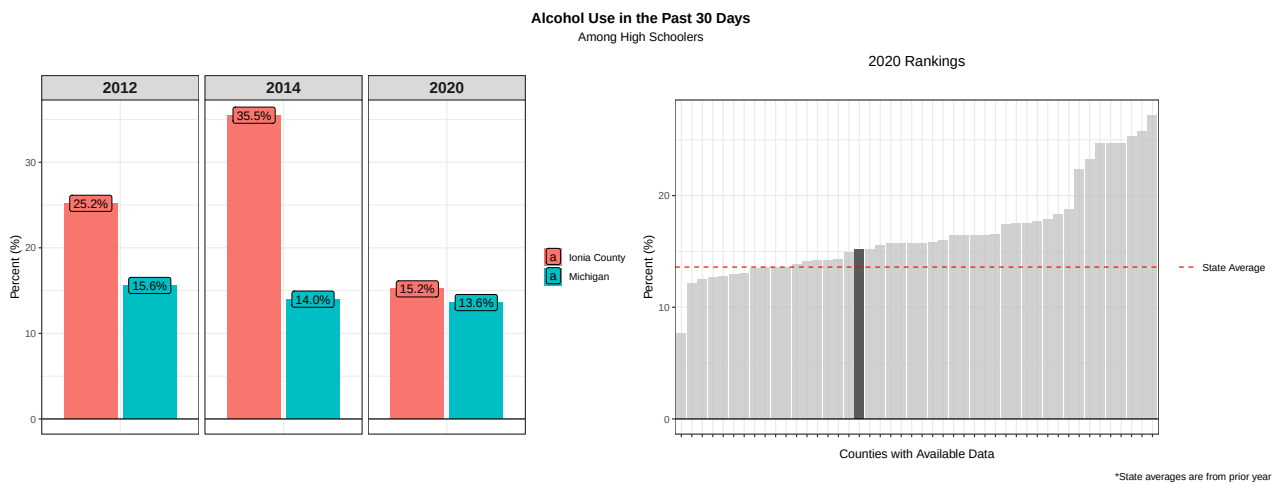


Figure 115: Alcohol Use Among High schoolers in Ionia County

9.2.4 Marijuana

In the 2012 survey, marijuana use among high schoolers in Ionia County was lower than the State of Michigan (15.0% vs. 18.6%). In the 2014 survey, marijuana use among high schoolers in Ionia County was greater than the State of Michigan (24.1% vs. 18.2%). Finally, in 2020, marijuana use among high schoolers in Ionia County was lower than the State of Michigan (15.5% vs. 21.6%).

Compared to other counties, Ionia County has the 18th highest rate of marijuana use in high schoolers in 46 counties with available data.

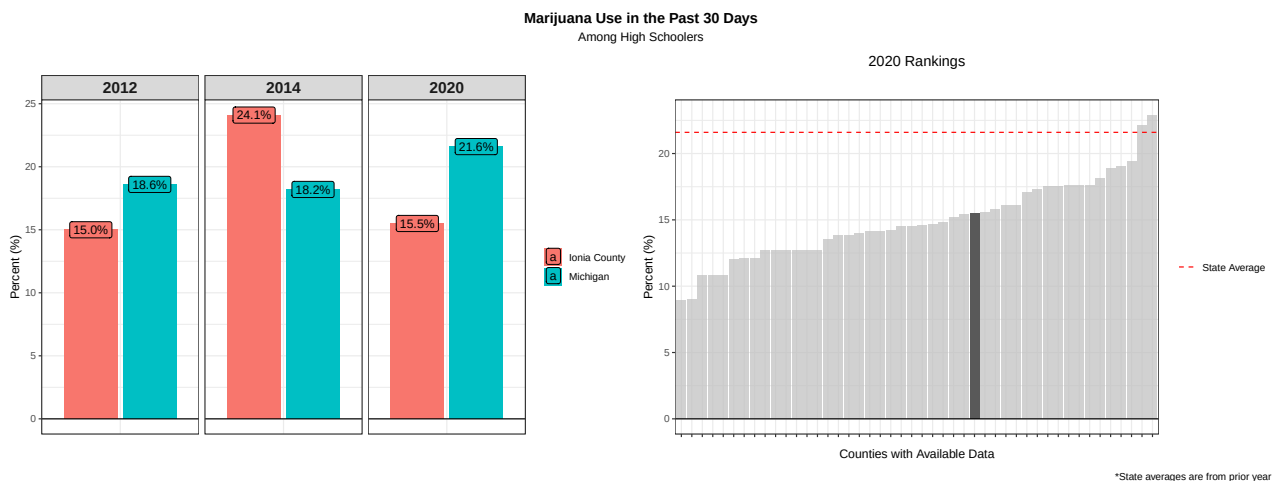


Figure 116: Marijuana Use Among High schoolers in Ionia County

9.2.5 Cocaine

In the 2012 survey, cocaine use among high schoolers in Ionia was 0.9%. This rate rose slightly in the 2014 survey to 1.2%. Lastly, in the 2020 survey, none of the surveyed high schoolers in Ionia County reported cocaine use.

Ionia County is among several counties with zero reported cocaine use in 2020.

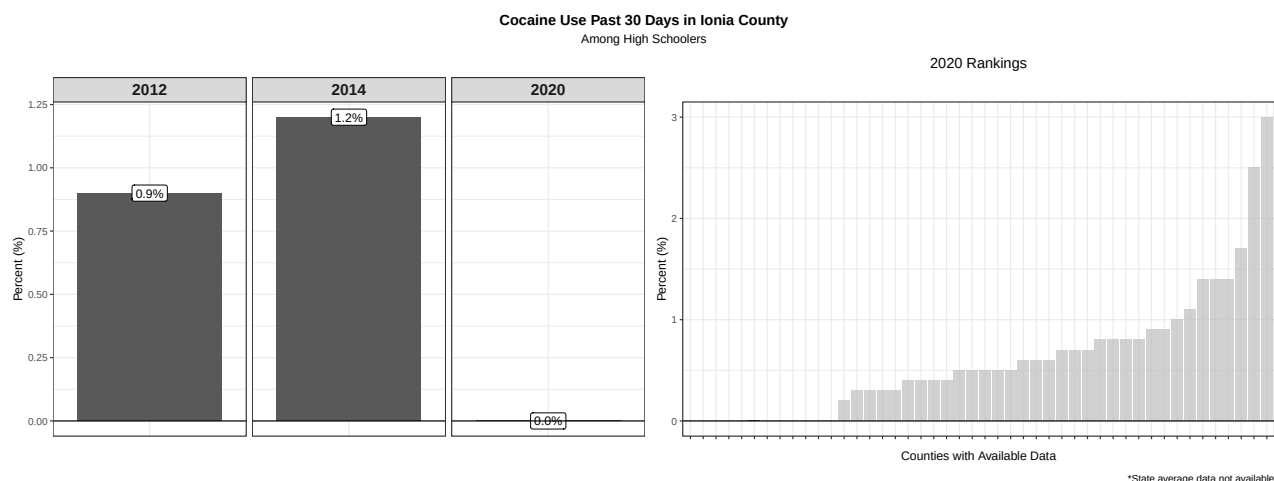


Figure 117: Cocaine Use Among High schoolers in Ionia County

9.2.6 Heroin

In the 2012 survey, heroin use among high schoolers in Ionia County was 0.2%. This rate rose to 0.4% in the 2014 survey. Lastly, in the 2020 survey, heroin use among high schoolers in Ionia County decreased to 0.3%.

Ionia is 13th-22nd (tied) out of 46 counties with available data in heroin use among high schoolers in 2020.

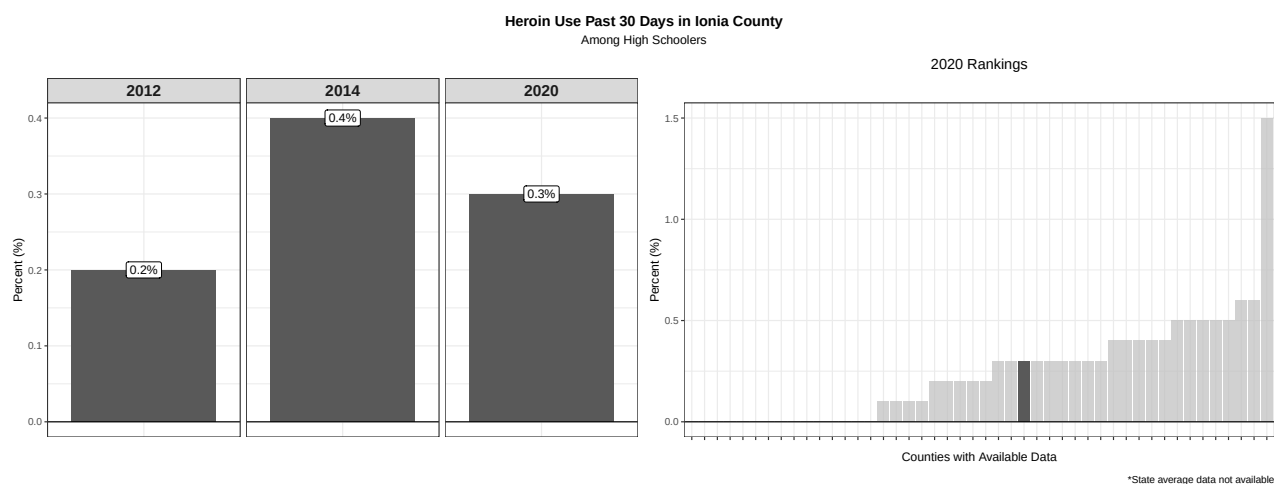


Figure 118: Heroin Use Among High schoolers in Ionia County

9.2.7 Methamphetamine

In the 2012 survey, methamphetamine use among high schoolers in Ionia county was 0.3%. This rate rose to 0.8% in the 2014 survey. Lastly, in the 2020 survey, methamphetamine use among high schoolers in Ionia County decreased to 0.3%.

Ionia County has the 19-26th (tied) highest out of 46 available counties with available data for surveyed methamphetamine use among high schoolers.

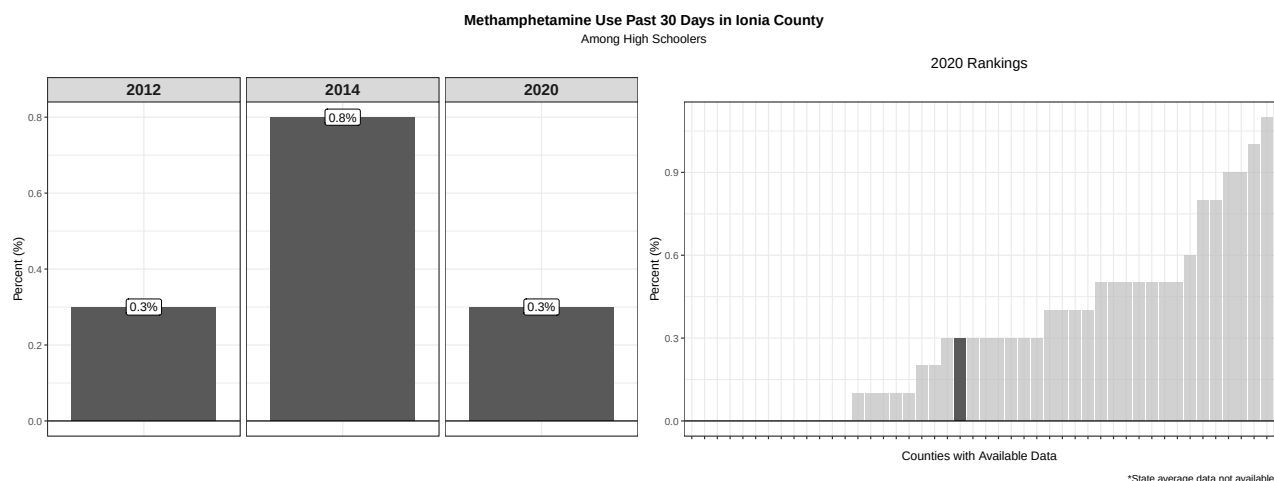


Figure 119: Methamphetamine Use Among High schoolers in Ionia County

9.2.8 Club Drugs

The MiPHY survey defines club drugs as being among the following: "ecstasy (E, adam, XTX, Roll), GHB (G, liquid ecstasy, grievous bodily harm), ketamine (K, special K), rohypnol (roofies), nitrous oxide (laughing gas, whippets), or LSD (acid, microdot, boomers)"

In the 2012 survey, club drug use among high schoolers in Ionia County was 1.2%. That rate rose to 2.7% in the 2014 survey. Lastly, in the 2020 survey, club drug use among high schoolers in Ionia County decreased to 0.9%.

Ionia County has the 32nd-35th highest surveyed club drug use among high schoolers in 46 counties with available data.

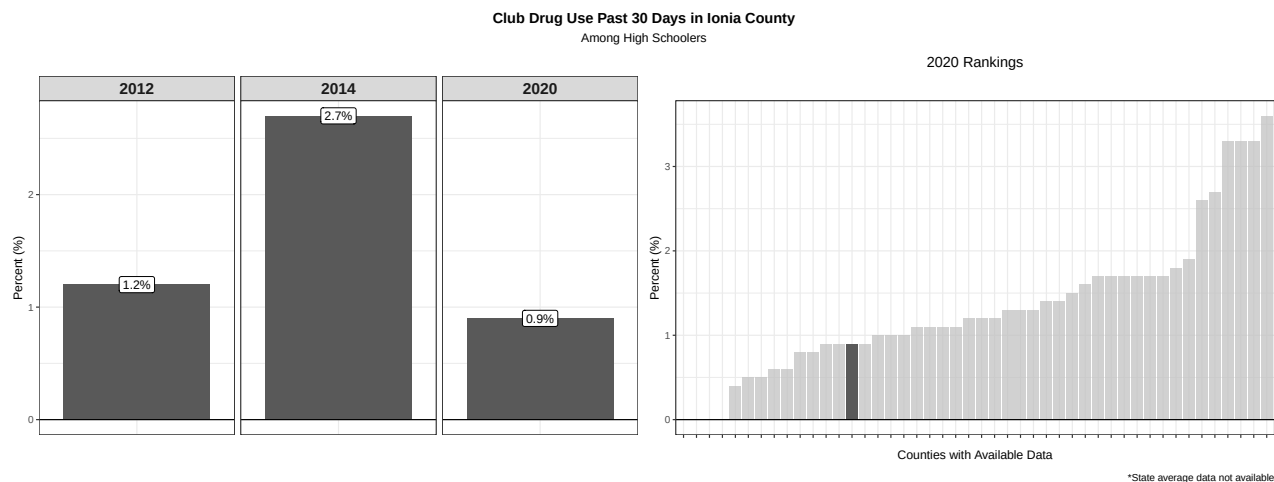


Figure 120: Club Drug Use Among High schoolers in Ionia County

9.3 Mental Health, Safety, and Bullying

9.3.1 Depression

In the 2012 survey, the percentage of high schoolers in Ionia County surveyed who experienced major depression was 30.2%. This rate rose slightly in the 2014 survey to 31.4%. In the 2020 survey, the percentage of high schoolers in Ionia County surveyed who experienced major depression increased to 40%.

Ionia County has the 23rd-24th (tied) highest surveyed rate of depression among high schoolers in 45 counties with available data.

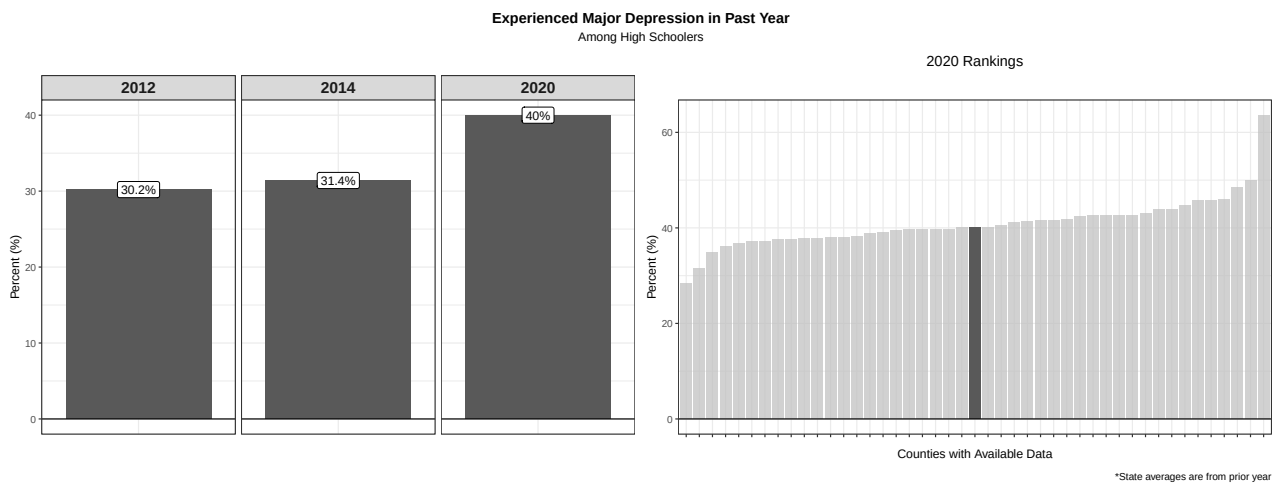


Figure 121: Depression Among High schoolers in Ionia County

9.3.2 Suicide

In the 2012 survey, the rate of attempted suicide among high schoolers in Ionia County was similar to the State of Michigan (8.4% vs. 8.1%). In the 2014 survey, the rate of attempted suicide among high schoolers in Ionia County was higher than the State of Michigan (12.9% vs. 8.9%). In the 2020 survey, the rate of attempted suicide among high schoolers in Ionia County decreased from the 2014 survey but was still higher than the State of Michigan (9.4% vs. 7.8%).

Ionia County has the 33rd highest rate of surveyed attempted suicide among high schoolers in 45 counties with available data.

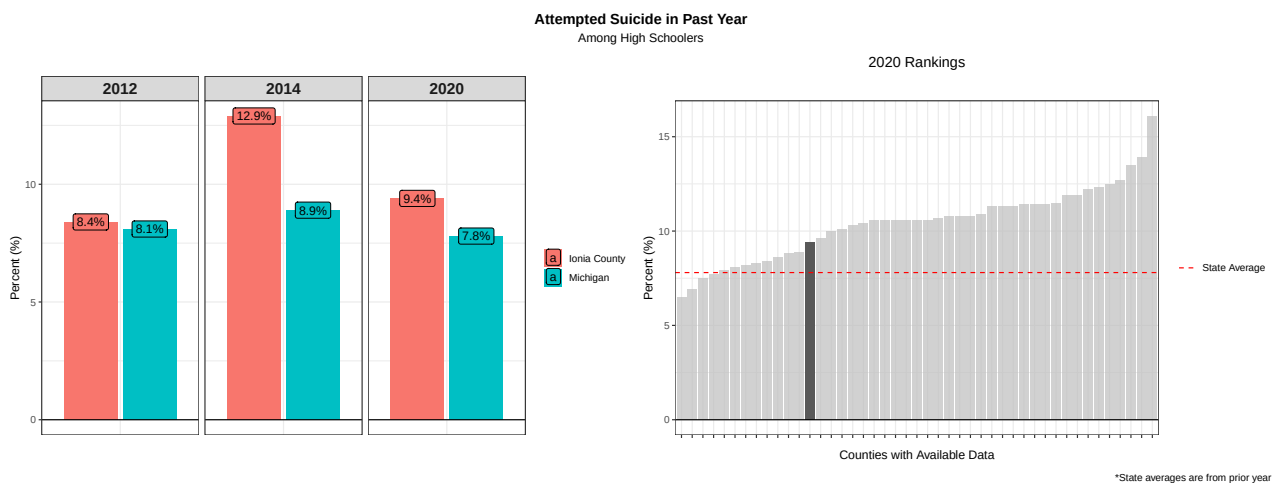


Figure 122: Attempted Suicide Among High schoolers in Ionia County

9.3.3 Feeling Unsafe at School

In the 2012 survey, the percentage of surveyed high schoolers in Ionia County who felt unsafe at school was 3.2%. This rate rose to 7.6% in 2014. In the 2020 survey, the percentage of surveyed high schoolers in Ionia County who felt unsafe at school increased to 10.1%.

Ionia County has the 5th highest rate of feeling unsafe at school in high schoolers in 45 counties with available data.

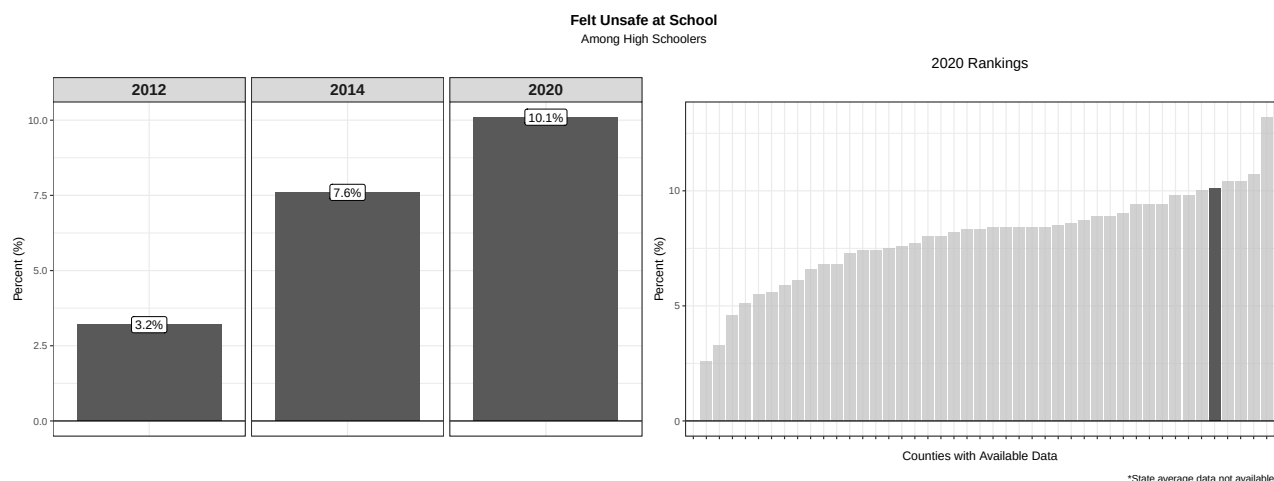


Figure 123: Feeling Unsafe at School Among High schoolers in Ionia County

9.3.4 Bullying

In the 2012 survey, the percentage of surveyed high schoolers in Ionia County who witnessed bullying was 76.4%. This rate decreased to 45.8% in the 2014 survey. In the 2020 survey, the percentage of surveyed high schoolers who witnessed bullying increased to 63.2%.

Ionia County has the 5th highest rate of witnessing bullying in high schoolers in 45 counties with available data.

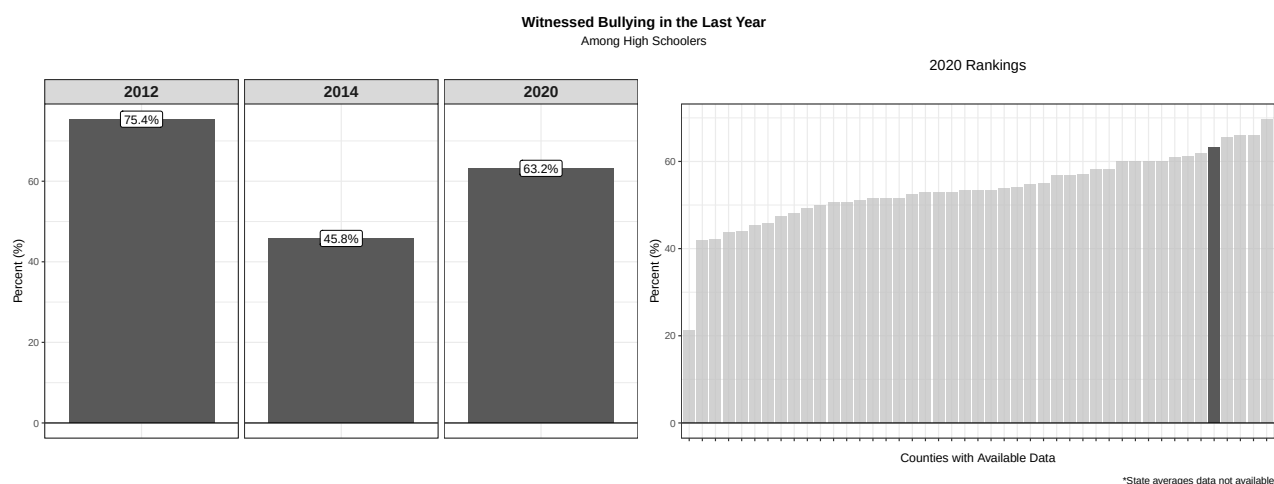


Figure 124: Witnessed Bullying Among High schoolers in Ionia County

9.3.5 Cyber Bullying

In the 2012 survey, the percentage of surveyed high schoolers in Ionia County who witnessed cyber bullying was 38.1%. This rate decreased in the 2014 survey to 30.1%. In the 2020 survey, the percentage of surveyed high schoolers in Ionia County who witnessed cyber bullying decreased to 24.7%.

Ionia County has the 28th highest rate of witnessing cyber bullying in 45 counties with available data.

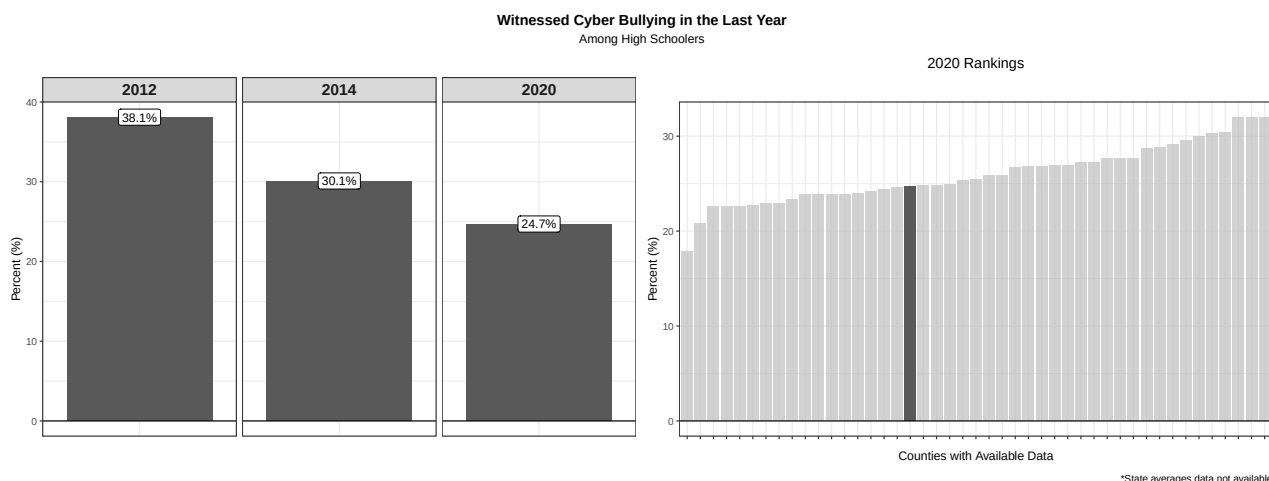


Figure 125: Witnessed Cyber bullying Among High schoolers in Ionia County

9.4 School Funding Inadequacy

Apart of the County Health Rankings data is a measure of School Funding Adequacy, defined as “actual per-pupil spending compared with an estimated amount that would need to be spent to achieve U.S. average test scores in each school district.”

School Funding Adequacy in Ionia County has decreased from 2020 (\$275) to 2022 (\$-1,940). This is in contrast to increasing trends both statewide and nationally from 2021 to 2022.

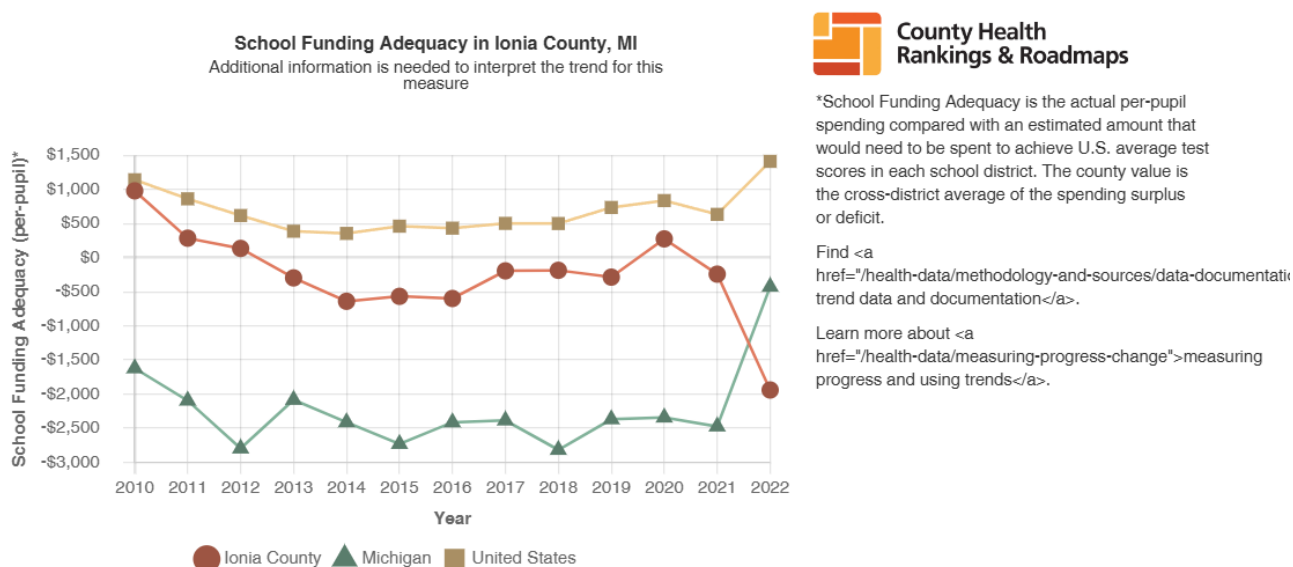


Figure 126: School Funding Adequacy in Ionia County Over Time

10 Survey of EightCap Program Recipients

EightCAP is a non-profit organization that serves the Michigan counties of Gratiot, Isabella, Montcalm, and Ionia. According to the organization, “EightCAP, Inc. improves our community by collaborating with private, government, and community organizations to deliver programs that alleviate the causes of poverty and its effects allowing individuals to become self-sufficient.”

A 2026 survey was conducted of recipients of the EightCAP programs, and the following shows results from those who live in Ionia county.

10.1 Participants' Backgrounds

The survey respondents are predominantly lower income households:

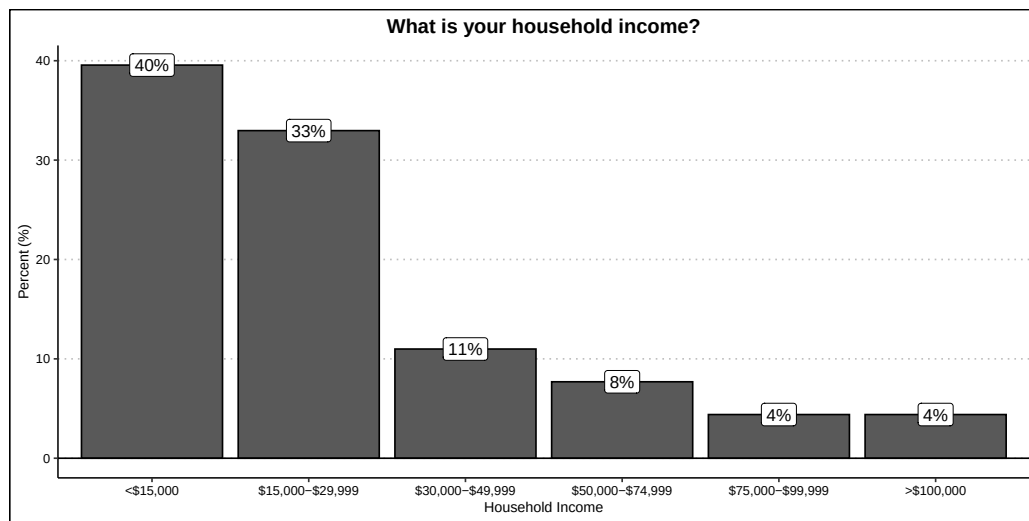


Figure 127: Household Incomes of Survey Respondents

Most respondents could not afford an unexpected \$500 expense:

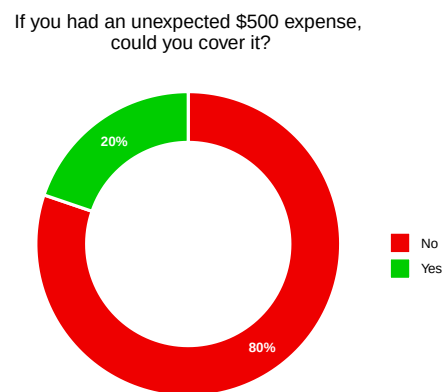


Figure 128: Ability of Survey Respondents to Pay Unexpected Expense

Many different employment types are common:

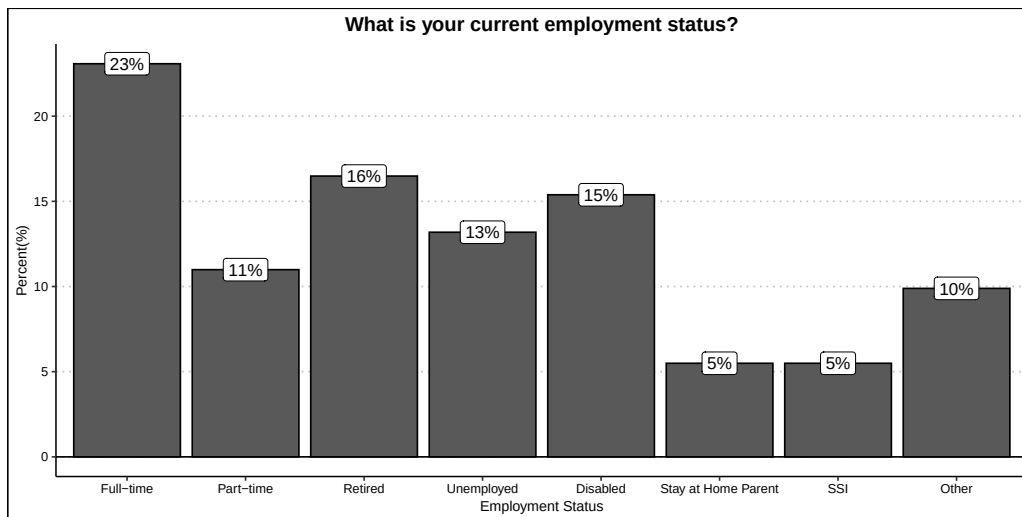


Figure 129: Employment of Survey Respondents

10.2 Challenges Faced by Respondents

The most frequently answered challenges faced by survey respondent households include food costs, housing costs, utilities, etc.:

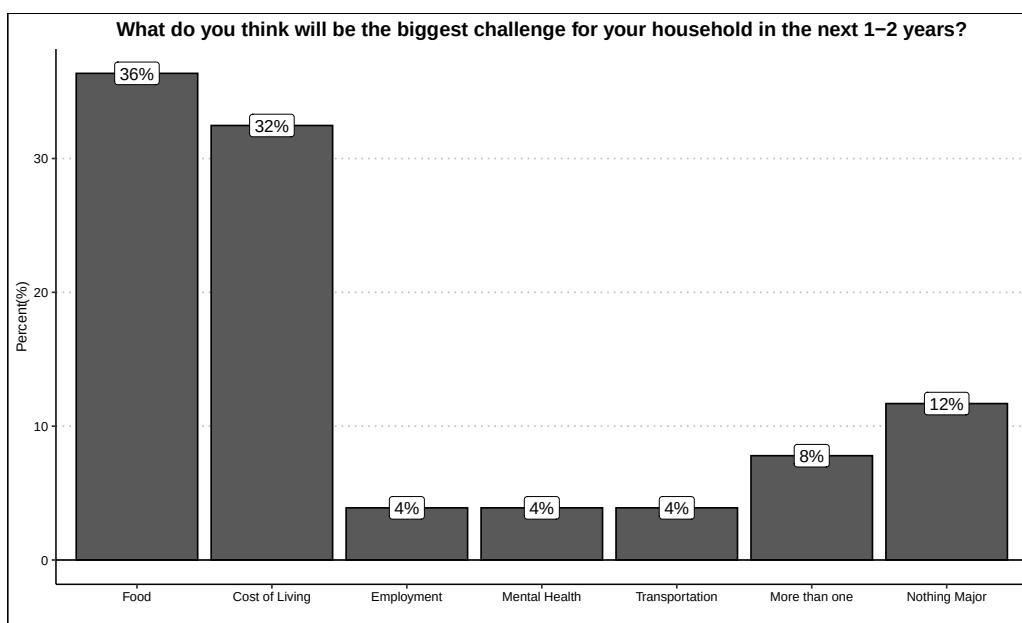


Figure 130: 5 Biggest Household Challenges of Survey Respondents

Household challenges are more varied for lower income households. Also, food costs stand out for almost every group:

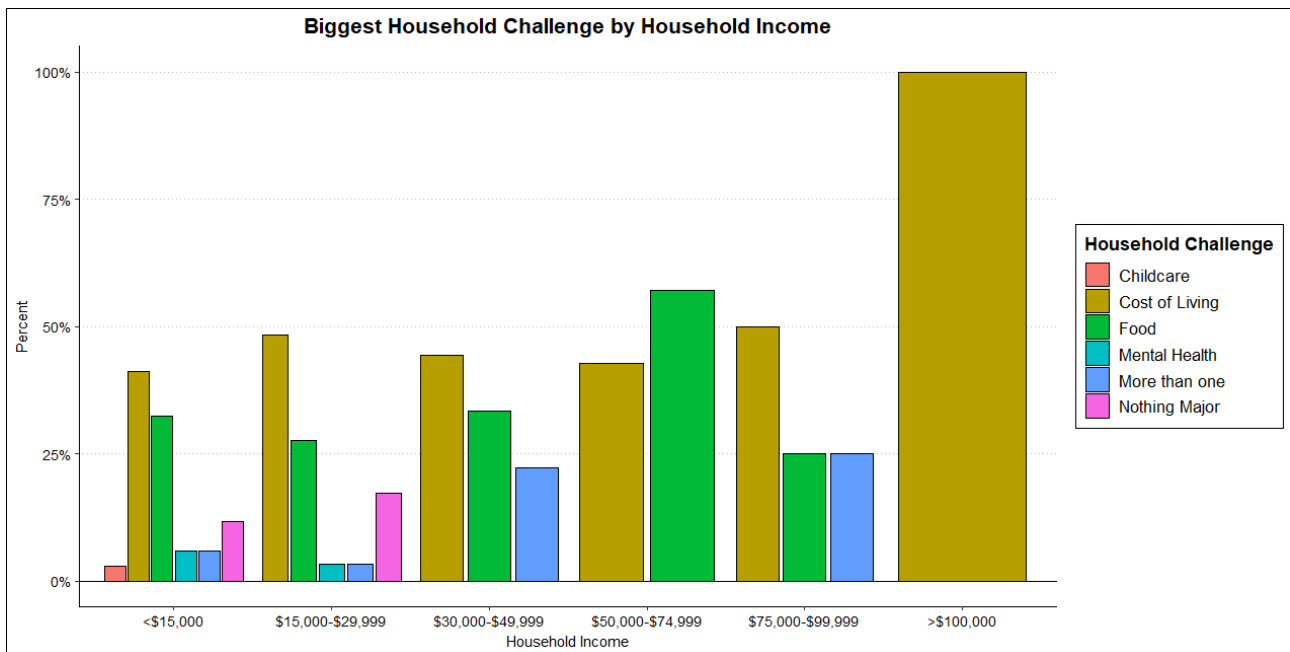


Figure 131: Biggest Household Challenges of Survey Respondents by Household Income

Household challenges are similar across age groups, with food costs still standing out:

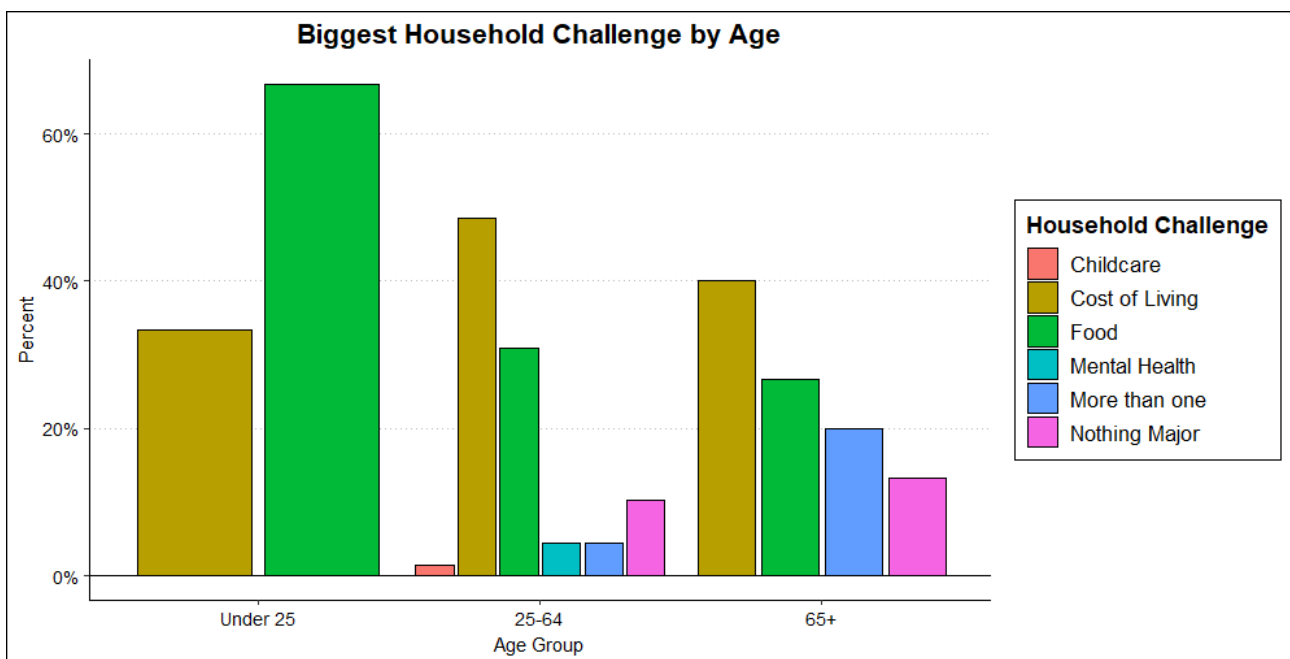


Figure 132: Biggest Household Challenges of Survey Respondents by Age Group

Household challenges are more varied for those with 'some college' being their highest level of educational attainment, while those respondents with a bachelor's or master's degree do not respond with food costs being a challenge as much:

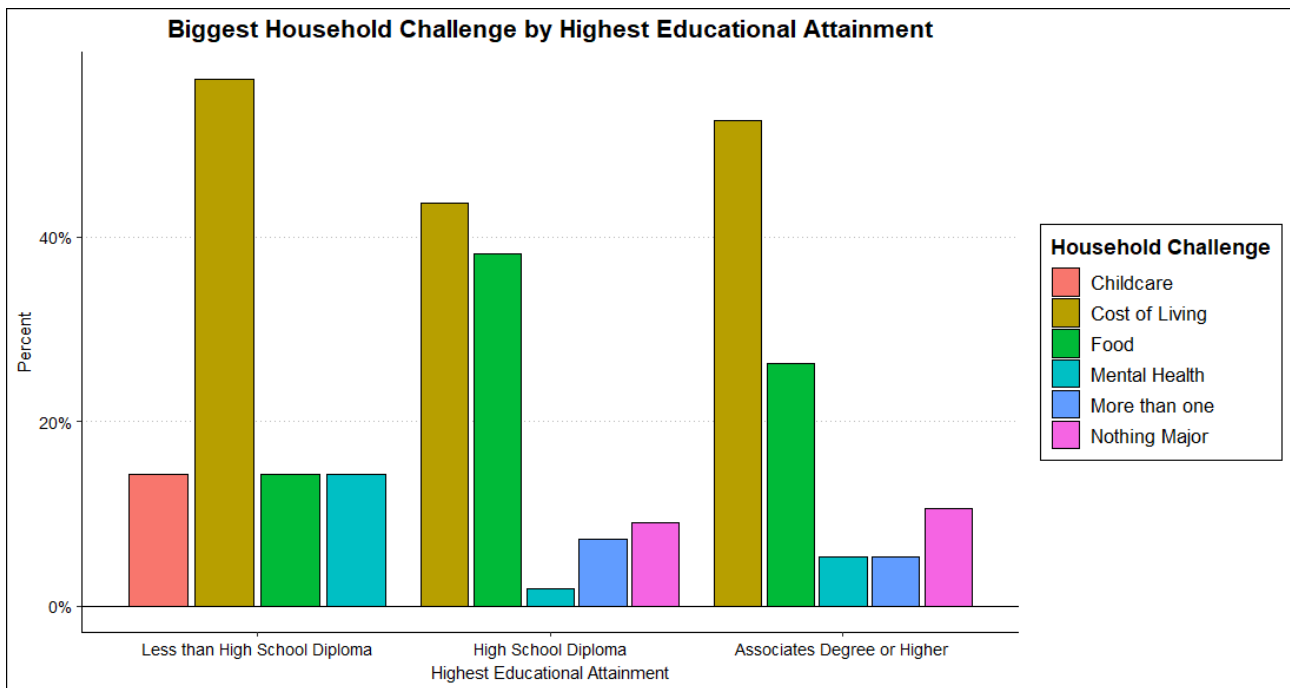


Figure 133: Biggest Household Challenges of Survey Respondents by Highest Educational Attainment

Lower income households frequently respond having difficulty having no vehicle, an unreliable vehicle, or no public transportation. Higher income households do not report as many transportation challenges:

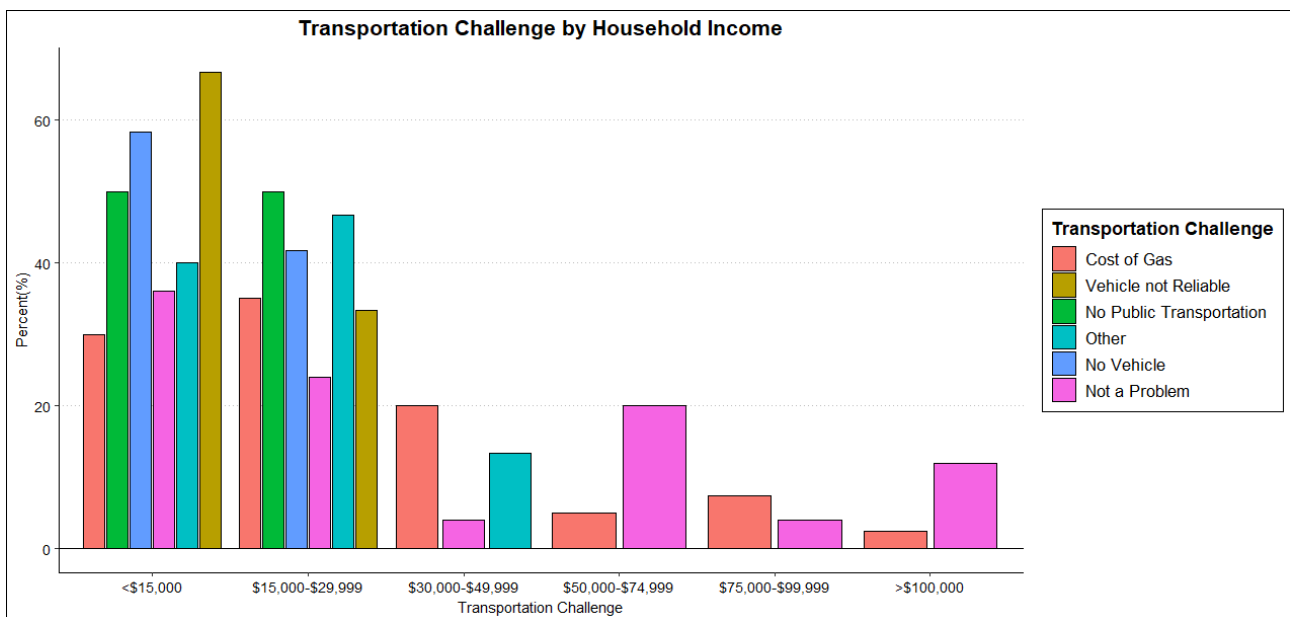


Figure 134: Biggest Household Challenges of Survey Respondents by Household Income

Survey respondents with household incomes between \$15,000 and \$50,000 are most likely to have to miss medical care due to costs.

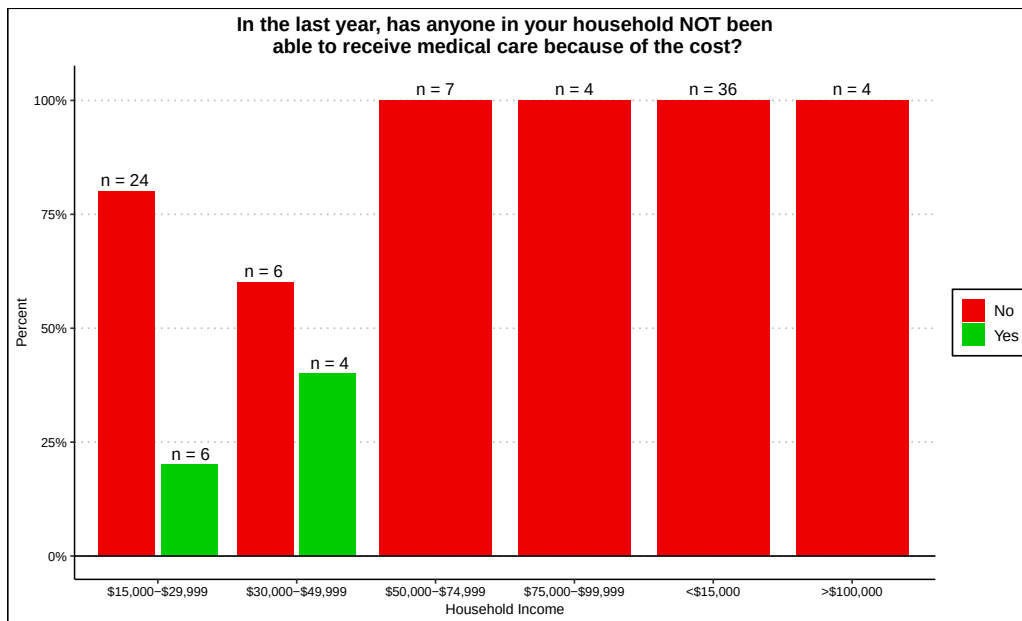


Figure 135: Survey Respondents Missing Medical Care Due to Cost

10.3 Response Themes

Along with survey questions where respondents were asked to identify themselves or their challenges in a set of given options, more open-ended questions were also asked to identify more unique or complex parts of their life experiences. Some of these questions were:

- What services were hard to find in your community that may be helpful to see in the future?
- Is there anything else you would like us to know about challenges in your community?
- Is there anything else you would like us to know about challenges in your community?

The sections below focus on several themes that appeared frequently in these open-ended questions.

10.3.1 Older Adult Care

The overall population trend in Ionia county as well as the State of Michigan is an aging population, with a higher percentage of older adults compared to children. Older adults often rely heavily on public services, and multiple survey respondents mention the lack of care for the older adult in Ionia County, including assisted living facilities and home repair services. This appears as a trend in the survey respondents as well as in patterns shown in data from the census.

10.3.2 Reliance on Personal Vehicles

While transportation is not *the* largest challenge for many, still a large amount of respondents reported difficulties with transportation as being a barrier to getting and keeping employment. Lower income household respondents frequently report issues with transportation. In the survey question titled: "What is the ONE thing that would make the biggest difference in your life right now?", over 10 respondents mentioned having trouble with transportation related to personal vehicles. Whether it's gas prices, vehicle reliability, or the lack of a personal vehicle, the lack of stability in a vehicle is providing many challenges for those in Ionia County. Several respondents also mention the lack of suitable public transportation options, which could be pushing people to use less reliable cars because they don't see another transportation option.

10.3.3 ALICE Population

Several respondents mention how a person making just over a certain amount of money might not qualify for public programs, but still struggle to cover their needed costs. The **ALICE** acronym (Asset Limited, Income Constrained, Employed) is often used to categorize these populations. The most important part of the acronym is "employed." This group of the population is actively working but is still not earning enough to have spare income to be dedicated to assets instead of immediate needs. We found for medical costs specifically, those that fall in the ALICE income range are more likely to miss medical care due to costs (Figure 135)

11 Discussion

11.1 Health Summary

11.1.1 Healthcare Access

Ionia County has fewer healthcare providers per person than the State of Michigan across general medicine, dentistry, pharmacy, and specialists.

Compared to the State of Michigan for general medicine access, Ionia County has fewer providers per person for physicians, emergency medicine, surgeons, physicians assistants, and pediatricians (Figure 40). There are more than twice as many providers per person for the State of Michigan compared to Ionia County in all general medicine categories (Figure 40). The number of primary care physicians per person has been decreasing in Ionia County since 2014 (Figure 41).

According to a 2002 article, only 9% of U.S. physicians practice in rural areas despite 20% of the U.S. population living in rural areas. Rural residents have fewer physicians available per person and are 10% less likely to visit the doctor for preventative care. There have been over 140 rural hospitals closed in the United States since 2010 according to the Rural Healthy People 2030 report. And while improvements in telehealth services can help those who struggle with transportation, rural areas tend to have fewer people with access to broadband compared to urban areas. Case in point, Ionia County has a higher percent of households with no broadband access than the State of Michigan (Figure 26).

Ionia County has fewer dentists per person compared to the State of Michigan (Figure 40). Those in urban areas in the United States are less likely to receive dental care, and poor oral health is associated with diseases like cancer, cardiovascular disease, birth complications, and dementia (Rural Healthy People 2030 report).

For specialty care, disparities between the State of Michigan and Ionia County are even wider. For most medical specialties, there are no reported providers within Ionia County, and for those that are reported, there is still a shortage compared to the State of Michigan (Figure 45). According to HRSA, Ionia County has no reported podiatrists, immunologists, dermatologists, Ob-Gyns, Ophthalmologists (eye-care), orthopedic surgeons, or ear-nose-throat doctors (Figure 45). For other specialties like chiropractors, optometrists, respiratory therapists, urologists, and anesthesiologists, there are fewer providers per person than the State of Michigan (Figure 45). Hospital closures in rural areas are associated with decreases in available specialty care (Rural Healthy People 2030 report). So while Ionia County has a similar number of hospitals per person than the State of Michigan (Figure 42), increased availability of specialty care has not followed with it. Many rural Americans who face a shortage of specialty care end up traveling into urban areas when needed. For those in Ionia County, this is likely to be areas around Grand Rapids and Lansing, for which travel times can be up to 30-60 minutes depending on where exactly someone lives within the county.

For mental health professionals and psychiatrists, Ionia County has fewer providers per person than the State of Michigan (Figure 46). Most rural areas in the United States are designated as having a shortage of mental healthcare professionals, even while rural populations are known to have a high prevalence of mental health conditions (Rural Healthy People 2030 report). A lack of treatment of mental health issues is often attributed to transportation issues for those in rural areas (Rural Healthy People 2030 report). The increasing availability of telehealth services for mental healthcare is seen a potential solution to nagging transportation, healthcare coverage, and other problems associated with disparities in in-person mental healthcare access (Rural Healthy People 2030 report).

11.1.2 Physical Health Conditions

Health conditions affect different parts of Ionia County more than others. Ionia County overall has higher rates of COPD (Figure 56) and Coronary Heart Disease (Figure 57) than the State of Michigan. For both conditions, the census tract most affected is tract 301.01, which contains Hubbardston and Palo.

The two ZIP codes most affected by reported poor physical health are Hubbardston and Palo (Figure 101). Census Tract 301.01 is also the tract in Ionia County most affected by arthritis (Figure 55), high blood pressure (Figure 59), high cholesterol (Figure 60), stroke (Figure 61), and the second most affected by asthma (Figure 58).

Heart conditions such as COPD and coronary heart disease are often associated with and coincide with other conditions and behaviors, such as aging, obesity, diabetes, and smoking.

Prevalence of heart disease increases with age (CDC), and Hubbardston has the highest percentage of individuals aged 65 and older (Figure 8). However, Palo has the lowest percentage of individuals aged 65 years and older (Figure 8), which conflicts with ascribing age to heart disease in both cases.

Obesity is associated with an increased risk of heart disease (Johns Hopkins). Every ZIP code in Ionia County is above the national average for obesity rate (Figure 103). Ionia City, Orleans, Hubbardston, and Palo have the highest rates of obesity of Ionia County ZIP codes (Figure 103). Ionia City, in tracts 317 & 321, Orleans, in tract 319, and Hubbardston & Palo, in tract 301.01 are all, except for tract 321, in census tracts

associated with higher rates of COPD (Figure 56). Five ZIP codes in Ionia County are above the national average for limited healthy food available (Figure 98). According to the Texas A&M Rural Healthy People 2030 report, grocery stores serving rural populations are more likely to serve unhealthy food than ones serving urban populations. For rural populations where large grocery stores aren't as available as corner stores, they're less likely to be able to buy fresh fruits and vegetables at smaller stores.

Diabetes is also associated with an increased risk of heart disease (American Heart Association). There are two ZIP codes associated with higher than average obesity nationally: Hubbardston & Palo (Figure 104). According to the Unmet Needs Report, despite being one of the least populated ZIP codes in Ionia County, only 3% of the population in Hubbardston is reported as having limited healthy food available (Figure 98).

Lastly, cigarette smoking is associated with increased heart disease (CDC). There are four ZIP codes with smoking rates of at least 20%: Ionia, Orleans, Hubbardston, and Palo (Figure 105).

11.1.3 Mental Health Conditions

According to the Rural Healthy People 2030 report by Texas A&M, mental health is the number one priority in rural health in a survey of rural stakeholders, and serious mental illnesses are significantly more prevalent in rural areas than urban ones.

While mental health conditions are often not as well tracked as physical health conditions, according to the HRSA Unmet Needs Report, every ZIP code within Ionia County has higher-than-average (nationally) reported rates of poor mental health (Figure 102). All the ZIP codes report a poor mental health rate of between 17% and 21%, which fits the ~ 20% rate for rural areas nationally, according to a 2010 literature review. It is possible the prevalence of mental health conditions in Ionia County is underestimated, as rural populations are often less likely to acknowledge their mental health conditions.

Suicide rates in rural areas tend to be higher than in urban areas, according to the same literature review. Ionia County has a higher suicide rate than the State of Michigan (Figure 73), including age-adjusted rates (Figure 72). Those with mental health conditions are at a much greater risk for suicide.

Depression, a mental health condition, often coincides with physical health conditions, according to the 2010 literature review. Physical conditions that often have depression alongside include "stroke (25 to 48 percent), coronary artery disease (8 to 44 percent), cancer (1 to 40 percent), Parkinson's disease (4 to 90 percent), and Alzheimer's disease (20 to 40 percent)." Nationally, less than 20% of adults living in rural areas with depression get treated by a mental health professional. Treatment is more likely to be done purely with medication by a non-specialized physician (Rural Healthy People 2030 report).

11.1.4 Opioid Abuse

According to the 2024 MDHHS Substance Use Vulnerability Report, Ionia County has a lower substance abuse vulnerability score compared to the State of Michigan average (Figure 89). Three ZIP codes within the county are above average for vulnerability score (Ionia, Orleans, and Hubbardston) and seven ZIP codes are below average for vulnerability score (Belding, Portland, Lake Odessa, Saranac, Lyons, Clarksville, and Palo) (Figure 90).

For Ionia, the city has a higher than average buprenorphine prescription rate (Figure 77), as well as having close to average rates of general opioid prescriptions (Figure 75), non-fatal & fatal overdose (Figure 79; Figure 81), and a lack of substance use treatment centers characteristic of the whole county (Figure 83).

For Orleans, the rate of opioid prescriptions per person is the highest of all the ZIP codes in the county (Figure 75). The buprenorphine prescription rate per person is largely above the state average (Figure 77). Additionally, The fatal overdose rate in Orleans is the highest of all the ZIP codes in the county (Figure 81). This combines to Orleans having the second highest substance use vulnerability metric rating of all the ZIP codes in Ionia County (Figure 90).

Lastly, for Hubbardston, their ZIP code also has a high rate of opioid prescriptions per person compared to the State of Michigan average (Figure 75). The non-fatal overdose rate is also highest in Hubbardston compared to all other ZIP codes in Ionia County (Figure 79). Mostly these metrics lead to Hubbardston having the third highest substance use vulnerability metric rating of all the ZIP codes in Ionia County (Figure 90).

In the Rural Healthy People 2030 report, addiction was ranked as the "second overall health priority for rural Americans for the coming decade." Overdose rates are much greater in rural areas than urban areas. Opioid overdose deaths doubled nationally between 2015 and 2016. Rural populations are uniquely affected by prescription opioid addiction. Compared with urban populations, rural populations are more likely to be prescribed prescription opioids, and less likely to be able to procure cheaply produced synthetic opioids that are more available to urban populations. Though, people in rural areas might start using synthetic opioids like heroin to cope with opioid withdrawal.

From the same Rural Healthy People 2030 report, Narcan/naloxone, an opioid overdose reversing medication, is less likely to be widely available to rural populations compared to urban ones. Based on current maps, there

appears to be a lack of reported Narcan vending machines in Ionia County (Figure 86). One limitation of Narcan vending machine maps is that they tend to require the medication to be available 24/7 to be listed, excluding other places where it is available mainly during the day.

Treatment centers for opioid and substance abuse disorders are not as widely available to rural populations as urban ones (Rural Healthy People 2030 report). There is only one reported substance use treatment disorder within Ionia County (Figure 83).

In the Rural Healthy People 2030 report, they acknowledge that policy changes in many rural areas meant to combat the opioid epidemic can have unintended consequences. For example, to address over-prescribing of opioids by physicians in the past, prescription limits, drug monitoring programs, and other things were introduced. Without proper alternatives however, those addicted to opioids often turned to heroin, fentanyl, or other drugs to supplement their addiction.

11.1.5 Vaccination Trends

Children in Ionia County are vaccinated at similar rates to children across the State of Michigan, though children in Michigan and Ionia County are less likely to be vaccinated than children across the United States (Figure 47; Figure 49). According to University of Michigan Medicine, many childhood immunizations have been declining in coverage since 2017, and have recovered following a sharp decline from the COVID-19 pandemic. Michigan had more than a dozen cases of measles in 2025, twenty five years after measles was declared to be eliminated in the United States. There were also more than 800 cases of pertussis (whooping cough) in Michigan in 2025. This coincides with vaccinations for measles and pertussis declining in recent years. In Ionia County, all vaccine coverages for newborns and children 19-35 months old have been on a decreasing trend since the beginning of 2024 (Figure 48).

Nationally, coverages for vaccinations available to receive in adulthood have been remaining steady or slowly increasing in recent years (CDC). In Ionia County, trends among adult vaccinations are similar to trends nationally, and have been slowly increasing since data has been tracked (Figure 52).

11.1.6 Pregnancy Conditions

Pregnant women in Ionia County are more likely to have excessive weight gain during pregnancy compared to pregnant women in the State of Michigan (Figure 64). According to Brown University Health, while healthy births are still possible, excessive weight gain during pregnancy is associated with gestational diabetes, sleep apnea, preeclampsia, and more need for a cesarean section. The children of excessive weight gain pregnancies are more likely who become overweight or obese. Excessive weight gain during pregnancies is also associated with postpartum weight retention. Doctors typically recommend losing weight before pregnancy, as well as eating a healthy diet with exercise during pregnancy.

Pregnant women in Ionia County have a higher rate of smoking cigarettes compared to pregnant women in the State of Michigan (Figure 65). According to the CDC, smoking during pregnancy carries risks for premature birth, lung and brain impairment in the baby, SIDS, small births, and abnormal bleeding during pregnancy. According to a 2025 article, smoking during pregnancy is more common in rural populations than urban ones. While Ionia County is higher than the State of Michigan average, it's still well below average for rural populations nationally.

11.1.7 Causes of Death

The leading cause of death in Ionia County is heart disease (Figure 72). This is consistent with the State of Michigan and for causes of death nationally. Compared to the age-adjusted rate of heart disease death in the State of Michigan, Ionia County is lower, but Ionia County's rate is still higher than the national rate (Figure 72). According to the American Heart Association, rural populations die from heart disease at higher rates than urban populations. The disparity between regions is seen most severely for adults under 65 years old. From 2010 to 2022, heart disease deaths in rural adults aged 25 to 64 increased by 21%, while heart disease deaths for adults 65 years and older *decreased* by 4.4%. This trend was not observed in urban populations, where heart disease death rates decreased for adults aged 25-64 and in adults 65 years and older. Researchers attribute these differences partly to high obesity rates in adults aged 40-59, as well as higher obesity rates for children in rural areas compared to urban areas. Other factors mentioned include disparate rates of smoking, poverty, and high blood pressure rates in rural areas.

The second leading cause of death in Ionia County is cancer (Figure 72). The age-adjusted cancer death rate in Ionia County is higher than the State of Michigan and national rates (Figure 72). According to the Texas A&M Rural Healthy People 2030 report, mortality from cancer are higher for rural populations than urban ones. Rural populations are more likely to be exposed to carcinogens such as smoke (primary or secondhand) and pollutants in the air and water. In Ionia County, there are many census tracts above the national average for percent of the population in proximity to a waste pollutant site, such as: six tracts for an EPA national priority

site, three tracts for an EPA Toxic Release Inventory (TRI) Site, 8 tracts for an EPA Treatment, Storage, and Disposal Facility (TSDF), and seven tracts for an EPA Risk Management Plan (RMP) site (Figure 93). In Ionia County, all the census tracts are above the national average for percent of the tract that intersects with an impaired or impacted watershed (Figure 94). According to the Texas A&M Rural Healthy People 2030 report, personal behavioral factors such as obesity, alcohol, and tobacco use are also associated with an increased risk of cancer. In Ionia County, all the ZIP codes have higher than average obesity rates nationally (Figure 103). Binge drinking rates in Ionia County are between 15% and 19% for all census tracts (Figure 62). Eight ZIP codes in Ionia County are also above the national average for smoking rates (Figure 105). Ionia County also has a lack of specialty care available beyond general medicine (Figure 45), which is consistent with a lack of specialty oncologists in rural areas nationally.

The third leading cause of death in Ionia County is stroke (Figure 72). The age-adjusted rate of stroke deaths is higher in Ionia County than the State of Michigan and national rates (Figure 72). Ionia County has 5 census tracts above the state average for stroke (Figure 61). According to the American Heart Association, incidents of stroke can be “up to 30% higher in rural areas compared to urban areas.” Those in rural areas tend to receive less thrombolytic treatment and tend to have slower treatment times compared to those in urban areas.

The rate of deaths by motor vehicle are 21% higher in Ionia County than the State of Michigan average (Figure 73). The rate of deaths by drunk driving are also 53% higher for Ionia County than the State of Michigan average (Figure 73). According to the Texas A&M Rural Healthy People 2030 report, deaths from motor vehicle crashes is higher in rural areas than urban ones. The fatality rate for vehicles in rural areas is nearly twice as high as the rate for urban vehicles. Also, crashes in rural areas tend to be more severe than crashes in urban areas. Even though driving fatalities are still higher for rural areas, the trend across the last decade has been a decrease of about 6% compared to a 49% fatality increase in urban areas. This trend is similar for drunk driving fatalities where the rate in rural areas decreased 7% across the 2010s, and increased in urban areas by 46%. In both urban and rural areas, while seatbelt use remained around 90%, around 50% of those who died in motor vehicle crashes in both areas were not wearing seatbelts.

Lastly, the age-adjusted rate of suicide death in Ionia County is 36% greater than the state of Michigan rate and 39% greater than the national rate (Figure 72). According to a 2026 article, rural areas have significantly higher age-adjusted suicide rates compared to urban areas, almost 50% higher. Across rural and urban areas, men have far greater suicide rates than women. Also across rural and urban areas, suicide rates are highest among adults 20-39 years old. This study identifies the lack of healthcare access common in rural areas compared to urban ones. A lack of mental health professionals, especially serving children, is associated with higher suicide rates. Youth suicide rates were found to be 4% higher in mental health shortage areas. In a 2020 survey of high schoolers in Ionia County, 40% of high schoolers reported experiencing major depression in the last year (Figure 121), and 9.4% reported attempting suicide in the last year (Figure 122). Chronic health conditions and behavioral health risks are often greater in rural areas compared to urban ones and are found to be associated with elevated suicide rates.

11.2 Socioeconomic Summary

11.2.1 Transportation

Ten ZIP codes in Ionia County are above the national average percentage of the population with no vehicle access (Figure 100). At least one in ten people in Ionia city & Hubbardston have no vehicle access (Figure 100). In a survey of EightCap program recipients, lower income respondents were more likely to report having problems with transportation, and the most common problems reported were cost of gas, not having a reliable vehicle or any vehicle, and a lack of public transportation (Figure 134).

While personal vehicles are the most common mode of transportation, according to the Rural Health Information Hub, some residents of rural areas cannot afford to buy or maintain a personal vehicle. People in rural areas are likely to travel farther distances to work than people in urban areas, and are therefore more affected by gas prices.

According to the Texas A&M Rural Healthy People 2030 report, those in rural areas are far less likely to use public transit to commute to work than those in urban areas. A variety of factors including low population density, geographically isolated populations, and underdeveloped infrastructure non-conducive to transportation infrastructure act as major barriers for public transportation in rural areas. Low population density in rural areas means you have fewer people paying taxes to fund infrastructure and services in a larger area compared to urban areas. However, demand for public transit in rural areas has been growing, especially for those who find it especially useful, such as older adults, people with certain types of disabilities, and low-income earners. Older adults are more likely to abstain from driving occasionally or entirely. Certain disabilities make driving more difficult or not possible, and accessibility measures may be less available in rural areas. Lastly, low-income earners are more likely to rely on public transportation because of cost concerns regarding the purchase and

maintenance of a personal vehicle.

According to the Rural Health Information Hub, transportation barriers in rural populations affect many parts of life, including healthcare, employment, and leisure. Most people in rural areas need transportation to and from medical services. Populations who need the most care might also face the most barriers with transportation, such as older adults and people with disabilities. Older adults and people with disabilities could also face more barriers to telehealth access, a strategy implemented to address transportation issues.

According to the Rural Health Information Hub, other ways in which rural areas face barriers to transportation include often worse road maintenance compared to urban areas, as well as inefficient communication between local, state, and federal agencies regarding transportation infrastructure and programs.

11.2.2 Food Insecurity

In a survey of EightCap program recipients, the most common household challenge was food costs (Figure 130), especially for respondents younger than 25 years old (Figure 132). 5 ZIP codes in Ionia County are above the national average for limited healthy food available (Figure 98). According to the Texas A&M Rural Healthy People 2030 report, “Rural communities are more likely to be disproportionately affected by food insecurity due to lower incomes, transportation barriers, and limited access to affordable nutritious foods.” According to Feeding America, there was an estimated food insecure population of over 9,900 people in Ionia County in 2024, about 14.9% of the population. Among the food insecure population, 45% are above the income threshold for the Supplemental Nutrition Assistance Program (SNAP). In a study of those on SNAP nationally, the majority felt that the assistance did not provide enough to feed their families. Feeding America projects that the food insecure population in Ionia County in 2025 would’ve needed an extra \$6.5 million to cover their basic food expenses, an average budget shortfall of ~\$650 per food insecure person for the year. After studying households receiving SNAP benefits, it was found that SNAP benefits help reduce poverty and food insecurity, whilst also helping boost the local economy through increased spending.

11.2.3 Housing

Only Ionia city and Palo are above the national average for housing stress, a metric that takes into account expenses, crowding, and incomplete maintenance (Figure 99). Only Lyons and Muir are above the national average for percent of households with high housing cost burden (Figure 19). According to the Texas A&M Rural Healthy People 2030 report, rural populations are more likely to have cheaper house costs and to own their own home compared to urban populations. These lower costs can be offset by a higher rate of high interest rate mortgages, according to the Office of Policy Development & Research. Nationally, houses in rural areas are more likely to be determined to be ‘physically inadequate’ because of structural and utility issues. In Ionia County, eight census tracts have a higher percentage of houses that were built before 1980 than the national average, potentially exposing people in those households to lead (Figure 95).

Ionia County has a smaller percentage of housing structures with ten or more units (Figure 16) than the State of Michigan average. Renters in rural areas, likely to live in housing structures with ten or more units, are more likely to be cost burdened than homeowners in rural areas nationally, according to the Office of Policy Development & Research. Also, it’s estimated that “only 54 affordable units exist for every 100 extremely low-income renter households in rural areas.”

Ionia County has a higher percentage of housing units that are mobile/manufactured homes than the state of Michigan average (Figure 24). Nationally, rural areas tend to have much higher percentages of housing structures that are mobile/manufactured units, according to the Office of Policy Development & Research.

Ionia County households are more likely to not have access to a computer with broadband internet than the state of Michigan average (Figure 26). According to the Texas A&M Rural Healthy People 2030 report, over half of the households without broadband internet access nationally are in rural areas, and urban households are getting broadband internet access at a faster rate than rural households.

11.3 Youth Trends

11.3.1 Substance Use

While the percent of high schoolers in Ionia County who smoked cigarettes has decreased to less than 5%, the percent who report using e-cigarettes (vapes) is over 20% (Figure 113; Figure 114). According to a 2020 article, using e-cigarettes can often lead to greater nicotine consumption compared to cigarette smoking. Adolescent nicotine consumption is associated with dysfunctional cognitive development and behavioral effects in animal studies. Also, adolescent e-cigarette usage is associated with higher rates of smoking later in life.

Fifteen percent of high schoolers in Ionia County reported recent alcohol consumption in a 2020 survey (Figure 115). This is lower than the 34% of high schoolers in Ionia County who reported recent alcohol consumption in 2014, but still greater than the 2020 state average rate of 14% (Figure 115). Alcohol is the

most common drug used by adolescents in the United States, according to the CDC. Underage (younger than 21 years old) alcohol use can alter a person's brain development. Higher rates of underage drinking are also associated with increased rates of drinking and driving. It's been shown that adolescents are four times as likely to drink if their parents regularly binge drink.

In a 2020 survey of high schoolers in Ionia County, 15%, or around 1 in 6, reported recent marijuana use, slightly higher than those who reported recent alcohol use (Figure 116; Figure 115). Marijuana usage among high schoolers is less for Ionia County than for high schoolers across the State of Michigan (Figure 116). According to the CDC, adolescent marijuana use is associated with impaired brain development with things such as memory, coordination, and attention span.

11.3.2 Mental Health & Safety

In multiple surveys of high schoolers in Ionia County, the rate of those reporting major depression has been increasing, from 30% in 2012, 31% in 2014, to 40% in 2020 (Figure 121). According to a 2025 article, rates of depression among youth populations in rural areas is higher than youth depression rates in urban areas. Youth in rural areas also have more barriers to access to mental health treatment, such as transportation, provider shortages, and stigma compared to youth in urban areas.

In a 2020 survey of high schoolers in Ionia County, 9.4% of high schoolers reported attempting suicide in the last year compared to 7.8% of high schoolers across the State of Michigan who reported attempting suicide in the last year (Figure 122). According to a 2025 article, youth in rural areas have suicide rates more than 70% higher than youth in urban areas.

Across multiple surveys of high schoolers in Ionia County, the rate at which high schoolers say they have 'felt unsafe' at school has risen from 3.2% in 2012, to 7.6% in 2014, finally to 10.1% (1 in 10) in 2020 (Figure 123). Students might avoid going to school because of feelings of unsafety from violence and bullying from their peers, fear of external violence in society, or other reasons. Feeling unsafe at school can cause students to avoid going to school, or be a distraction from succeeding in school.

12 Sources

12.1 Data & Maps

- GPS Map of Ionia County taken from Google Maps
- GIS Map made using QGIS and GIS data from Michigan Open GIS Data
- Demographic, income, poverty, veteran, disability, educational attainment, and health insurance data sourced from the United States Census Bureau
- Housing data & Substance Use Vulnerability Index data sourced from the Michigan Department of Health and Human Services
- Healthcare Access & Unmet Needs Report Data sourced from the Health Resources and Services Administration (HRSA)
- Healthcare & School Funding Time Series Data sourced from countyhealthrankings.org
- Vaccination data sourced from the Michigan Department of Health and Human Services
- Pregnancy and infancy data sourced from the Michigan Department of Health and Human Services Division for Vital Records and Health Statistics, specifically the 2024 Geocoded Michigan Birth Certificate Registry
- Health Condition Data sourced from Population Level Analysis and Community Estimates (PLACES) at the CDC
- HIV & STI data sourced from Michigan Department of Health and Human Services
- Leading Causes of Death data sourced from Michigan Department of Health & Human Services
- Substance Use Vulnerability Index Data sourced from Michigan.gov
- Syringe Service Program & Narcan Vending Machine maps sourced from Michigan Department of Health and Human Services
- All data on environmental health and climate was sourced from the Centers for Disease Control and Prevention and Agency for Toxic Substances Disease Registry: Environmental Justice Index (2024)

- High School Behavior Survey Data sourced from the Michigan Profile for Healthy Youth (MiPHY) online survey conducted bi-annually by the Michigan Department of Education and Michigan Department of Health and Human Services
- Survey of EightCAP Program Recipients provided by EightCAP
- Food Insecurity Measures sourced from Feeding America

12.2 Analysis

All analysis done in RStudio with *R* (4.6.0) and all plotting done with *ggplot2*.

12.3 Discussion

Sources used in the discussion are linked to directly in the body where they are referenced.